The Effects of Tipping on Workers and Firms

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Tipping is large and growing, but we know little about its consequences for workers and firms. We link point-of-sale and payroll records at 20,000 U.S. firms, which lets us observe tips from customer payment to worker earnings. We find that tips are a nearly constant share of revenue, which means hourly pay moves with revenue. Firms that begin collecting tips claw back between 23 and 33 cents of each dollar of tips through lower base wages. Tips reward effort: workers at tipping firms earn more by serving more customers. We show this with demand shocks constructed from customer spending at other businesses. A demand shock raises revenue nearly twice as much at tipping firms as at non-tipping firms, while base wages and prices do not respond. Output per hour rises more despite a larger increase in hours. Effort drives the gap because each worker is more productive, so the firm expands and recovers about a third of the tips. Yet not every firm adopts tipping: the gain is smaller where customers tip less. A model calibrated to the demand-shock estimates implies that tipping benefits workers and firms, raising job value by about half and firm profit by about 30 percent in firms where customers are prone to tip. These gains may explain why tipping has recently grown in many industries.

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Seven in ten U.S. adults report being asked to tip in more settings than five years prior (Pew, 2023).[1] In the point-of-sale data we use, the share of firms collecting tips in sectors where tipping is customary, such as coffee shops, bars, and restaurants, rose from 44 percent in 2013 to 70 percent by 2018, as checkout systems made tips easier to request (Figure 1). Since 2025, federal law has exempted the first $25,000 of qualified tips from income tax, so whether tips add to workers' pay or replace base wages is now also a question for tax policy (Balla-Elliott and Norwich, 2026).

Tipping splits the financing of labor costs between the firm, which pays the base wage, and customers, who pay tips directly to workers. Research on tipping has mostly asked why customers tip (Conlin, Lynn and O'Donoghue, 2003; Haggag and Paci, 2014; Azar, 2020; Thakral and Tô, 2026), not how tipping affects workers' earnings or firms' labor costs. How are the gains from tipping shared between firms that collect tips and their workers, and where do they come from? One possibility is that nothing changes: base wages fall one for one with tips (Wessels, 1997), the job is unchanged, and firms produce the same. Because tips are customarily calculated as fractions of per-transaction revenue, tipping could make hourly pay vary with revenue, thereby functioning as performance pay (Lazear, 2000; Lemieux, MacLeod and Parent, 2009). If tipping incentivizes effort and therefore raises output, both workers and firms could gain. However, output alone cannot reveal the extent to which tipping incentivizes effort, since output also varies with demand. The closest empirical work observes either the tips customers pay (Kohlhepp, 2025) or the tips workers report for taxes (Talesara and Wickard, 2025), but not both, so neither can measure the tradeoff between base wages and tips.

This paper studies how the gains from tipping are shared between firms and workers, and how much, if any, of those gains comes from the effort tipping induces. We use novel administrative point-of-sale and payroll records from a business technology platform covering 20,000 firms and 500,000 workers nationwide from 2022 to 2025. These records let us observe each tip dollar twice: when it is collected as a payment from the customer and when it is distributed to the worker as earnings. We can therefore verify that workers receive the tips customers pay; tax records, which understate tips, cannot.

We find that the gains from tipping are shared between workers and firms: base wages fall at firms that collect tips, but by less than the full value of tips, and nearly all of the gain comes from the effort tipping induces. First, among firms that begin collecting tips during the sample period, base wages fall by 33 cents per dollar of tips at the firm level, a share we call the *clawback*. Second, we develop a model in which tips function as performance pay, tying pay to revenue so that a rise in demand raises the return to effort at tipping firms. Third, a one-standard-deviation shock to customer demand raises hours more and output per hour about 47 percent more at tipping firms, consistent with demand raising effort. Finally, the model, calibrated without the adoption estimates, implies a clawback of 34 cents per dollar of tips at adopters' tip percent, close to the

[1]The spread of tipping has even reached the writers room of *The Simpsons*: in "The Tipping Point" (2024), Homer is annoyed that every cashier he pays asks him for a tip (Burns, 2024).

adoption estimate. In the model, tipping raises the value of a job by about half and the firm's profit by about 30 percent, against about 3 and 2 percent with effort held fixed.

Firms in our sample are small, mostly counter-service businesses that hire from the same pool of workers but differ in whether they collect tips. The sample spans food and beverage (53 percent of firms), retail (23 percent), and other small-business industries (23 percent). The median firm employs seven workers in a month, 13 over the year, and most are single establishments. We classify firms as tipping or non-tipping based on whether their median monthly tip percent (tips as a share of pre-tax, pre-tip revenue) is at least one percent. Food and beverage accounts for 78 percent of tipping firms and 10 percent of non-tipping firms, while retail accounts for 8 percent of tipping firms and 50 percent of non-tipping firms. In our sample, 98 percent of workers and 97 percent of firms are in sub-industries connected to one another by workers who move between them.[2] Within this labor market, workers earn comparable amounts each month ($1,817 at tipping firms versus $1,741 at non-tipping firms).

First, we estimate how base wages respond to tips from an event study of adoption, in which firms begin collecting tips during the sample period. Advances in technology have made it easier for firms to collect tips at the point of sale, including at the counter of a coffee shop, fast-food restaurant, or bakery, and firms in our data adopt at different times. We compare firms that adopt in a given month with firms that adopt later, which separates variation in the timing of adoption from variation in the decision to adopt. Our identifying assumption is that, absent adoption, outcomes at firms that adopt at different times would have evolved in parallel. Adopting firms are similar to non-tipping firms in size and base wages before they adopt and to established tipping firms in workers' monthly earnings afterward. The estimates apply to firms that already collect tips if the effect of adoption does not depend on when a firm adopts.

When firms begin collecting tips, base wages fall by about a third of the tips workers receive, so workers' hourly pay rises and adopting firms expand through hiring. At adopting firms, the tip rate (tip earnings per hour worked) rises by $1.47 per hour averaged over the five months after adoption, while the base wage falls by 48 cents. Firms therefore claw back 33 cents of each dollar of tips (standard error 9.6 cents).[3] Averaged over the adoption month and the three months that follow, hours and employment rise about 10 percent. Restricting to the 880 incumbent workers present before and after adoption holds the firm-worker match fixed. For these workers, firms claw back 23 cents over the five months after adoption, and by a year, the clawback is unchanged. A second design compares workers who move between tipping and non-tipping firms, netting out the pay change from moves between firms of the same type, and gives a clawback of about 39 cents for workers who move to tipping firms and 44 cents for workers who move to non-tipping firms.

[2]The connected set is the largest set of sub-industries that stays linked by workers observed in more than one when any single worker is dropped (Kline, Saggio and Sølvsten, 2020). Treating the same sub-industry in different states as separate, 82.3 percent of workers and 75.4 percent of firms remain in the connected set.

[3]Clawback refers to economic, not statutory, incidence. Tips are mechanically paid to workers in full, but if tip earnings make workers willing to accept lower base pay, part of the benefit accrues to firms.

The clawback ranges from 23 to 44 cents per dollar of tips across these designs and is 28 cents in related work studying firms that began collecting tips after the federal tip-income tax exemption (Balla-Elliott and Norwich, 2026).

Second, to explain why firms claw back only part of the tips and to quantify the gains from tipping, we develop a model in which tips function as performance pay, inducing workers to supply effort (Lazear, 2000; Lemieux, MacLeod and Parent, 2009). Although tips are discretionary customer payments, we find that they are a nearly fixed share of a firm's revenue from month to month, so they make hourly pay move mechanically with revenue. This share, the tip percent, differs across firms but is stable within them—a 10 percent increase in revenue raises tips by 9.7 percent. We assume that tipping and non-tipping firms have the same production function, in which output depends on hours and worker effort. Effort makes each worker more productive, and because the firm faces an upward-sloping labor supply, it expands to take advantage of higher output per worker, which requires keeping base wages above the level a one-for-one clawback would set. When demand rises, additional effort generates more revenue, so the marginal return to effort rises and workers at tipping firms supply more effort, raising both output per hour and hours.

Third, we isolate the demand response using a shock constructed from each firm's customers' spending at other businesses. Within a firm, hours and revenue move with demand but also with other factors, so their comovement alone does not isolate the demand response. Growth in a firm's revenue decomposes into growth in its customers' total spending and growth in the firm's share of that spending, and only the firm's share depends on its prices, hours, and effort. We observe the spending of approximately 200 million cards used at firms in our analytic sample, including their spending at businesses outside our analytic sample. We construct each firm's monthly demand shock as the sum of its customers' spending growth at other businesses, weighted by each customer's share of the firm's revenue in the previous month. Following Borusyak, Hull and Jaravel (2022), we identify the shock's effects from variation in the card-level spending changes rather than in the shares. The design requires that whatever drives customers to spend more elsewhere reaches these firms only as higher demand, not through a direct change in their workers' pay or productivity. Comparing level responses across tipping and non-tipping firms also requires that a one-standard-deviation shock represent the same change in demand at both, an assumption Section 5 states and supports.

Tips make hourly pay at tipping firms rise with demand, and tipping firms respond to the same shock with larger increases in hours and in output per hour. A one-standard-deviation shock raises hourly pay at tipping firms by 0.18 symmetric percentage points (sym pp) through tips alone, and has no effect at non-tipping firms.[4] Base wages and prices are rigid at both compensation structures, consistent with the model's assumption that base wages are set before demand is realized. The same shock raises revenue nearly twice as much (2.18 versus 1.14 sym pp) and hours nearly four times as

[4]A symmetric percent change is $200(x_t - x_{t-1})/(x_t + x_{t-1})$. It approximates a log change and is defined when the outcome is zero in one of the two months.

much (0.75 versus 0.21) at tipping firms.[5] Despite the larger increase in hours, output per hour rises more at tipping firms (1.43 versus 0.94 sym pp), not less. Most of this gain comes from serving more customers: transactions per hour rise by 1.32 versus 0.86 sym pp, a difference of 0.46 ($p = 0.01$).

The pattern at tipping firms is not explained by a larger shock, which would raise revenue, hours, and output per hour in the same proportions as at non-tipping firms. Instead, the hours response is 34 percent of the revenue response at tipping firms and 17 percent at non-tipping firms, a difference of 17 percentage points (standard error 8, $p = 0.03$). In the model, effort that rises with demand explains both the larger hours response and the larger output-per-hour response at tipping firms.[6]

Finally, the effort that raises output per hour when demand rises is enough to explain why firms claw back only part of the tips. We calibrate the model to hours, headcount, wage-bill shares, and the responses to demand shocks, but not to any estimate from adoption. The ratio of the hours response to the revenue response differs between tipping and non-tipping firms, and that gap pins down how much effort contributes to production and how steeply its cost rises. The more effort rises with demand, the more productive each worker at a tipping firm becomes, so the firm expands, which keeps the clawback below one for one. The model implies a clawback of 34 cents of each dollar of tips, against 33 cents at the firm level and 23 cents for incumbents.

In the model, tipping raises the value of a job by about half and the firm's profit by about 30 percent, and most of both gains comes from the effort it induces. With effort held fixed, the gains fall to about 3 and 2 percent. About four fifths of the firm's gain comes from employing more workers, because the effort each one supplies makes expansion profitable. The clawback understates workers' gain: the job's value counts the added hours, which carry no cost in the model, and nets out the effort cost. The gain is smaller at lower tip percents, so adoption is not profitable for every firm. Whether customers tip, and how much, depends on norms that differ across sectors (Conlin, Lynn and O'Donoghue, 2003; Azar, 2005).

We contribute to four literatures. First, we contribute to the empirical study of tipping by measuring tips where customers pay them and again where workers receive them, alongside base wages and hours, which prior work has not observed together. Most research on tipping studies the customer side: how defaults, menu options, prices, and social norms shape tips (Conlin, Lynn and O'Donoghue, 2003; Haggag and Paci, 2014; Azar, 2020; Hoover, 2022; Chandar et al., 2024*a*,*b*; Cross and Zhang, 2024; Donkor, 2026; Thakral and Tô, 2026). Studies of tipped workers' pay measure tips from W-2 tax records and either lack hours (Talesara and Wickard, 2025) or divide annual pay by hours reported in a survey (Jones, 2016). Tax records also understate tips: workers report about 55 percent of tips among jobs that report any, and more than a quarter of restaurants report none, with underreporting worse at small employers (Basker, Foster and Stinson, 2024, 2026). Basker,

[5] At non-tipping firms, hours rise despite no change in hourly pay, consistent with workers accepting the additional hours their employer assigns at their existing wage (Lachowska et al., 2026).

[6] The response to demand at tipping firms holds whether workers keep their own tips or share them with coworkers. Firms that pool tips respond at least as much as firms that pay each worker the tips from the transactions they serve. Transactions per hour rise 1.31 versus 1.06 sym pp, so the larger response does not require direct attribution.

Foster and Stinson (2024) end their sample in 2018 before "checkout systems that request tips spread across businesses"; our sample begins in 2022, as this technology spreads. At the median tipping firm, 97 percent of the tips customers pay at the register appear in workers' payroll records, which also report hours each pay period. Measuring tips, base wages, and hours at the same firm lets us estimate how base wages respond to tips and whether tips raise output.

Second, we contribute to the literature on the incidence of tipping on wages. Within tipping, Jones (2016) and Talesara and Wickard (2025) estimate how the tipped minimum wage affects tips (see also Wessels, 1993; Even and Macpherson, 2014; Allegretto and Nadler, 2015; Neumark and Yen, 2023, for effects on earnings and employment); we estimate substitution in the opposite direction. On platforms, Chandar et al. (2019) observe both tips and driver pay in a ten-day experiment on Uber, too short to precisely estimate whether base pay adjusts. More broadly, a large literature studies the incidence of payroll taxes, mandated benefits, and subsidies on wages (Summers, 1989; Gruber, 1994, 1997; Rothstein, 2010; Saez, Schoefer and Seim, 2019), where the gain is always set by the government. The closest case is Azmat (2019), where a tax credit paid through employers led firms to cut claimants' wages by 30 percent of the credit. In a setting where customers, not the government, determine the gain, we find that firms still recover about a third through lower base wages.

Third, we contribute to the literature on performance pay (Prendergast, 1999; Lemieux, MacLeod and Parent, 2009). Evidence that tying pay to output raises effort comes mainly from piece rates and commissions, where the employer designs the incentive and ties pay to a worker's own output (Oettinger, 1999; Lazear, 2000; Shearer, 2004). Most closely related is Oettinger (1999), who finds that workers on commissions put in more effort when demand is higher; tips are similar in tying pay to revenue but differ in that customers pay the incentive. Kohlhepp (2025) asks whether tipping is incentive relevant and finds that tip percents largely do not reward service quality (Lynn and McCall, 2000), reflecting defaults and norms instead (Conlin, Lynn and O'Donoghue, 2003; Haggag and Paci, 2014). We find that tips are incentive relevant through a different channel: because tip percents are stable within firms, workers at tipping firms earn more by serving more customers, so the incentive operates through revenue rather than through the tip percent. Incentive pay is hard to sustain when the measure it rests on cannot be verified (Malcomson, 1984; Baker, Gibbons and Murphy, 1994; MacLeod, 2003). A third-party point-of-sale system records every transaction, which makes tying pay to revenue feasible, as computerized cash registers and outside verification did for revenue sharing in video rental (Mortimer, 2008, p. 169). The classic account is that tips work because customers monitor service (Jacob and Page, 1980; Azar, 2004). Our evidence points instead to pay scaling with revenue verified by the point-of-sale system, not with service assessed by customers.

Finally, we contribute to the literature on the pass-through of firm-level shocks to workers (Guiso, Pistaferri and Schivardi, 2005; Kline et al., 2019; Lamadon, Mogstad and Setzler, 2022; Garin and Silvério, 2024). Prior work regresses wages on firm output or value added to estimate labor supply

elasticities (Lamadon, Mogstad and Setzler, 2022) or rent sharing (e.g., Garin and Silvério, 2024). Given the stable tip percents we document, pass-through at tipping firms reflects the tip percent as well as how base wages are set. We therefore estimate the response of each outcome to a demand shock directly. We construct the shock from customers' spending at other businesses, a shift-share design with the same structure as shocks built from export markets (Aghion et al., 2024; Garin and Silvério, 2024; Hoeck and Renkin, 2026) or house prices in other regions (Giroud and Mueller, 2019; Hyun and Kim, 2026). To our knowledge, no prior paper builds a demand shock from a customer spending network; the closest construction is Einav, Guido and Klenow (Forthcoming), who measure customer overlap across merchants but do not use it as a shock. We find that base wages and prices do not respond to demand at either compensation structure, but where tips tie pay to revenue, demand passes through to hourly pay.

The remainder of the paper proceeds as follows. Section 1 describes the linked point-of-sale and payroll records and the construction of the analytic and adoption samples. Section 2 documents that pay at tipping firms differs in composition, not level, and that tip percents are stable within firms. Section 3 estimates the clawback from firms that adopt tipping and shows that adopting firms hire more workers. Section 4 develops a model in which tips reward effort, linking compensation structure to the base wage, hours, and effort responses to demand. Section 5 constructs a demand shock from customers' spending at other businesses and estimates the response of base wages, hours, and output per hour. Section 6 calibrates the model to the demand responses and reports the incidence of tipping on workers and firms. Section 7 concludes.

1 Linked Transaction and Payroll Records at 20,000 Firms

We use administrative point-of-sale (POS) and payroll records from a large business technology platform, linked at the firm level. We access the data as unpaid independent contractors of the platform. The platform placed no restrictions on the topics, methods, or conclusions of the analysis. Its review of the paper is limited to confirming that no individual firm, worker, or the platform itself can be identified; it cannot require other changes or block publication. Together, these records let us observe firm revenue and prices alongside worker hours, wages, and tip earnings at the same firm. The linkage lets us trace a dollar from the customer's payment through the firm's payroll to the worker's hourly pay. Our tip measures therefore do not rely on the self-reports or tax filings that substantially understate tip earnings (Basker, Foster and Stinson, 2024).

1.1 Measures and data quality

The payroll data record individual worker paystubs: base wage rates, hours worked, overtime, and tip earnings. We aggregate paystubs to the worker-firm-month, and count every worker who received a paycheck as employed in that month. Paystubs are dated by pay date, which can fall up to four days after the end of the period worked, so a paycheck dated in month t can cover days worked in month $t-1$. The *base wage* is the hourly rate attached to the job. Base-wage responses use

a matched-worker index rather than the firm's average base wage. Each month's change is the average change in log base wage among workers the firm also paid the month before, weighted by their regular hours in that earlier month. Changes in which workers are paid therefore do not move the index, paralleling the price index, which compares the same items across consecutive months. The *tip rate* is tip earnings divided by hours worked, putting tips per hour on the same footing as the wage. The *combined rate* is the base wage plus the tip rate, the worker's effective hourly pay.[7] We distinguish worked hours, which exclude paid leave, from total hours; all per-hour measures use worked hours. Observing the base wage and tip earnings on the same paystub lets us estimate how much of each tip dollar firms recover through lower wages.

The POS data record itemized transactions: each record contains the items sold, their prices and quantities, the payment method, and the tip amount if any.[8] The platform also processes card payments, so card transactions carry anonymized identifiers that let us follow the same card's spending across firms, from which we construct a demand measure in Section 5. Cash transactions appear in the data only when the business enters them at the register. Revenue is constructed from all transactions net of tips and taxes. The *tip percent* is the share of pre-tip, pre-sales tax revenue that customers pay as tips; we construct it from the POS data. The itemized receipts let us construct a firm-level price index, which we pair with the demand measure to test whether firms adjust prices. When we restrict to card transactions, we say so. Section A.1 provides the full variable definitions and construction.

To be able to compare firms where customers tip to firms where they do not, we classify firms into the two compensation structures: tipping and non-tipping. To ensure the classification reflects stable behavior, we select each firm's longest window of at least eight months in which its compensation structure does not change. We classify a firm as *tipping* when its median monthly tip percent is at least one percent, measured as the larger of card tips and tip earnings divided by card revenue.[9] The analytic sample contains 13,000 tipping and 7,000 non-tipping firms.[10] Tips are 10.4 percent of revenue at tipping firms and 0.01 percent at non-tipping firms. Positive tip earnings appear in under three percent of non-tipping firm-months, likely reflecting occasional customer tips at businesses that do not solicit them. At non-tipping firms, the combined rate is therefore effectively the base wage. Section A.4 reports the denominator and alternative cutoffs.

The data linkage lets us check each record against the other. We apply two screens to keep only firms whose transaction and payroll records are internally consistent. The first compares monthly tip earnings in payroll to tip revenue in the POS data. It aims to catch firms where some or all of the tips collected are missing from payroll, or where payroll reports more tips than the transaction data

[7] The combined rate captures usual hourly pay. Overtime premiums, double-time premiums, and paid leave are infrequent and add noise to hourly measures without shifting the level.

[8] A point-of-sale system is the software and hardware a business uses to record sales; the data are what the register records.

[9] Since tips are mostly recorded via card transactions, we use that subset here. Switching to all-tender transactions reclassifies fewer than 50 firms.

[10] All counts are rounded for confidentiality purposes, and components need not sum to their rounded totals.

show, a sign of incomplete transaction records. The median removed firm passes under two percent of its POS tips through payroll. At the median tipping firm in the analytic sample, payroll tip earnings are 97 percent of point-of-sale tips (interquartile range 89 to 100 percent). A second screen removes firms whose wage bill routinely exceeds revenue, a sign of incomplete records regardless of whether the firm collects tips. Section A.2 details the thresholds and the distribution of pass-through ratios before and after screening.

1.2 Industries and workers at tipping and non-tipping firms

We observe 20,000 firms in three broad sectors: food and beverage (53 percent of firms), retail (23 percent), and other small-business industries (23 percent), such as personal care. Food and beverage accounts for 78 percent of tipping firms and 10 percent of non-tipping firms. Retail accounts for 8 percent of tipping firms and 50 percent of non-tipping firms (Table A1). The median firm employs seven workers in a month, eight at tipping firms and five at non-tipping firms. Ninety percent of food and beverage firms and 97 percent of retail firms in the data have fewer than 20 employees, compared with 77 and 91 percent in the same industries nationally (U.S. Census Bureau, 2024). The platform's payroll service is aimed at small firms, and larger firms either process payroll in house or use other providers (Grigsby, Hurst and Yildirmaz, 2021), so they do not appear in our data. The sample contains 500,000 workers at firms in all 50 states and the District of Columbia.

The hours, wages, and earnings in the data are consistent with national benchmarks for workers at small firms in these industries. Food and beverage and retail workers both average about 21 hours per week; the Current Employment Statistics (CES) reports 24 hours in food services and 30 in retail, reflecting large chains with longer schedules that are absent from our data (Bureau of Labor Statistics, 2026). The base wage at non-tipping retail firms ($16.40 per hour) falls between Bureau of Labor Statistics estimates for cashiers ($15.29) and retail salespersons ($17.90; Bureau of Labor Statistics, 2025). Monthly earnings at tipping food and beverage firms in 2022 ($1,953) are close to payroll per employee at small food-service firms nationally ($1,890; U.S. Census Bureau, 2024; Table A2).

Tipping and non-tipping firms hire in the same labor market. In the analytic sample, 98 percent of workers and 97 percent of firms are in sub-industries connected to one another by workers who move between them.[11] The sample includes workers who move between, for example, coffee shops and retail stores. At tipping firms, the base wage is $14.47 per hour and tips bring the combined rate to $20.61; at non-tipping retail firms, the base wage alone is $16.44. The difference in base wages is the subject of Section 3. Half of non-tipping firms are retailers, whose workers do the same counter and sales-floor work as workers at tipping firms. The demand-shock design in Section 5 uses a subset of these firms, mostly counter-service businesses in food and beverage and retail.

[11] We track workers across employers using anonymized identifiers and define two sub-industries as linked when at least one worker appears in both. The connected set is the largest group of sub-industries in which any two are linked directly or through a chain. We restrict to the *leave-one-out connected set* of Kline, Saggio and Sølvsten (2020), which is the largest subset that remains connected when any single worker is removed.

1.3 Sample construction

A firm enters the sample when both its transaction and payroll records are present, consistent with each other, and span a sustained window. The restrictions described above and in Section A.2 select firms with multiple employees, a sustained presence in the data, and records complete enough to support the per-hour, per-dollar comparisons the paper relies on. The analytic sample contains 20,000 firms observed from January 2022 through March 2025. January 2022 avoids any impact of the pandemic; March 2025 ends the sample before the "No Tax on Tips" provision of the One Big Beautiful Bill Act altered the incentives for reporting tip income (Balla-Elliott and Norwich, 2026).

The point-of-sale data extend back to 2013 and cover more firms than the analytic sample, which lets us document how tipping has spread. Figure 1 plots the share of firms collecting tips of at least one percent of revenue since 2013, separately for customarily tipped sectors, such as coffee shops, bars, and restaurants, and other sectors.[12] In the customarily tipped sectors, the share rose from about 45 percent in 2013 to about 70 percent by 2018 and has been flat since. In the other sectors, it rose from about 6 to about 11 percent by 2026 and is still rising.

We construct a separate sample for the adoption design in Section 3, which studies firms that begin collecting tips during the panel. Adopting firms meet the same size and presence restrictions as the analytic sample. Because they change compensation structure by definition, we select them before the quality screens and the stable-window requirement are applied; the tip-match screen in particular would select on a post-treatment outcome.[13] The adoption sample identifies 300 firms and 880 incumbent workers during the same sample period. Before they begin collecting tips, adopting firms resemble the non-tipping firms in the data in size, base wages, and monthly earnings; after adoption, their earnings rise close to the median at established tipping firms (Table A3). Food and beverage accounts for 32 percent of adopters, retail for 38 percent, and other sectors for 30 percent (Table A1).

2 Pay at Tipping and Non-Tipping Firms

Tipping splits the financing of labor costs between the firm, which pays a base wage, and customers, who leave tips at the point of sale. At tipping firms in the sample, 25 percent of worker pay arrives as tips; at non-tipping firms, the share is near zero. This section establishes three facts about pay at tipping and non-tipping firms. Total pay is similar at tipping and non-tipping firms despite a $3 per hour base wage gap; the tip percent is stable enough to function as a fixed link between revenue and worker pay; and tipped workers face substantially more hourly pay variation, yet monthly earnings dispersion barely differs.

[12] The classification uses each sector's average share of tipped firms in 2013–2015; sectors above 30 percent are classified as customarily tipped.

[13] After adoption, payroll tip earnings at the median adopting firm are 96 percent of the tips it collects at the point of sale, close to the 97 percent in the analytic sample.

2.1 Total compensation is similar at tipping and non-tipping firms

Workers at tipping firms earn about $3 per hour less in base wages than workers at non-tipping firms ($14.45 versus $17.61) but earn roughly the same amount each month. Mean monthly earnings are $1,817 at tipping firms and $1,741 at non-tipping firms, a difference of $76, or 4 percent. This difference decomposes into three components (Figure 2). Base earnings are $342 per month lower at tipping firms, reflecting the $3 hourly wage gap applied over slightly fewer hours.[14] Tip earnings at tipping firms are $453, about a quarter of total pay and more than the base-earnings gap. The residual from overtime, bonuses, and paid leave is $33 lower at tipping firms, small and unsystematic. The net difference is about one-fifth of the $342 gross base-earnings gap. Pay at tipping firms differs from pay at non-tipping firms far more in composition than in level.

Workers at tipping firms earn more per hour and work 6 percent fewer hours, so the rate premium and hours deficit partially offset in monthly pay (Table 1). The combined rate (base wage plus tips per hour) averages $21.09 at tipping firms and $18.53 at non-tipping firms, a 14 percent premium. Monthly hours average 83.4 at tipping firms and 88.4 at non-tipping firms. At the median, the earnings gap is larger: $1,361 versus $1,187, a 15 percent difference, because non-tipping firms employ more high-earnings workers in the right tail (Figure A2).

Every comparison in this subsection is across firms, so none says what happens to workers' pay when a firm begins collecting tips. Across firms, a dollar of monthly tip earnings ($452) is associated with base earnings roughly 76 cents lower ($342). This pattern is consistent with four stories: base wages are lower because tips replace part of them; tips induce effort, and workers must be paid a higher combined rate to supply it; tipping firms employ different kinds of workers; or tipping firms operate in industries with structurally lower wages. The first two are effects of tipping; the last two are selection. The 76-cent offset, the $2.56 rate premium, and the 4.9-hour gap are each a mix of the four. The cross-section can overstate or understate what tips add to workers' earnings. If tipping firms employ more productive workers, the comparison overstates what tips add. If tipping firms operate in industries with lower pay, it understates what tips add. Separating the effects of tipping from selection requires variation in tipping status within firms.

2.2 Tip percentages are stable across firms and over time

Tips are proportional to revenue: the median tipping firm collects about ten cents in tips per dollar of revenue, and the proportion holds across firms of very different sizes (Figure 3a). The slope is 0.094: a firm with $100,000 in monthly revenue collects roughly $9,400 in tips. The median tip percent, tips collected at the point of sale divided by revenue, is 10.2 percent across tipping firm-months. This average includes transactions where no tip is left; on transactions where customers do tip, they tip 18.2 percent of the pre-tax bill.

[14] Base wages at tipping and non-tipping firms alike bunch at statutory minimums and round dollar amounts; the tipping distribution has more mass below $15 per hour (Figure A1).

Across firms, tips are proportional to revenue, but this pattern could reflect sorting if higher-revenue firms attract more generous customers. Within a firm, the relationship is nearly the same. Residualizing log tips and log revenue on firm and calendar-month fixed effects gives an elasticity of 0.974 (standard error 0.006, clustered by firm), statistically below one but within 3 percent of one, the value implied by a fixed tip percent (Figure 3b).[15] A 10 percent revenue increase in a given month raises tips by 9.7 percent. Revenue explains 68 percent of the within-firm variation in tips. The R^2 rises with tipping intensity, from 0.72 at moderate-tipping firms to 0.83 at high-tipping firms, while low-tipping firms with a median tip percent of 2.5 percent pull the pooled figure down (Figure A3a).[16]

Month-to-month movements in the tip percent are small relative to its level. In 78 percent of firm-months the tip percent moves by less than one percentage point, and in 93 percent by less than two (Figure 3c). For the median tip percent of 10.2 percent, a one-percentage point change is a 10 percent fluctuation in the tip percent, and most months have a smaller change. The typical firm's tip percent varies by 0.79 percentage points around its own mean.[17] Nor does the tip percent drift over the sample period: the median is flat within each tipping category from 2022 through 2025 (Figure A3b).

Any single tip left by a customer is discretionary: the customer chooses whether to tip and how much. Aggregated over a month of transactions, that discretion averages out, and what remains is proportional to revenue. A firm that can forecast its revenue can therefore forecast the tips its workers will earn; tips introduce no additional pay variation beyond the demand variation the firm already faces. Tipping converts a share of each transaction into worker pay, and that share is stable enough that tips are a mechanical link from firm output to worker pay.

2.3 Tipping firms have less volatile revenue but more volatile pay

Tipping firms have less volatile revenue than non-tipping firms, consistent with tipping being viable where demand is relatively stable. For each firm, we compute the standard deviation of log monthly revenue (net of tips and taxes) over the firm's panel, after removing month effects common to all firms.[18] The unconditional difference in SD(ln Revenue) between tipping and non-tipping firms is -0.123 ($p < 0.001$). Conditioning on firm age as measured in the data, months in the analytic sample, sub-industry seasonality, state, sub-industry, firm size, transaction volume, and bill-size dis-

[15] Without fixed effects the log-log elasticity is 1.02, so the two estimates bracket one.

[16] The category medians divide tips by revenue, as in the rest of this subsection. The classification that defines the categories instead divides the larger of card tips and payroll tip earnings by card revenue, which places the same firms above the tipping threshold but at slightly higher tip percents.

[17] In relative terms, the within-firm coefficient of variation has a median of 0.083. Firms with low tip percents show higher coefficients of variation (median 0.27 in the low-tipping category versus 0.06 among high-tipping firms), but this is a denominator effect: the within-firm standard deviation grows only slowly with the level (slope 0.031 against the median). Because the standard deviation rises with the level, high-tipping firms show the largest changes in percentage points: the tip percent moves by less than one percentage point in 71 percent of their firm-months, against 86 percent at low-tipping firms (Figure A3c).

[18] The revenue measure is pre-tip, pre-tax transaction revenue. This ensures that tip income, the object of interest on the worker side, does not enter the revenue volatility comparison.

persion reduces the gap to −0.041 (Figure 4a, full specification in Table A4). Firm age and months in the sample leave the gap almost unchanged; sub-industry seasonality reduces it to −0.097, and sub-industry to −0.065. Revenue variance decomposes almost entirely into transaction-count variance, which accounts for 113.9 percent of total revenue variance at tipping firms; price variance accounts for 15.0 percent at tipping firms and 23.3 percent at non-tipping firms, well below transaction-count variance at both (Table A5). Tipping is more common among firms where this stability makes tip income a predictable pay component.

Turning to the worker side, tipping firms expose workers to substantially more hourly pay variation even though their revenue is less volatile. Within a worker's spell at a single firm, the combined hourly rate deviates from its own average by a median of $0.85 per month at tipping firms and $0.00 at non-tipping firms (Figure 4b; quartiles in Table 2).[19] The non-tipping 75th percentile is $0.50, still below the tipping median. Over half of non-tipping spells (53 percent) show no deviation in the combined rate at all, compared with 7 percent of tipping spells. A zero median is consistent with the prevalence of nominal wage rigidity: Grigsby, Hurst and Yildirmaz (2021) find that 35 percent of U.S. job-stayers receive no nominal base wage change over a full year, and Ganong et al. (2025) report that hourly workers' base wages change in only 12 percent of months. At tipping firms, tips add a pay component tied to each month's transaction volume, and this component generates the $0.85 gap.[20]

Despite the large gap in hourly pay variation, monthly earnings dispersion is nearly identical at tipping and non-tipping firms. The within-worker MAD of monthly base-plus-tip earnings, expressed as a share of the worker's mean, is 30.0 percent at tipping firms and 28.0 percent at non-tipping firms (Table 2). Most monthly income variation comes from the quantity margin (how many hours a worker is scheduled), not the price margin (what they earn per hour). The rate variation that tipping introduces is small relative to the hours variation that tipping and non-tipping firms share; on its own, it barely moves the monthly total.

The small difference in monthly earnings dispersion conceals a structural difference in how rate and hours variation interact at tipping firms. The identity $\ln E = \ln w + \ln h$, for monthly base-plus-tip earnings E, combined hourly rate w, and hours h, gives $\mathrm{Var}(\ln E) = \mathrm{Var}(\ln w) + \mathrm{Var}(\ln h) + 2\,\mathrm{Cov}(\ln w, \ln h)$. Total mean within-spell earnings variance is 15 percent higher at tipping firms (0.341 vs. 0.296). Rate variance is 2.86 times higher at tipping firms (0.015 vs. 0.005) while hours variance is similar (0.316 vs. 0.290), confirming the margins behind the MAD comparison (Fig-

[19] A spell is the longest consecutive run of months with positive paid hours (≥ 5) at a single employer. Hourly rates divide by regular plus contractor hours, since the hourly rate applies to both. We require a minimum of three months, which retains 74 percent of worker-firm spells. Before this restriction, median spell length is five months; among retained spells, it is seven months at tipping firms and eight at non-tipping firms.

[20] Including all pay components (overtime, bonuses, paid time off), the non-tipping median MAD rises to $0.22 per hour, so the tipping median is then roughly four times the non-tipping median. "Other" earnings account for 50 percent of within-worker earnings variance at non-tipping firms and 12.2 percent at tipping firms. Because these components are lumpy and unrelated to tipping, all volatility measures in this section and the rest of the paper use base wages plus tips only.

ure 4c).[21] The covariance between the rate and hours is an order of magnitude larger at tipping firms (0.010 vs. 0.0005). At tipping firms, months with higher hourly pay are also months with more hours scheduled; the two sources of variation compound rather than offset.

The rate-variance gap narrows with tenure as non-tipping workers accumulate infrequent base wage changes, not because tipped workers stabilize. The rate-variance ratio falls from 2.86 at a three-month minimum spell length to 1.84 at twelve months (Table A6). Tipping rate variance is roughly flat across thresholds (0.015, 0.015, 0.016), while non-tipping rate variance rises from 0.005 to 0.009 as longer spells allow mid-spell raises to register. This reflects the timing of base-wage adjustment: Hurst et al. (2026) show that most firms concentrate annual raises in a single month, and 78 percent of workers who receive a raise get exactly one per year. A five-month spell will often miss this annual adjustment entirely; a twelve-month spell nearly always spans it. Tip income varies at a higher frequency. At a twelve-month minimum spell length, the median combined-rate MAD is \$1.14 at tipping firms and \$0.48 at non-tipping firms; the non-tipping median rises from zero as the observation window spans the annual adjustment.

Tipping creates a specific, high-frequency source of hourly pay variation that a fixed base wage would otherwise absorb. The combined rate moves \$0.85 per month at tipping firms and not at all at non-tipping firms, yet monthly earnings dispersion barely differs (30 percent vs. 28 percent of the worker's mean). Rate and hours variation compound at tipping firms rather than offset, a pattern absent at non-tipping firms. What the cross-section cannot establish is why rate and hours move together, or whether the rate variation that tipping introduces is a cost to workers or a benefit. Answering those questions requires within-firm variation in tipping and a framework that connects hourly pay to the demand conditions firms face.

3 The Base-Wage Response to Tipping Adoption

Firms recover about a third of each dollar of tips through lower base wages, and workers keep the rest. Among 300 firms that adopt tipping during the sample period, base wages fall by 33 cents per dollar of tips ($\kappa = 0.33$, standard error 0.10); among incumbent workers at those firms, by 23 cents ($\kappa = 0.23$, standard error 0.07). Incumbent workers' combined hourly rate rises 7 percent, and their hours do not change after the adoption month, so the tips they keep represent a per-hour raise. Adoption also expands the firm: labor hours rise 9.8 percent in the five months after adoption, almost entirely through headcount, and transactions and revenue rise with them, at unchanged prices and an unchanged labor share of revenue. Tipping expands the firm's scale: revenue increases, with workers paid in part by customers.

[21] The 0.045 log-variance gap between tipping and non-tipping firms splits into 0.010 from rate variance, 0.026 from hours variance, and the remainder from the covariance.

3.1 The incidence of tips

Section 2.1 established that workers at tipping firms earn about the same each month as workers at non-tipping firms, despite a base wage that is $3 per hour lower. Monthly pay parity is consistent with two arrangements. In the first, base wages fall dollar for dollar when tips arrive: tips replace wages, and nothing about the job changes when the firm starts collecting them. In the second, collecting tips changes the job, workers are compensated for the difference, and base wages fall by less than the tips. Whether workers earn rents on that difference or only a compensating differential for it, the firm recovers only part of the tips. How tips are split between workers and firms is an incidence question: the statutory flow is to workers, but base wages may adjust, just as they do for payroll taxes (Gruber, 1997) and tax credits (Rothstein, 2010). We define the clawback as the share of each dollar of tips the firm recovers through lower base pay:

$$\kappa \equiv -\frac{\Delta w_{\text{base}}}{\Delta(\text{tip rate})} \tag{1}$$

where both quantities are in dollars per hour. At $\kappa = 1$, each dollar of tips displaces a dollar of base wages and expected total compensation is unchanged. At $\kappa = 0$, tips are a transfer from customers to workers with no base wage response.

The cross-sectional earnings comparison from Section 2.1 implies $\kappa \approx 0.76$, but this estimate conflates the base wage response with differences in hours, workforce composition, and the types of firms that choose to tip. To isolate the base wage response, this section uses firms that adopt tipping during the sample period. At the firm level, base wages fall by 33 cents per dollar of tips ($\kappa = 0.33$, standard error 0.10). Among the 880 incumbent workers at adopting firms observed throughout the event window, base wages fall by 23 cents per dollar of tips ($\kappa = 0.23$, standard error 0.07). The gap between the two is composition. The firm-level base wage falls in the adoption month itself, while incumbents' base wages are unchanged in that month and fall over the following five. New hires enter at lower base wages than the workers already there. Under either estimate, workers keep at least 70 cents of each dollar of tips. Tips are not, for the most part, a substitute for wages the firm would otherwise have paid.

The magnitude of κ is informative about what workers are being paid for. At full offset, expected compensation is unchanged, but tipping still exposes workers to demand-linked pay variability and effort incentives, since tips are a stable share of what customers spend. A worker who supplies more effort when paid in tips, or who dislikes pay that varies with demand, is worse off under this arrangement than with the same compensation paid as a fixed wage, and would not accept it without being compensated for the difference. If the base wage fell too far, workers would leave for non-tipping employers offering a higher guaranteed rate. That κ is well below one indicates workers are being paid for something that tipping introduces. It also raises a question about the firm: if it recovers only a third of the tips, why begin collecting them at all? The explanation this paper develops is that tips induce effort, and the firm's gain from adopting is the additional output that

effort produces.

Compensation is not all that changes at adoption. Adopting firms also hire, serve more customers, and charge the same prices, so that the labor share of revenue is unchanged.

3.2 Identification and estimation of tipping adoption effects

We estimate the effects of adoption on pay, labor, and sales from a stacked difference-in-differences event study of firms that begin collecting tips during the sample period and their incumbent workers (Deshpande and Li, 2019; Wing, Freedman and Hollingsworth, 2024). A firm adopts tipping in the first month in which payroll records tip earnings, after at least six months with none; Section B.1 gives the full definition. The firm-level sample contains 300 firms that adopt tipping during the sample period, with later-adopting firms as controls and an event window from five months before adoption to five months after. The worker-level sample is the 880 incumbent workers who are observed in every month from five months before adoption to five months after. The adopting firms begin collecting tips years after the share of firms in customarily tipped sectors that collect tips leveled off at about 70 percent in 2018 (Figure 1).

For each adoption month, we build a *slice* containing either the firms that adopt that month or the incumbent workers at those firms, as well as the later-treated units. The firm-level March 2023 slice, for example, covers October 2022 to August 2023 and contains the firms that adopt tipping in March 2023 as well as any firm that adopts from April 2023 onwards in the months before it adopts so long as it meets the treatment definition described in Section B.1. We stack the slices and estimate the following equation twice, once with firms and once with incumbent workers as the unit of observation:

$$Y_{its} = \psi_{is} + \phi_{ts} + \sum_{\ell \neq -1} \beta_{\ell s} \, \mathbb{1}\{c(i) = s, \; t - s = \ell\} + \varepsilon_{its} \tag{2}$$

where Y_{its} is the outcome for firm or worker i in month t of slice s, indexed by its adoption month; $c(i)$ is the month in which i's firm adopts; ψ_{is} and ϕ_{ts} are unit and calendar-month effects specific to the slice; and ℓ runs from -5 to 5. The event-study coefficient β_ℓ averages $\beta_{\ell s}$ across slices, weighting each slice by its treated observations at month ℓ. Standard errors are clustered by firm, since a firm's workers share shocks and a firm serves as a control in every slice for an earlier adoption month. Section B.2 details which workers and firms enter each slice.

Under the standard common trends and no anticipation assumptions, β_ℓ is the effect of adoption on the outcome ℓ months after adoption. The common trends assumption requires common untreated outcome trajectories across all cohorts. No anticipation requires that outcomes do not respond before payroll records tips—that firms do not adjust operations in ways that impact pay, labor, or sales systematically before then.

3.3 When firms adopt tipping, base wages absorb a third of the tips

Before they begin collecting tips, adopting firms resemble the non-tipping firms in the data: the median adopting firm employs six workers at a base wage of $15.50 per hour, with monthly earnings of $1,124, against five workers, $16.00, and $1,183 at non-tipping firms. After adoption, median monthly earnings rise to $1,300, within four percent of the $1,361 median at established tipping firms (Table A3). In the month before adoption, the median tenure across incumbent workers is 14 months of active employment at the firm, against 16 months for workers at non-adopting firms who are present throughout the same calendar windows.[22] Base wages at adopting firms sit far above the tipped subminimum wage in every state, so tip-credit reporting incentives of the kind documented by Basker, Foster and Stinson (2026) cannot drive the wage response below. Tip earnings in payroll track tips collected by the firm: at adopting firms, payroll tip earnings are 96 percent of POS tips collected at the median firm after adoption, so the tip rate is not understated by tips paid outside payroll.[23]

After adoption, tips at adopting firms average about 8 percent of pay and 2.7 percent of revenue. About half of the workers at an adopting firm earn any tips in a given month after adoption, while 31 percent have tips account for more than 10 percent of their combined pay. Figure 5 shows that adoption is a sharp change in how workers are paid: the share of workers whose tips exceed 5 percent of pay rises from zero to 31 percent in the adoption month and 40 percent (standard error 2) over the five months after.[24]

Figure 6 shows the raw compensation series around adoption at both the worker and firm level. Controls receive steady raises of about 12 cents per month at the worker level (Figures 6a and 6b) and 7 cents at the firm level (Figures 6c and 6d). Treated workers' base wages flatten at adoption: by the fifth month, controls have risen 70 cents above their pre-adoption level while treated workers have risen only 32 cents, a gap of 38 cents (Figure 6a). The firm-level base wage (Figure 6c) diverges from controls in the adoption month itself; incumbents' base wages (Figure 6a) diverge one month later. The combined rate rises sharply in both samples, reaching about $1.10 above controls within two months (Figures 6b and 6d).

Figure 7b plots the base wage, the tip rate, and the combined hourly rate for incumbent workers around adoption. Both compensation series are flat before adoption, with no pre-period coefficient on the base wage exceeding $0.07 in absolute value. The tip rate rises by $0.86 per hour in the adoption month and by $1.43 per hour on average over the following five months. The base wage does not move in the adoption month and then falls steadily, reaching $0.41 per hour below the counterfactual by the fifth month; averaged over months one through five, the decline is $0.33 per

[22] Each adoption cohort's eleven-month window is applied to non-adopting firms, tenure is measured in the month before that cohort's adoption, and cohorts are weighted by their number of incumbents.

[23] The mean ratio exceeds one because payroll can record tips a month after the point-of-sale data does; the median is therefore more informative.

[24] The share earning any tips jumps from zero to 56 percent (standard error 1.6) at the adoption month. Over the five months after the adoption, 59 percent of workers earn tips in at least one month.

hour. The combined rate rises by $1.09 per hour, or about 7 percent, over the same window. The implied clawback is $\kappa = 0.33/1.43 \approx 0.23$ (standard error 0.07): for each dollar of tips incumbent workers begin to earn, the firm recovers 23 cents through lower base pay, unchanged over the first year after adoption.[25] In Balla-Elliott and Norwich (2026), we estimate a clawback of 28 cents per dollar of tips induced by the 2025 No Tax on Tips provision, using policy-driven adoption with the same data source.

At the firm level (Figure 7a), which includes workers hired after adoption, the base wage falls by $0.48 per hour against a tip rate of $1.47, so $\kappa = 0.33$ (standard error 0.10). Unlike the incumbent series, the firm-level base wage falls in the adoption month itself, by 27 cents per hour. The timing and the new-hires result in Section 3.4 suggest that workers hired after adoption enter at a lower base wage than incumbents.

Workers who move between tipping and non-tipping firms face a clawback of about 40 cents per dollar of tips, and its confidence interval includes both within-firm estimates. Section B.4 reports the stacked difference-in-differences and triple-difference designs behind that number.

3.4 Adopting firms expand: more workers, more customers, same prices

The tips the firm does not claw back attract additional labor: in the five months after adoption, firms increase total hours worked by 9.8 percent. Figure 8 summarizes eight firm-level outcomes, each as the average event-study coefficient before and after adoption: hours worked, employment, new hires, and hours per worker in the top row; transactions, revenue, the price index, and the labor share of revenue in the bottom row. As the combined hourly rate rises by 6.3 percent, labor hours rise by 9.8 percent over months 1 to 5 after adoption, while employment rises by 7.5 percent. If the added labor is workers responding to higher pay, the ratio of the two is the elasticity of labor supply to the firm, 1.6 for hours and 1.2 for employment, within the range of estimates for individual employers (Falch, 2010; Naidu, Nyarko and Wang, 2016; Caldwell, Dube and Naidu, 2026) and the level expected for a response measured five four months rather than several years (Kroft et al., 2025; Caldwell, Dube and Naidu, 2026).[26]

The response to higher pay is on the employment margin: adopting firms hire more workers, and the workers already there keep the same hours. Hours per worker stay within two percent of their pre-adoption level over the five months after adoption. We cannot rule out a modest hours response among incumbents: the confidence interval on their post-adoption hours spans −5.4 to +1.3 percent, so an elasticity of hours to pay of up to about 0.2 is consistent with the data, below the intensive-margin Hicksian elasticity of 0.33 in Chetty, Raj (2012). What the data rule out is that adopting

[25] Both averages exclude the adoption month because its payroll records can cover days worked before the firm began collecting tips (Section A.1). The standard error uses the delta method, with the covariance of the two averages from estimating Equation (2) with the base wage minus the tip rate as the outcome.

[26] Each elasticity is the ratio of the labor response to the combined rate response, both averaged over months 1 to 5 after adoption. Hours worked is estimated in levels and expressed as a percent of the month −1 mean; employment and the combined rate are in logs.

firms met their demand for additional labor by lengthening incumbents' hours. New hires rise from 1.8 per month to 2.3 over months 1 to 5, while separations do not change, among all workers or among incumbents. Employment is 7.5 percent higher because more workers were hired, not because fewer left.

The new workers serve more customers at unchanged prices. Transactions rise by about 200 per month over months 1 to 5, against a pre-adoption mean of 1,220 (Figure 8). Revenue rises in step with labor, 9.9 percent on average over months 1 to 5 against 9.8 percent for hours. Prices do not respond to adoption: the average coefficient over months 1 to 5 is -0.028 index units (standard error 0.049), and no individual post-adoption coefficient is distinguishable from zero. The revenue increase is entirely a quantity increase. The firm sells more at its existing prices, which means the constraint on output before adoption was not customer demand at those prices but the labor available at the pre-adoption wage. The labor share of revenue confirms this: the wage bill excluding tips divided by revenue is 0.29 at the median adopting firm before adoption, and the post-adoption change is -4.6 percent (standard error 4.3). Because the tip percent is stable (Section 2.2), the tip rate tracks revenue per hour worked, and it reaches its post-adoption level within two months and is flat over the four months after that.

Adoption changes both what workers earn and how much the firm produces. Base wages absorb about a third of tips; the firm hires, output rises at unchanged prices, and the labor share of revenue does not move. Because the labor share is unchanged, the 9.9 percent increase in revenue is not absorbed by a larger wage bill: revenue net of base wages rises by the same amount. Tipping does not change the firm's revenue per unit of labor cost; it lets the firm produce more, with customers funding part of the wage bill.

4 Wage Setting and Production with Demand-Contingent Pay

In this section, we develop a model of wage setting and production with demand-contingent pay. Firms choose a compensation structure, post a base wage, and set hours after observing demand. Workers choose effort and which employer to work for. The model determines base wages, hours, effort, and employment under each compensation structure, given a common demand distribution. We use it to assess the value of tipping to workers and firms and to link compensation structure to the base wage, hours, and effort responses to demand.

Tips make hourly pay move with demand, giving workers at tipping firms an incentive to supply effort that a fixed base wage does not provide. The firm posts a base wage and recruits workers before observing demand. Each month the firm draws demand, sets hours, and the worker chooses effort. At tipping firms, a fraction of per-worker revenue is paid to the worker as tips, so tips act as performance pay (Lazear, 2000): the worker has an incentive to raise output that workers at non-tipping firms lack. Tips are a revenue share whose rate customers set; because the point-of-sale system records every transaction, tying pay to revenue is feasible (cf. Mortimer, 2008, p. 169).

Because tips reward effort and base wages do not, the two compensation structures differ in the marginal return to effort. The effort tips induce raises the revenue each worker generates, which raises profit per worker. Because each firm faces an upward-sloping labor supply, the higher profit makes the firm want to hire more workers, and attracting them requires a base wage that does not fall one for one with tips.

4.1 Environment

Technology. Firm j employs N_j workers with identical productivity and effort costs. Each worker works h hours and chooses effort e, producing per-worker revenue[27]

$$y = a g(h, e) \qquad g(h, e) = \left[\alpha h^{-\theta} + (1-\alpha) e^{-\theta}\right]^{-\nu/\theta} \tag{3}$$

The demand state a scales the revenue generated by a given combination of hours and effort. Demand can mean more customers or larger orders; a shifts total revenue at given hours and effort. The production function g is increasing in both arguments, homogeneous of degree ν, with elasticity of substitution $1/(1+\theta)$ between hours and effort. When $\theta > 0$, both inputs are essential: $g(0, e) = g(h, 0) = 0$. Revenue at given a, h, and e is the same with or without tipping.[28] Adding workers replicates per-worker production, so total production is linear in employment at given h and e. There are no non-labor variable costs.

Compensation. A fraction τ_j of per-worker revenue goes to the worker as tips; the firm retains $(1-\tau_j)y$, and pays the hourly base wage w_j. Tipping firms have $\tau_j = \tau_T > 0$; non-tipping firms have $\tau_j = 0$. Within each firm type, we assume that the tip percent is fixed, consistent with the stability documented in Section 2.2. Tips relative to sales net of tips, which Section 2.2 measures, equal $\tau_j/(1-\tau_j)$. Operating profit per worker is therefore $(1-\tau_j)y - w_j h$, the revenue the firm retains net of the base wage bill. The production function, demand distribution, worker preferences, and recruitment environment are common across the two compensation structures. Each firm draws a fixed cost F_j that remains constant over time and is paid each month the firm collects tips. Variation in F_j can explain why some firms adopt tipping while others do not; the adoption decision is in Section 4.4. Conditional on τ_j, all firms face the same problem; we drop the j subscript and index by τ when solving it.

Preferences. Workers differ in their preferences over employers. Worker i's payoff from working at firm j is

$$U_{ij} = \underbrace{\mathbb{E}_a\left[w_j h_j + \tau_j a\, g(h_j, e_j) - c(e_j)\right]}_{\equiv V_j} + \lambda \varepsilon_{ij} \tag{4}$$

The taste shock ε_{ij} is type-I extreme value, independent across workers and firms. Each worker

[27] Lowercase letters denote per-worker quantities; $Y_j = N_j y$ is firm revenue and $H_j = N_j h$ is total hours.

[28] That is, the revenue function does not depend on whether the firm collects tips. Prices, the number of customers, and how customers value effort can all vary with hours and effort; the assumption rules out tipping changing revenue at given a, h, and e.

chooses the firm with the highest U_{ij}. The parameter $\lambda > 0$ is the dispersion of workers' idiosyncratic preferences, in dollars; $1/\lambda$ is the semi-elasticity of recruitment to job value (the percent increase in recruitment from a dollar increase in job value). Workers are risk neutral and have no disutility of hours. The no-hours-disutility assumption follows Lachowska et al. (2026), who document that most hourly workers would accept more hours at their current wage. If extra hours were costly to workers, the gain from added hours at tipping firms would be smaller.

Effort costs the worker

$$c(e) = \frac{k}{1+\gamma}\left(\frac{e-e_0}{e_0}\right)^{1+\gamma} \tag{5}$$

with $c(e_0) = c'(e_0) = 0$. The minimum e_0 normalizes the effort scale: it is the effort a worker chooses with no incentive to do more. Effort is measured in total effective hours; e/h is the intensity of an hour. The cost is zero at e_0 and its marginal cost is zero there, so any positive return to effort above e_0 raises effort above the minimum. The curvature $\gamma > 0$ governs how steeply marginal cost rises and $k > 0$ scales the cost in dollars.

Each firm is atomistic, so recruitment at firm j is

$$N_j = \exp\left\{\xi_N + \frac{V_j}{\lambda}\right\} \tag{6}$$

The intercept ξ_N sets the scale of recruitment. The employer-choice model follows Card et al. (2018), who decompose the value of a job into a common firm component and an idiosyncratic worker taste shock, giving each firm an upward-sloping labor supply curve.

Timing and information. The model has three stages. In Stage 1, the firm observes F_j, chooses whether to collect tips, posts a base wage w_j, and recruits N_j workers. In Stage 2, the firm draws demand a from a distribution G, common to all firms, and chooses hours h for each worker, anticipating the worker's effort response. The worker then chooses effort e. Revenue is produced and wages and tips are paid. The base wage and employment remain fixed during this demand realization. Workers accept the hours the firm assigns at the base wage, so the job is a wage-hours contract: the firm chooses hours in each demand state anticipating the worker's optimal effort choice, and workers know this when they are hired.

Hours are verifiable. Effort is observable to the firm but not contractible, so the firm cannot condition pay on effort directly. Customers pay tips directly to workers; tips depend on the revenue the worker generates, which depends on effort.

4.2 Effort and hours

The worker's effort choice. The worker chooses effort to maximize total compensation net of effort cost, taking assigned hours as given:

$$e^\star \in \arg\max_{e \geq e_0} \{wh + \tau a g(h, e) - c(e)\} \tag{7}$$

At an interior solution, the worker raises effort until the tips earned from an additional unit of effort equal its cost.

$$\tau a g_e(h, e^\star) = c'(e^\star) \tag{8}$$

Without tips, $\tau = 0$, so effort above e_0 does not increase earnings for the worker, and the worker therefore chooses e_0. With tips, the worker receives a fraction τ of the revenue that additional effort generates. Because $c'(e_0) = 0$ and $\tau a g_e(h, e_0) > 0$ at any positive hours, effort exceeds e_0. The effort floor therefore binds only at non-tipping firms.

The firm's hours choice. The firm chooses hours to maximize per-worker operating profit, anticipating the worker's effort response:

$$\pi(a, w; \tau) = \max_{h \geq 0} \{(1 - \tau) a g(h, e^\star) - wh\} \tag{9}$$

The first-order condition is

$$(1 - \tau) a \left[g_h + g_e \frac{\partial e^\star}{\partial h} \right] = w \tag{10}$$

The firm equates the retained marginal revenue product of an hour to the base wage. At non-tipping firms, effort is e_0 regardless of hours, so the induced-effort term $g_e \partial e^\star / \partial h$ is zero and the condition reduces to $a g_h = w$. At tipping firms, an additional hour raises revenue directly and through the effort it induces: higher hours raise the marginal product of effort, which raises effort, which raises revenue.

Under CES, hours and effort are complements when $\theta > 0$. Combined with $\nu > 0$ and $1 + \gamma > \nu$, this implies $\partial e^\star / \partial h > 0$. Effort per hour $e^\star / h$ falls as hours rise: each additional hour raises effort, but by less than proportionally. Section D.1 shows how effort responds to demand and hours under the CES specification.

4.3 Wage setting

The job value V defined in Equation (4) depends on the base wage through the firm's hours and effort solutions. The firm chooses w to maximize expected total profit before the fixed cost:

$$w^\star(\tau) \in \arg\max_{w} \; N(w; \tau) \, \mathbb{E}_a[\pi(a, w; \tau)] \tag{11}$$

A higher wage attracts workers but reduces profit per worker. Although workers are risk neutral, the variance of demand affects wages and recruitment: optimized profit is convex in a, so a more dispersed demand distribution raises expected profit.[29] At an interior optimum,

$$\frac{V_w(w;\tau)}{\lambda}\,\mathbb{E}_a[\pi(a,w;\tau)] = \mathbb{E}_a[h(a,w;\tau)] \tag{12}$$

The left side is the profit the firm earns from the additional workers a higher wage attracts; the right side is the cost, the additional hours it must pay for. The effort response is the reason the wage clawback is below one-for-one. The clawback is an outcome of Equation (12), not an assumption. Section D.5 derives the condition, the implied elasticity of recruitment to the wage, and how tips change the posted wage.

4.4 Compensation choice

Let $\Pi(\tau) \equiv N(w^\star(\tau);\tau)\,\mathbb{E}_a[\pi(a,w^\star(\tau);\tau)]$ denote maximized expected monthly profit before the fixed cost. The firm chooses to collect tips when

$$\Pi(\tau_T) - F_j \geq \Pi(0) \tag{13}$$

Because F_j is additive and independent of employment, wages, hours, and effort, it affects which compensation structure the firm chooses but not the optimal choices within either structure. Heterogeneity in F_j allows tipping and non-tipping firms to coexist. Firms with low F_j adopt, and firms with high F_j do not. As the mean of F_j falls over time, more firms adopt.

4.5 Responses to demand

Tipping and non-tipping firms respond differently to demand because tips give workers a return to effort that a fixed base wage does not. At a tipping firm, higher demand raises the return to effort, workers supply more of it, and the additional effort raises revenue and the combined hourly rate. At a non-tipping firm, effort is fixed and hours are the only margin of adjustment. The propositions below formalize these comparative statics with respect to a at the posted wage and employment. Proofs are in Section D.7.

The three propositions share the following restrictions on the production function and effort cost.

Assumption 1. *Returns to scale are positive* ($\nu > 0$). *Workers choose finite effort* ($1 + \gamma > \nu$). *Hours and effort are sufficiently complementary and effort responds sufficiently little to incentives that revenue per hour rises with demand at tipping firms* $\left(\frac{1}{1+\theta} + \frac{1}{\gamma} \leq 1\right)$.

[29] The firm chooses hours after observing demand, so it increases h in high-demand states and scales down in low-demand states. At non-tipping firms, profit at given hours is linear in a, and the pointwise maximum is convex. At tipping firms, profit at given hours need not be convex in a because effort responds to demand, but the firm's re-optimization of hours makes optimized profit convex (Section D.7). By Jensen's inequality, a more dispersed demand distribution raises expected profit.

The first two conditions make the worker's effort choice unique and finite: the effort cost curves more steeply than revenue as effort rises. The third condition implies $\theta > 0$, so hours and effort are complements: an increase in one raises the marginal product of the other. Without it, higher effort can raise the marginal product of an hour enough that the firm adds hours faster than revenue grows.

The first two conditions are imposed in estimation. The third is a sufficient condition that is not imposed; Section E.1 discusses its status at the estimated values. Proposition 1 uses only the first two conditions and $\theta > 0$, and Propositions 2 and 3 also hold under a weaker sufficient condition that holds at the estimated values (Remark 1).

Proposition 1 (Effort)**.** *Suppose Assumption 1 holds. An increase in a raises effort at a tipping firm. At a non-tipping firm, effort is e_0 regardless of a.*

Higher a raises the return to effort: the worker earns more tips per unit of effort when demand is high, because $\tau a g_e$ in Equation (8) is increasing in a. The rising marginal cost of effort limits the response, but effort increases until the higher return is absorbed. At a non-tipping firm, $\tau = 0$, so a does not enter the worker's problem and effort stays at e_0.

Proposition 2 (Revenue per hour)**.** *Suppose Assumption 1 holds. An increase in a raises revenue per hour y/h at both tipping and non-tipping firms.*

At non-tipping firms, effort is fixed, so revenue grows faster than hours: an additional hour has a diminishing marginal product when effort does not adjust. At tipping firms, higher effort raises revenue at given hours but also raises the marginal product of an hour, so the firm adds hours; under Assumption 1, the first effect dominates. With effort held fixed at e_0, the ratio of the hours response to the revenue response at a tipping firm takes the same form as at a non-tipping firm, $1/[1+\theta(1-s_h)]$, evaluated at the tipping firm's hours share. The gap between this effort-fixed ratio and the actual tipping ratio isolates the contribution of endogenous effort.

Proposition 3 (Combined rate)**.** *Suppose Assumption 1 holds. An increase in a raises the combined hourly rate $w + \tau y/h$ at a tipping firm. At a non-tipping firm, the combined rate is the base wage and does not respond to a.*

By Proposition 2, y/h rises with a, so $\tau y/h$ rises and the combined rate rises with it. At a non-tipping firm, $\tau = 0$ and the combined rate equals the base wage.

With effort held fixed, revenue per hour rises with demand under either compensation structure when $\theta > 0$. At tipping firms, endogenous effort changes both the revenue response and the hours response.

5 Demand Shocks and the Effort Response

Section 3 established that tipping changes the level of compensation when firms adopt it. Whether tipping also changes how compensation responds to demand is a separate question, and the one the model is built to answer. The model assumes that base wages and prices are set before demand is realized and that both do not respond to it. Given these assumptions and the stable tip percent documented in Section 2, the combined hourly rate at tipping firms responds mechanically to demand while at non-tipping firms it does not. The model's prediction is about quantities: hours and output per worked hour should respond more at tipping firms. This section confirms both: base wages and prices do not respond to short-run changes in demand, and hours and output per worked hour respond more at tipping firms. The magnitudes calibrate the model's production function and effort parameters. More than half of tipping firms *pool* tips across workers, so a worker's tips depend on coworkers' output as well as their own. Output per worked hour responds at least as much at these firms as at firms where each worker keeps their own tips. When pay moves with revenue, firms respond to demand with more hours and more output per hour, and the response holds when workers share that pay with their coworkers.

5.1 Customer spending at other firms traces out demand variation

The ideal experiment raises demand at a firm while holding its production function, workforce, and compensation structure fixed, and compares the response of wages, hours, and output across demand states. In the model, firm revenue is $Ny = Nag(h, e)$: the level of demand a times each worker's output $g(h, e)$ at their hours and effort, summed over N workers. The ideal experiment varies a. We approximate this with the spending of each firm's own customers at other businesses. A firm's revenue can change for two reasons: its customers spend more in total, or the firm captures a larger share of what they spend. The share moves with the firm's own prices, hours, and effort, which are the responses we aim to measure, so the shock uses only the first source. Spending at other businesses measures each customer's total without the firm's own revenue in it.

Let m_{ct} be customer c's total card spending at firms in the point-of-sale data in month t and ϕ_{jct} the share of it spent at firm j, so that firm j's revenue is $\sum_c \phi_{jct} m_{ct}$. Its revenue growth is approximately

$$\Delta(Ny)_{jt} \approx \sum_c s_{j,c,t-1} \Delta m_{ct} + \sum_c s_{j,c,t-1} \Delta \phi_{jct} \tag{14}$$

where $s_{j,c,t-1} = \phi_{j,c,t-1} m_{c,t-1} / (Ny)_{j,t-1}$ is customer c's share of firm j's previous-month revenue. Every Δ denotes a symmetric percent change. The first term is the customer side and corresponds to Δa; the second is the firm side, $g(h, e)$ moving. The shock uses only the first term, with $m_{-j,c,t}$

in place of m_{ct}.[30] The point-of-sale data assign a tokenized identifier to each credit card, so we observe each card's spending at other firms, not only at the firm in question.

For each firm j in month t, the demand shock is a sum of its customers' spending growth elsewhere, weighted by each customer's share of the firm's previous-month revenue:

$$\Delta z_{jt} = \sum_{c} s_{j,c,t-1} \cdot \Delta m_{-j,c,t} \tag{15}$$

where $\Delta m_{-j,c,t}$ is the growth in c's total card spending in the point-of-sale data other than j in month t. Growth rates are symmetric percent changes (sym pp), $200 \cdot (x_t - x_{t-1})/(x_t + x_{t-1})$, so the measure captures both the intensive margin (existing customers spend more) and the extensive margin. A customer who spent nothing elsewhere last month but does this month registers as a large positive change rather than a division by zero. Intuitively, the shock is high when a firm's customer base is spending more everywhere, for reasons that originate outside the firm. Each firm accounts for a small share of its customers' total spending, so its own prices and hours do not move what those customers spend elsewhere.

We track approximately 200 million card identifiers used at firms in our analytic sample, including their spending at firms outside it.[31] The point-of-sale data tokenize credit card numbers, so a customer with multiple cards appears as multiple identifiers. A customer who switches cards across months can generate spurious spending changes at the card level. We treat each card as a customer; a customer split across cards, or switching cards between months, adds measurement error to the shock.

The shock is constructed as a leave-out measure: $\Delta m_{-j,c,t}$ excludes customer c's spending at firm j itself, so the firm's current outcomes never enter its shock. The firm's lagged revenue enters only through the predetermined shares. Spending is measured net of tips and taxes, so tips do not enter the shock's construction either. Without the leave-out, the firm's own revenue appears on both sides of the regression, creating a mechanical positive correlation between the shock and any revenue-linked outcome. The leave-out removes this channel entirely. Spending is also measured net of tips and taxes, so tips appear nowhere in the shock's construction.

The identifying assumption follows the shift-share logic of Borusyak, Hull and Jaravel (2022): conditional on controls, the leave-out spending changes of firm j's customers are uncorrelated with unobserved determinants of firm j's labor market outcomes.[32] The controls X_{jt} absorb two sources

[30] Lamadon, Mogstad and Setzler (2022) face the same two-piece structure in firm value added: productivity, and an amenity term that also reflects worker composition. They identify labor supply from the productivity piece by assuming amenities do not move over the estimation window, or, with external instruments, that the instrument is uncorrelated with changes in amenities. Here the customer-side piece is observed rather than inferred; the identifying assumption below is still required.

[31] Measuring spending at other firms requires only card transactions, not linked payroll records.

[32] The increase in effort that is due to the demand shock is an effect we aim to estimate, not a threat to identification. The assumption requires that changes in effort that would occur without the shock are uncorrelated with the shock (Garin and Silvério, 2024, p. 1679 and footnote 16).

of variation in the shock. The first is a month effect for each compensation structure $\mu_{T_j,t}$, where T_j indicates tipping firms; it removes anything common to all firms in a month, such as seasonal and aggregate spending, and anything common to firms of the same compensation structure in a month. The second is $S_{jt} = \sum_c s_{j,c,t-1}$, the share of firm j's previous-month card revenue from cards with spending elsewhere in month $t-1$ or t, with a separate coefficient for each month and compensation structure. Cards with no spending elsewhere in either month have no symmetric growth rate and drop from the sum, so the shares do not sum to one. A common monthly component in customer spending growth therefore enters each firm's shock in proportion to S_{jt}. Following Borusyak, Hull and Jaravel (2022), we absorb this component by allowing the coefficient on S_{jt} to vary by month, separately for each compensation structure:[33]

$$X_{jt} = \mu_{T_j,t} + \gamma_{T_j,t} \cdot S_{jt} \tag{16}$$

After recentering, the conditional mean of the shock is flat in S_{jt} for both compensation structures (Figure A4a). The shock aggregates many card-level changes that are uncorrelated with one another net of the common monthly component, which means the estimates average over many independent sources of variation rather than a few. We standardize the shock by its pooled cross-sectional standard deviation, so a one-unit move corresponds to the same absolute change in customer spending growth for both compensation structures.[34]

The same change in the shock need not be the same change in demand a. We assume that, conditional on the controls X_{jt}, a one-unit move in the shock is the same change in a at tipping and non-tipping firms. Comparisons of the size of a response across compensation structures rely on this assumption; ratios of two responses within a compensation structure do not, because the scale of the shock cancels. The distribution of S_{jt} overlaps substantially across compensation structures, although its mean differs (Figure A4b). The interaction of S_{jt} with compensation structure, like that of the month effects, lets the recentering differ between tipping and non-tipping firms.

Figure 9 shows how the shock operates in event time. In every calendar month, we split tipping firms at that month's median of the S_{jt}-residualized shock, following Lamadon, Mogstad and Setzler (2022). Firms above the median are the treated group; firms at or below are controls. Because the split is redone each month, the same firm can be treated in one month and control in another. For each monthly cohort we track outcomes from six months before to six months after and average the treated-control gap across cohorts at each event time. Outcome paths are normalized to zero at event time -1.

Figure 9a plots the cumulated shock, forward-summed from event time -6 and net of its mean over event times -6 to -2. The treated-control gap is flat before assignment (-0.24 sym pp at

[33]The shock enters Equation (17) directly as the treatment rather than as an instrument; Borusyak and Hull (2023) show that the same recentering applies in this case.

[34]Standardizing within each compensation structure instead leaves each structure's own t-statistics unchanged but changes the differential β_2 and the ratio of responses, which is why we use the pooled standard deviation.

event time −1), jumps to 7.96 at event time 0, and stays roughly level afterward (7.17 at +1, 8.62 at +6). A firm's treatment status carries little information about its shock in neighboring months. The level shift means the impulse to the growth rate lasts one month: demand rises once and stays higher. Figure 9b plots the tip rate. The pre-period gap is negligible, and the 95 percent confidence interval includes zero at all five pre-period event times. The shock is a lasting increase in the level of customers' spending, and the tip-rate response persists over at least six months: a single month of above-median customer spending raises the tip rate by about 6 cents per hour, and the gap does not close within the event window.

The main specification tests whether tipping and non-tipping firms respond differently to the same demand variation:

$$\Delta y_{jt} = \beta_1 \Delta z_{jt}^{\text{std}} + \beta_2 \Delta z_{jt}^{\text{std}} \cdot T_j + X_{jt}\delta + \varepsilon_{jt} \tag{17}$$

where $\Delta z_{jt}^{\text{std}}$ is the shock standardized by its pooled cross-sectional standard deviation and T_j is an indicator for tipping firms. All responses are per one-standard-deviation change in customer demand. β_1 is the response at non-tipping firms, β_2 is the differential response at tipping firms, and $\beta_1 + \beta_2$ is the response at tipping firms. Standard errors are clustered at the firm level.

In the model, a positive demand shock raises the marginal product of an additional hour, so the firm demands more hours at its posted base wage. At tipping firms it also raises the return to effort, and the effort it induces raises the marginal product of an hour further. The results below test how firms and workers respond.

5.2 Demand raises revenue and hours more at tipping firms

Tipping firms respond to the same demand shock with nearly twice the revenue increase and close to four times the hours increase of non-tipping firms. Revenue rises 2.18 sym pp at tipping firms (standard error 0.10) against 1.14 at non-tipping firms (standard error 0.11), and hours rise 0.75 (standard error 0.09) against 0.21 (standard error 0.08). Scaled per unit of revenue, the hours response is 0.34 at tipping firms and 0.18 at non-tipping firms. Lafontaine and Sivadasan (2009) find a weekly labor-cost elasticity to revenue of approximately 0.34 at fast-food outlets, declining toward 0.18 in countries with stricter employment protection, the same as our 0.18 at non-tipping firms where hourly pay does not vary with demand. Our ratios are not labor demand elasticities: revenue embeds the firm's own hours choice and, at tipping firms, the workers' effort choice.[35] In the model, effort stays at the floor at non-tipping firms, so their ratio depends only on the production function. It equals $1/[1 + \theta(1 - s_h)]$, where s_h is the hours share in production, so it reflects how substitutable hours and effort are; at tipping firms it also contains the effort response (see Section D.6 for a derivation). We present reduced-form responses per one-standard-deviation shock, where each outcome is separately identified from the exogenous demand variation.

[35] Revenue is co-determined with hours: when the demand shock raises customer traffic, the firm's hours response itself generates additional revenue. At tipping firms, the endogeneity is compounded because effort also responds to demand and feeds back into revenue through higher output per hour.

Prices and wages are rigid at monthly frequency, so the demand shock operates entirely through quantities and tips (Figure 10a). A firm-level price index responds −0.01 (standard error 0.04) at tipping firms and 0.08 (standard error 0.06) at non-tipping firms.[36] These estimates are consistent with retail price microdata, in which regular prices for food away from home change in about 5 to 6 percent of months (Nakamura and Steinsson, 2008) and last close to a year (Bils and Klenow, 2004). Base wages are equally rigid: −0.003 (standard error 0.013) at tipping firms and 0.013 (standard error 0.008) at non-tipping firms. Grigsby, Hurst and Yildirmaz (2021) document that nominal hourly wages at similar firms rarely change within a month. With prices and base wages fixed in the short run, the revenue differential between tipping and non-tipping firms operates through hours, effort, and tips.

5.3 Tipping firms add four times the hours and produce more per hour

Hourly pay and hours both move with demand at tipping firms; at non-tipping firms, only hours do. At non-tipping firms, hours rise 0.21 sym pp (standard error 0.08) despite no change in hourly pay (combined rate unchanged at 0.01, standard error 0.02). This is consistent with underemployment at the prevailing wage: Lachowska et al. (2026) document that hourly workers would accept more hours than they receive at their existing wage. The demand shock shifts labor demand outward, and workers supply the additional hours without requiring higher pay. Because workers accept the hours the firm assigns at an unchanged base wage, as in the model's wage-hours contract, the hours response at non-tipping firms is the shift in labor demand.

At tipping firms, tips make hourly pay move with demand, and hours move nearly four times as much as at non-tipping firms. The combined rate rises 0.18 sym pp (standard error 0.03) while the base wage does not move (Section 5.2).[37] Tips are a stable percent of revenue (Section 2.2), so the tip rate rises whenever revenue per worked hour does. Worked hours rise 0.75 sym pp (standard error 0.09) against 0.21 at non-tipping firms, a differential of 0.54 sym pp.

Hours move nearly four times as much at tipping firms because labor demand shifts out by more, and the model attributes that to effort. Under the model's wage-hours contract, workers accept the hours the firm assigns at both compensation structures, so the hours response is the shift in labor demand. The shock is the same change in a under both compensation structures, so the larger shift comes from the response to it at tipping firms. A higher a raises the return to effort at tipping firms, and workers supply more of it. Because hours and effort are complements, the additional effort raises the marginal product of an hour, and labor demand shifts out further than it does from a alone. At non-tipping firms, additional effort earns no additional pay, effort stays at e_0,

[36] The index is a chained Laspeyres measure constructed from item-level log price changes, weighted by lagged revenue shares. Section A.1 details the construction.

[37] Because tip income exists only at tipping firms, we estimate tip outcomes on the tipping sample using Equation (17) without the interaction term. The tip rate rises 0.80 sym pp (standard error 0.09). Tips account for approximately a quarter of monthly earnings at tipping firms (Section 2), so the implied combined-rate response is $0.25 \times 0.80 \approx 0.20$ sym pp, close to the direct estimate of 0.18. At non-tipping firms, the combined rate is effectively the base wage, and neither moves: 0.01 (standard error 0.02).

and labor demand shifts only with a. When demand falls, effort falls, and the response reverses.[38] Tipping firms respond symmetrically: output per worked hour moves 1.48 sym pp when demand rises and 1.39 when it falls ($p = 0.77$), and hours 0.74 and 0.75 ($p = 0.93$).

The demand shock raises both pay and hours at tipping firms, but it raises revenue by more than either. Combined earnings rise 0.90 sym pp at tipping firms (standard error 0.09) against 0.21 at non-tipping firms (standard error 0.08), while revenue rises 2.18 against 1.14.[39] The gap between revenue and pay widens more at tipping firms when demand increases. Output per worked hour is what drives this gap. Because prices do not respond, revenue per worked hour measures output per worked hour, and it rises 1.43 sym pp at tipping firms (standard error 0.12) against 0.94 at non-tipping firms (standard error 0.13). If additional hours face diminishing returns (as in the CES production function in Section 4), a firm that adds more hours should see output per hour rise less, and tipping firms add roughly four times the hours yet see output per hour rise more. If instead the demand shock raises the productivity of each hour, output per hour can rise even as hours increase, but this force operates at both compensation structures facing the same shock and does not by itself explain the differential. Card transactions per worked hour confirm this in physical units: 1.32 sym pp at tipping firms (standard error 0.11) against 0.86 at non-tipping firms (standard error 0.12). The additional hours alone cannot produce the gap in revenue responses: so long as a one percent increase in hours raises revenue by no more than one percent, the additional hours account for at most 0.75 of the 1.04 sym pp gap. The differential in output per hour is consistent with effort responding to demand at tipping firms, as the model predicts.

5.4 Tips rise from more customers, not more generous customers

Demand transmits to worker pay one transaction at a time: more customers walk in, each tips at nearly the same percent as before, and volume accounts for almost the entire tip-rate response. This supports treating the tip percent as fixed in the model. The transaction data record every sale and tip at the same firms where we observe worker hours, so we can decompose the tip rate into its constituent margins and trace demand from the customer's payment to the worker's hourly pay:

$$\underbrace{\frac{Tips}{H}}_{\text{tip rate}} = \underbrace{\frac{n_{trx}}{H}}_{\substack{\text{trx per} \\ \text{worked hr}}} \times \underbrace{\frac{n_{tipped}}{n_{trx}}}_{\substack{\text{share} \\ \text{tipped}}} \times \underbrace{\frac{R_{tipped}}{n_{tipped}}}_{\substack{\text{bill size} \\ \text{(tipped)}}} \times \underbrace{\frac{Tips}{R_{tipped}}}_{\substack{\text{tip} \\ \text{percent}}} \tag{18}$$

[38] The response at tipping firms is also consistent with a kinked hours supply curve common to both compensation structures, with non-tipping firms on the flat portion and tipping firms on the upward-sloping portion for increases and decreases in demand. Under this reading, labor demand shifts out such that hours rise 0.75 sym pp at tipping firms against 0.21 at non-tipping firms, and workers supply the additional hours because the combined rate rises 0.18 sym pp. Remaining on the supply curve requires an hours supply elasticity of $0.75/0.18 \approx 4$. This would explain why workers supply the additional hours when labor demand shifts, though it still requires that the same shock move labor demand more at tipping firms.

[39] The demand shock is symmetric and residualized, so it averages zero, and tipped workers earn more in high-demand months and less in low-demand months. Whether this variation is a cost to workers depends on risk preferences and whether workers are compensated for bearing it.

where H is worked hours at the firm-month, n_{trx} counts transactions, and R_{tipped} is revenue on tipped transactions. The first two terms are the extensive margin: how many tipped transactions each worked hour handles. The last two are the intensive margin: how much revenue each tipped transaction carries and what percent of it the customer tips. The components multiply, so their responses to the demand shock approximately sum in symmetric percent changes; Equation (18) attributes the tip-rate response margin by margin. We estimate each component's response on the tipping sample, where tips exist, using card transactions, which account for nearly all recorded tips.

The firm meets demand by serving more customers with the workers on hand: transaction volume accounts for essentially the entire tip-rate response. Tipped transactions per worked hour rise 1.21 sym pp (standard error 0.13), while tip per tipped transaction is statistically indistinguishable from zero (0.02, standard error 0.06); the extensive margin carries the tip rate (Figure 11).[40] Within the extensive margin, transactions per worked hour rise 1.34 sym pp (standard error 0.12) while the share of transactions tipped is unchanged (-0.11, standard error 0.07). When demand rises, each hour worked at a tipping firm processes more transactions, about two thirds of them tipped, and the tip per tipped transaction does not change. This capacity-utilization margin is what raises output per worked hour in Section 5.3: the workers on hand handle more transactions per worked hour, which the model attributes to effort; the stable tip percent converts the additional volume directly into worker pay.

The intensive margin is flat because its two components move slightly in offsetting directions. Bill size on tipped transactions rises 0.14 sym pp (standard error 0.06) and the tip percent on those transactions falls 0.12 sym pp (standard error 0.03). The offset is consistent with customers tipping a smaller percent of larger bills, a pattern documented in Conlin, Lynn and O'Donoghue (2003). Both are an order of magnitude smaller than the 1.21 sym pp volume response.

We find that the additional revenue comes from more customers, not more spending per customer. The number of unique card identifiers rises 2.07 sym pp (standard error 0.10) at tipping firms, accounting for nearly all of the 2.18 sym pp card-only revenue response. Revenue per customer (0.13, standard error 0.07) and transactions per customer (-0.03, standard error 0.03) barely move.[41] The share of revenue from cards absent the previous month also rises (0.16 sym pp, standard error 0.05), yet the shock is built entirely from the spending of the firm's *existing* customers at other businesses. The increase in revenue from customers who contribute nothing to the shock's construction demonstrates that the shock measures demand changes at the firm beyond those from its existing customers.

[40] The payroll tip rate in Section 5.3 rises 0.80 sym pp. The gap is consistent with timing. A paycheck dated in month t might cover a pay period that partly falls in month $t-1$, so payroll tip earnings absorb a month-t demand shock with delay, attenuating the contemporaneous response relative to the transaction measure. Section 1 shows that tips collected from customers and recorded in the POS data closely match tips paid out through payroll at these tipping firms, so the two measures differ in timing, not in whether workers receive the tips.

[41] Cash transactions carry no card identifier, so this analysis uses card transactions only. Card transactions account for approximately 84 percent of revenue at these firms.

Demand reaches worker pay almost entirely through the number of customers: more customers arrive, each transaction resembles the ones before it, and the stable tip percent converts the added volume into pay. The average tip percent across all card transactions falls 0.23 sym pp (standard error 0.09).[42] This is the dynamic counterpart of the cross-sectional stability in Section 2, and it supports treating the tip percent as approximately fixed when the model maps demand into compensation.

5.5 Firms that pool tips respond at least as much to demand

The larger response to demand at tipping firms holds whether workers keep the tips their own customers leave or share them with their coworkers. Some tipping firms pay each worker the tips their own customers leave, which we call *direct*; others *pool* tips and divide them among the workers. Pooling could matter because splitting tips with coworkers means a worker receives only part of the tips their own effort adds. In the model, each worker receives τ of their own output, and effort satisfies Equation (8). In a pool of N workers with equal shares who each choose effort separately, a worker receives τ/N of the revenue their effort adds, and the condition becomes

$$\frac{\tau}{N} a g_e(h, e^\star) = c'(e^\star) \tag{19}$$

Pooling divides tips that customers have already paid, so it leaves the tip percent τ unchanged. At a given level of demand and hours, effort falls unless it is already at its minimum; this is the free-rider problem of Holmstrom (1982).[43]

Firms that pool tips respond at least as much to demand as direct firms (Figure 12). We classify each tipping firm as direct or pooled and estimate Equation (17) on tipping firms, with an indicator for pooled firms in place of T_j. Section C.2 describes the classification and sample. About 60 percent of tipping firms pool, and both settings are common at full- and quick-service firms. A one-standard-deviation demand shock raises revenue 2.58 sym pp at pooled firms (standard error 0.16) against 1.49 at direct firms (standard error 0.18).[44] Output per worked hour, which Section 5.3 interprets as evidence of effort, rises at least as much at pooled firms: revenue per worked hour rises 1.66 sym pp (standard error 0.19) against 0.90 at direct firms (standard error 0.20). Hours rise 0.91 sym pp at pooled firms against 0.61 at direct firms, a difference that is not statistically distinguishable ($p = 0.14$). The combined rate move is the same under both settings, rising 0.19 sym pp at pooled firms against 0.19 at direct firms ($p = 0.999$), and prices do not move under either. The classification matches how pay is divided within the firm. When demand rises, pooled firms split the additional

[42] The average is dollar-weighted: each transaction is weighted by its revenue. Weighting customers equally gives a similar decline (-0.20 sym pp, standard error 0.10), suggesting that the decline is not driven primarily by large bills.

[43] Lower effort need not mean a smaller percentage response to demand. The response depends on the mix of hours and effort at which the firm operates, and it can be larger or smaller under pooling (Section D.3).

[44] Among firms observed for at least three months under each setting, the estimates point the same way and are less precise: revenue rises 2.17 sym pp in pooled months (standard error 0.49) against 0.39 in direct months (standard error 0.66).

tips among coworkers, so earnings across a firm's workers should become more equal, and more so the longer the period over which tips are pooled (a transition, a workday, or a workweek). The coefficient of variation of earnings across a firm's workers falls 0.33 at pooled firms (standard error 0.17) against 0.01 at direct firms (standard error 0.18), and the decline is stronger the longer the pooling period, although these differences are imprecise (Figure A5). The response at tipping firms therefore does not require that each worker keep the tips from their own customers, even though pooling cuts each worker's share of the tips their own effort adds.

In the model, pooled firms behave like direct firms if workers choose effort jointly, to maximize their combined earnings net of effort costs. In that case, workers collectively receive the full $\tau a g_e$ from a unit of any worker's effort, and the pooled effort condition is Equation (8). Workers may internalize how their effort affects their coworkers' pay. Bandiera, Barankay and Rasul (2005) document this in the opposite direction: fruit pickers paid relative to coworkers' productivity were less productive than under piece rates, when higher effort lowered their coworkers' pay. Under pooling the externality reverses—a worker's effort raises coworkers' tips. Peer pressure can also sustain effort when workers split what they produce (Kandel and Lazear, 1992; Hamilton, Nickerson and Owan, 2003). Small workforces may facilitate this internalization: the average pooled firm pays 10 workers in a month. The estimates above do not establish that workers choose effort jointly or identify the mechanism behind it; the model requires only that effort at pooled firms satisfies Equation (8). We assume workers at pooled firms choose effort jointly, so a single effort condition describes all tipping firms regardless of their tip-distribution setting. Section 6 estimates the model using the responses of all tipping firms.

6 From Demand Responses to the Incidence of Tipping

In the model, tipping raises the value of a job by 50 percent, or $728 per worker per month, and firm operating profit by 28 percent before the fixed cost of tipping. The comparison is a tipping firm against the same firm without tipping: under tipping the firm re-solves its base wage and workforce, and workers supply the effort tips reward. The calibration connects two sets of estimates. Section 3 measures how much of the tips workers retain at adoption. Section 5 measures how pay and production move with demand under each compensation structure. This section calibrates to the demand responses and cross-sectional levels (Section 6.1), reports fit (Section 6.2), compares the implied clawback with the adoption estimates (Section 6.3), reports the value of tipping to workers and firms (Section 6.4), and examines how the gains vary with the tip percent (Section 6.5).

6.1 Calibration

Six parameters are free: the production parameters ν, θ, and α, the effort-cost parameter $c'(1.05e_0)$, mean demand $\bar{a}$, and the recruitment scale λ (Table 3). Log demand is normal with mean $\log \bar{a} - s_a^2/2$ and standard deviation $s_a = 0.28$. The effort unit is fixed at $e_0 = 82.95$, the tip percent at $\tau_T = 0.088$, and the recruitment intercept so that a non-tipping firm employs five workers.

Nine moments are targeted (Table A7). Three are the demand responses from Section 5: the ratio of the hours response to the revenue response at each compensation structure, and the ratio of the tipping to the non-tipping revenue response. These inform the production function and the effort cost: because effort is fixed at non-tipping firms, their hours response reflects mainly the substitutability of hours and effort, and the two tipping ratios mainly the curvature of effort cost and the weight of effort in production. The remaining six are levels and ratios: the base-payroll share of revenue at each compensation structure, non-tipping worked hours and base wage, and the tipping to non-tipping ratios of worked hours and employment. These identify demand scale, the level of effort cost, and the recruitment scale. The six parameters are fitted jointly to all nine moments. The non-tipping levels are raw medians; the ratios are reweighted to the sector composition of tipping firms. Using these ratios as targets treats the cross-sectional differences between compensation structures as the differences a single firm would show under the two compensation structures, which is what the model assumes when firms share all primitives except F_j. The parameters minimize a weighted sum of squared log deviations from the targets, with moment weights from the bootstrap covariance of the estimates. Section E.1 states the model counterpart of each moment, the construction of the targets, the objective, and the fit.

6.2 Fit and untargeted comparisons

The model fits the cross-sectional levels it targets more closely than the demand responses, and the calibrated parameters make hours and effort strong complements. It matches both base-payroll shares and the non-tipping base wage within one percent. The three demand-response ratios fall 4, 12, and 38 percent below their targets, non-tipping worked hours 5 percent below, and the tipping to non-tipping hours ratio 4 percent above; the tipping to non-tipping employment ratio is 1.23 against a target of 1.33 (Table A7). Table 4 shows compensation and labor allocation under each compensation structure beside the raw cross-sectional values from Section 2.[45] The calibrated parameters imply strong complementarity between hours and effort, with an elasticity of substitution of 0.15. Because hours and effort are complements, the effort that tips induce raises the marginal product of hours, making additional hours more valuable to the tipping firms. At median demand, this reinforcement raises the marginal revenue of an hour by about 52 percent. Expected effort cost is \$39 per worker per month at tipping firms and zero at non-tipping firms, where effort stays at its floor.

6.3 Clawback

Neither base wages at tipping firms nor the tip rate enters the estimation. The model produces a tip rate of \$5.53 per hour and base wages \$0.37 lower at tipping firms, a clawback of 7 cents per dollar of tips ($\kappa = 0.07$). The base wage gap is smaller than the raw cross-sectional \$3.16. The adoption estimates are the incidence comparison: κ is 0.23 (standard error 0.07) among incumbent workers

[45] The calibration targets reweight the ratios to the sector composition of tipping firms (Section 6.1); the table reports raw values. The targets and the fit are in Section E.1.

and 0.33 (standard error 0.10) at the firm level (Section 3), and the model's 0.34 at adopters' tip percent lies inside both confidence intervals. The comparison is at adopters' tip percent: the model is solved with tips at 2.66 percent of pre-tip revenue, the firm-level post-adoption estimate, rather than the calibrated 9.7 percent. In the model, the clawback is 0.34 at 2.66 percent of pre-tip revenue and 0.07 at the calibrated 9.7 percent. Table 5 shows the model's tipping-minus-non-tipping differences at adopters' tip percent beside the adoption estimates.

6.4 The value of tipping to workers and firms

The counterfactual compares a tipping firm with the same firm when the gratuity is folded into the posted price: a $10 price plus a $1 tip becomes an $11 price paid entirely to the firm. The customer pays the same total for the same service, the firm receives all of it, and the worker receives only base wages. In the model this sets τ_T to zero and leaves the production function, the demand distribution, effort preferences, and the recruitment intercept unchanged; the firm resets its wage and recruitment, and hours and effort are re-solved in every demand state. Realized revenue can change because production and employment change. The experiment assumes that folding the gratuity into the price has no further effect on customer demand. The experiment also limits the firm to hourly pay: it cannot replace tips with a bonus paid out of revenue, which would restore the incentive for effort that tips provide.

The model's solution at $\tau = 0$ and the same parameter vector provides this counterfactual. We report every change as tipping minus non-tipping, the direction of adoption. Expected value per worker differs by

$$V_T - V_{NT} = \underbrace{\mathbb{E}[w_T h_T] - \mathbb{E}[w_{NT} h_{NT}]}_{\text{base earnings}} + \underbrace{\mathbb{E}[\tau_T y_T]}_{\text{tips}} - \underbrace{\big(\mathbb{E}[c(e_T)] - \mathbb{E}[c(e_{NT})]\big)}_{\text{change in effort cost}} \tag{20}$$

The firm counterpart per worker is $\mathbb{E}[\pi_T] - \mathbb{E}[\pi_{NT}]$; total firm profit also reflects the change in employment.

Tipping lowers the hourly base wage but raises monthly base earnings, because hours rise (Table 6). The wage reduction contributes −$40 per worker per month and the additional hours $215. With $592 in tips and $39 in additional effort cost, expected job value rises by $728 per worker per month. Tips account for 81 percent of the gain; the effort required to earn them costs 7 cents per dollar of tips. The clawback understates the worker's gain because job value counts the additional hours, which add $215 per month in base earnings and carry no cost in the model, and nets out the effort cost ($39). The recruitment equation gives the same difference:

$$V_T - V_{NT} = \lambda \log\left(\frac{N_T}{N_{NT}}\right) \tag{21}$$

Under the recruitment specification in Equation (6), the gain in job value and the gain in employment are the same object, scaled by λ.

The firm gains as well. Per worker, tipping raises customer payments by $955 per month, of which $592 goes to the worker as tips, so the firm's pre-tip revenue rises by $362. The firm pays $175 of this as additional base earnings and keeps $187 as operating profit before the fixed cost per worker. For firm j, the net monthly gain is $\Pi(\tau_T) - \Pi(0) - F_j$.

Firm profit rises $6,096 per month, and employment rises from 5.00 to 6.15 workers. Between 81 and 85 percent of the firm's gain comes from the added employment, depending on the order of the decomposition; the rest comes from higher profit per worker ($187). The effort each worker supplies makes expansion profitable. Most of both gains come from the effort tipping induces. With effort held fixed at its floor e_0 and the same tip percent, job value rises only 2.9 percent ($42 per worker per month) and firm profit 2.1 percent; the base wage falls $5.72 per hour, so the clawback exceeds one ($\kappa = 1.11$) and the combined rate falls 3.5 percent.

6.5 Gains at lower tip percents

At adopters' tip percent of 2.66 percent of pre-tip revenue, job value rises 10.7 percent ($157 per worker per month) and firm profit 5.9 percent. At the calibrated tip percent of 9.7 percent the gains are 50 and 28 percent. The gain rises with the tip percent, so where customers tip less the gain may not cover the fixed cost of tipping, and those firms do not adopt. Whether customers tip, and how much, depends on sector norms (Conlin, Lynn and O'Donoghue, 2003; Azar, 2005).

7 Conclusion

We use point-of-sale and payroll records at 20,000 U.S. firms to study how the gains from tipping are shared between workers and firms, and where those gains come from. Firms that begin collecting tips claw back 23 to 33 cents of each dollar of tips through lower base wages, well below the one-for-one offset that was available to them. A demand shock constructed from customers' spending at other businesses raises revenue nearly twice as much at tipping firms, and output per hour rises more despite a larger increase in hours, while base wages and prices do not respond. In a model calibrated to the demand responses, effort makes each worker more productive, so the firm expands and recovers only about a third of the tips. These gains may explain why tipping has recently grown in many industries. We treat customers' tipping as mechanical: the tip percent differs across firms but is stable within them, consistent with customers tipping out of habit or warm glow rather than to reward service quality (Kohlhepp, 2025). The within-firm stability of the tip percent turns tips into performance pay, and explaining that stability is the question our findings leave unanswered.

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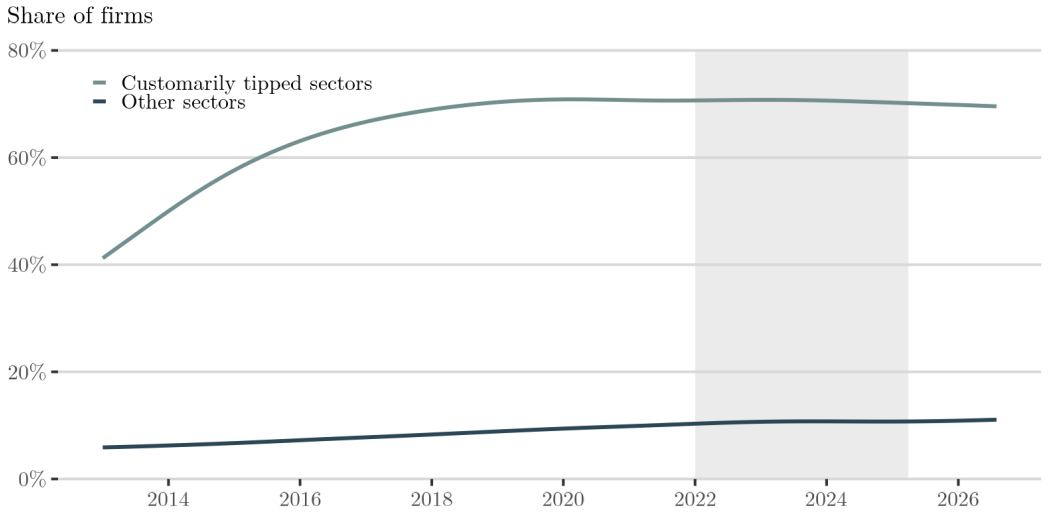
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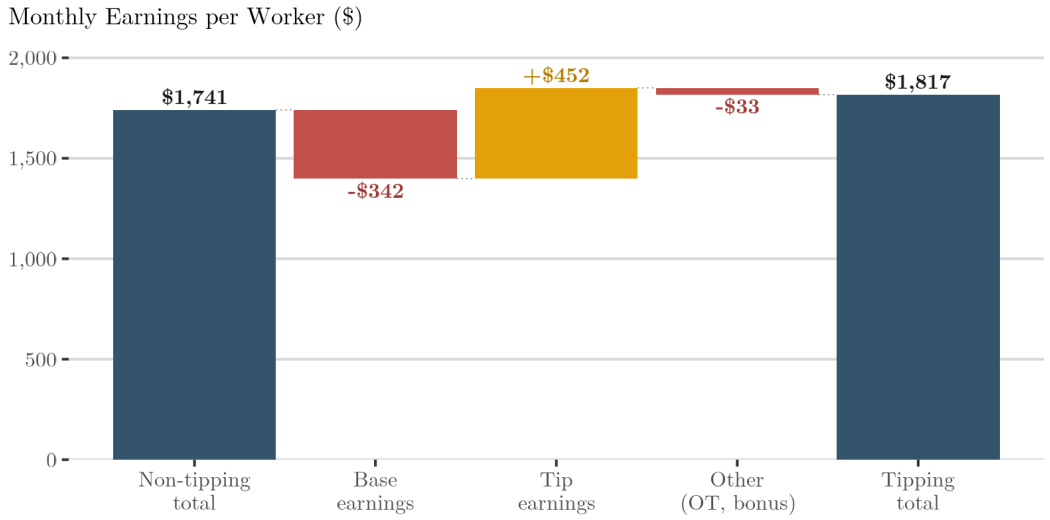
Main Figures

Figure 1: Tipping Spread Rapidly Through Customarily Tipped Sectors



Notes: This figure plots the monthly share of U.S. firms in the point-of-sale data that collect tips of at least one percent of revenue, separately for customarily tipped sectors and other sectors. Customarily tipped sectors are mostly personal care and food service. The largest of the other sectors is retail. The series covers all firms in the point-of-sale data from 2013 onward, not only the analytic sample. A firm-month enters the denominator if the firm has at least ten transactions in the month. Each firm's tip percent is smoothed with a three-month trailing average to remove month-to-month noise. Each sector is classified by the share of its firms collecting tips, averaged over 2013–2015; sectors above 30 percent are customarily tipped. The classification is fixed over time. Each series is a cubic smoothing spline with eight degrees of freedom, fit to the monthly share excluding March through May 2020, when pandemic closures sharply reduced transaction volume. In customarily tipped sectors, the share rises from 45 percent in 2013 to 70 percent by 2018 and is flat thereafter; in other sectors, it rises from about 6 percent to about 11 percent by 2026. The shaded region marks the sample period (2022–2025).

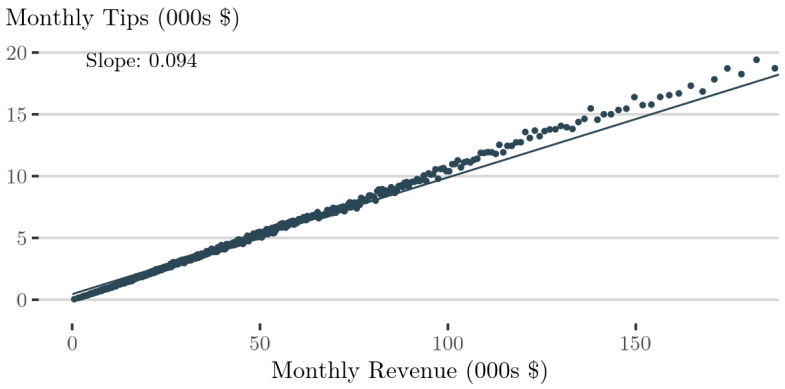
Figure 2: Tipping Firms Pay Lower Base Earnings but Similar Total Earnings



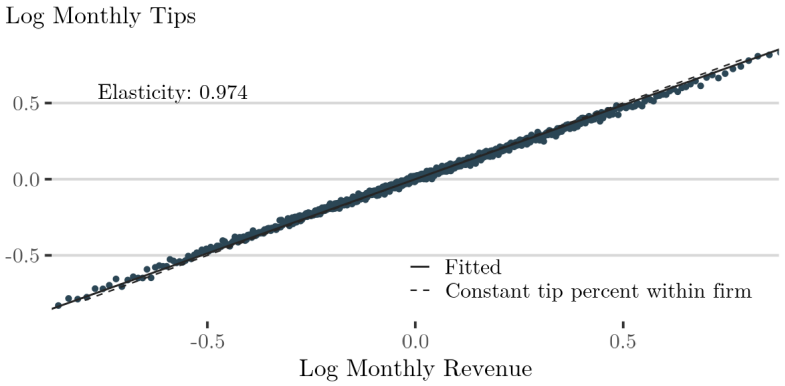
Notes: Worker-month sample with paid hours of at least 5, 2022 to 2025. Worker-level decomposition of the $76 gap in average monthly earnings between tipping and non-tipping firms into a base earnings component, a tip earnings component, and a residual (overtime, bonuses, and paid leave). Base earnings are $342 lower per month at tipping firms; tip earnings are $452 higher; the residual is $33 lower. The comparison is across firms and is not the effect of adopting tipping. Differences in industry composition and in which workers each firm employs enter every bar, and they can make the gap larger or smaller than what workers would gain if their firm began collecting tips.

Figure 3: The Tip Percent Is Stable Across Firms and Over Time

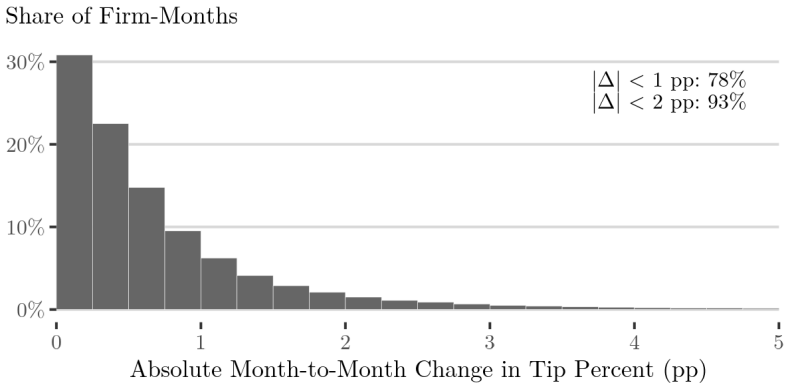
(a) Tips vs. revenue



(b) Within-firm elasticity



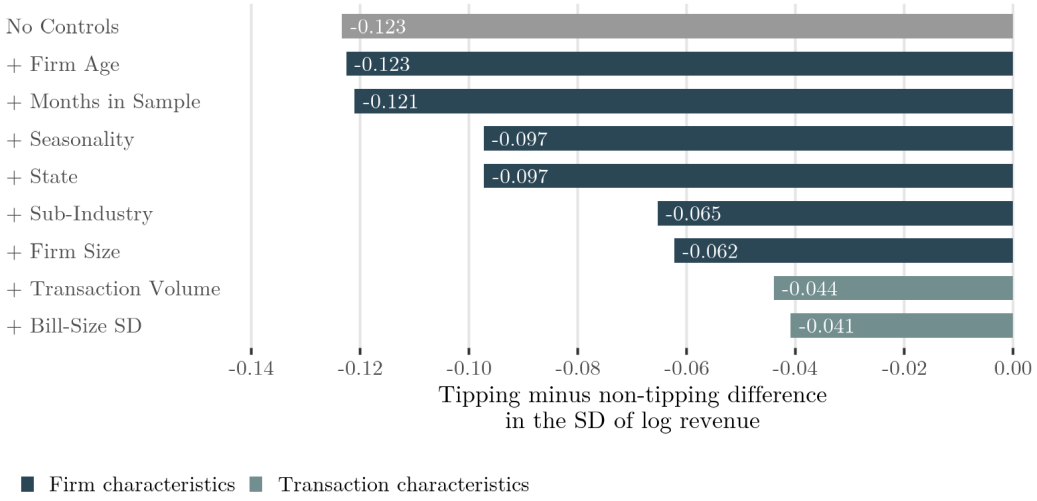
(c) Month-to-month change



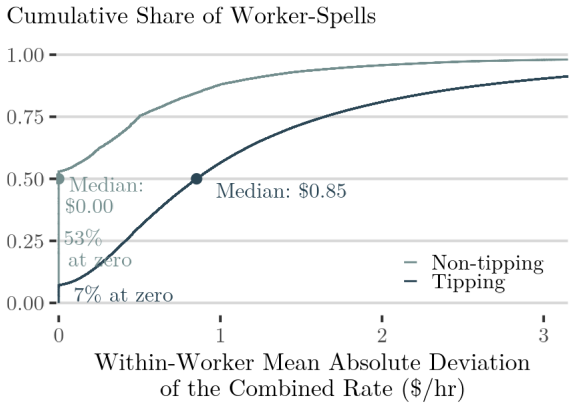
Notes: Panel (a) plots tips against revenue at the firm-month level for tipping firms, 2022 to 2025 (slope 0.094). Panel (b) plots log tips against log revenue, both residualized on firm and calendar-month fixed effects (elasticity 0.974, $R^2 = 0.683$); the dashed line is the unit-elasticity benchmark. Panel (c) shows the distribution of the absolute change in the tip percent between consecutive months within a firm, in percentage points. The change is below one percentage point in 78 percent of firm-months and below two in 93 percent. The horizontal axis is truncated at five percentage points, which excludes 0.78 percent of changes.

Figure 4: Tipping Firms Have More Stable Revenue but More Volatile Hourly Pay

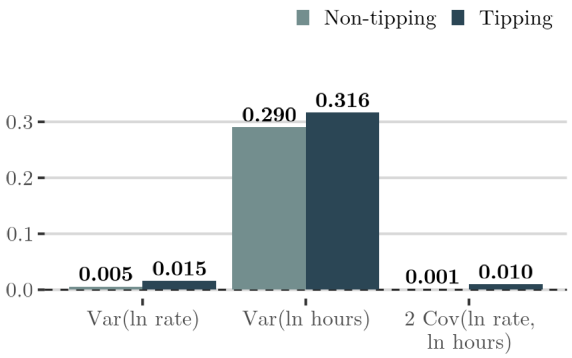
(a) Revenue volatility gap, sequential controls



(b) Within-worker MAD of the combined rate

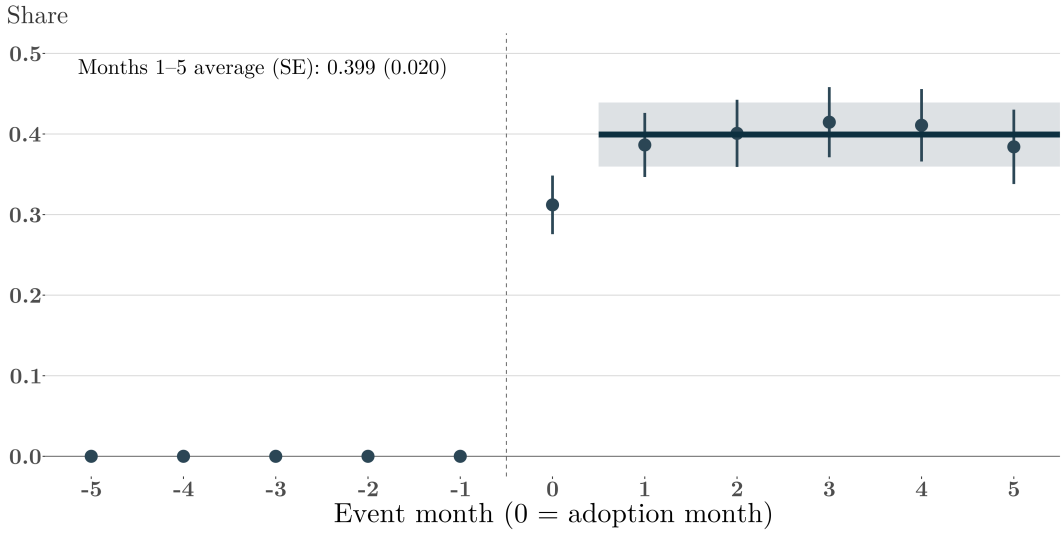


(c) Variance decomposition of monthly earnings



Notes: Panel (a) shows the tipping minus non-tipping difference in the firm-level standard deviation of log monthly revenue, after removing month effects common to all firms, as controls are added cumulatively from top to bottom. Each bar is the coefficient on tipping status in a cross-firm regression on the control in its row and all controls above it, for 12,600 tipping and 7,100 non-tipping firms. Firm age (whole years since the firm's first transaction), months in sample, state, and sub-industry enter as fixed effects; the remaining controls enter linearly. Seasonality is how much the average revenue of other firms in the same sub-industry varies across months of the year, measured over the firm's own sample months. Firm size is log mean monthly revenue, transaction volume is log mean monthly transactions, and bill-size SD is the log standard deviation of the firm's monthly average bill. Panel (b) shows the cumulative distribution of the within-worker mean absolute deviation (MAD) of the monthly combined rate (base wage plus tips per hour), computed within spells of at least three consecutive months (median \$0.85 at tipping firms, \$0.00 at non-tipping firms; the horizontal axis is truncated at \$3, which excludes 9.6 percent of tipping spells and 2.1 percent of non-tipping spells). Panel (c) decomposes $\mathrm{Var}(\ln E) = \mathrm{Var}(\ln w) + \mathrm{Var}(\ln h) + 2\,\mathrm{Cov}(\ln w, \ln h)$ for monthly earnings E, combined rate w, and hours h; rate variance is 2.86× higher at tipping firms, hours variance is similar, and the rate-hours covariance is an order of magnitude larger at tipping firms (0.010 versus 0.0005).

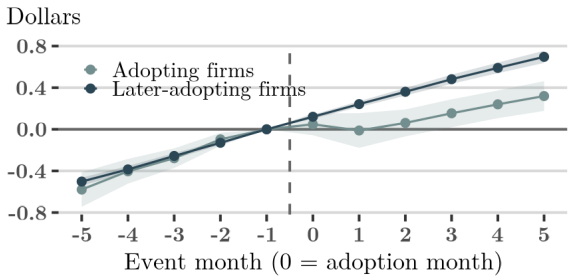
Figure 5: Share of Workers Earning Tips After Adoption



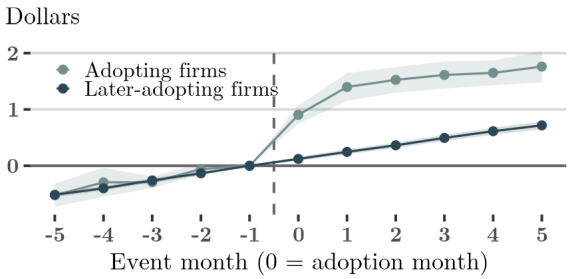
Notes: Stacked difference-in-differences event study of the 300 firms that begin collecting tips during the sample period, with later-adopting firms as controls and event month −1 as the reference. The outcome is the share of the firm's workers whose tip earnings exceed 5 percent of their base plus tip earnings that month. The share is zero at every pre-adoption horizon, rises to 0.31 in the adoption month, and averages 0.40 (standard error 0.02) over the five months that follow. A firm adopts in the first month with positive payroll tip earnings, provided card tips are positive in that month or the month before, following at least six consecutive months with zero payroll tip earnings, positive card revenue, and pooled card tips at or below 0.05 percent of pooled card revenue (Section B.1).

Figure 6: Raw Event-Time Means at Adopting and Not-Yet-Adopting Firms

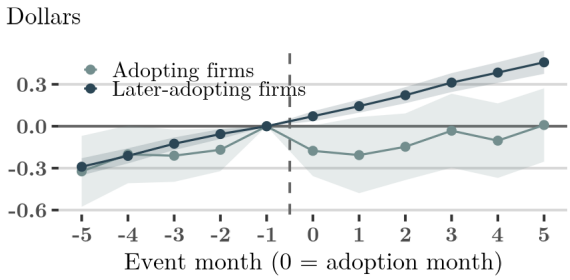
(a) Incumbent workers, base wage



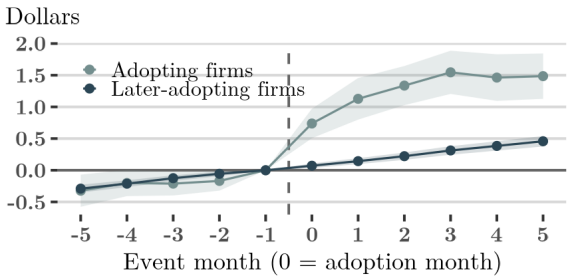
(b) Incumbent workers, combined rate



(c) Firm-level average, base wage



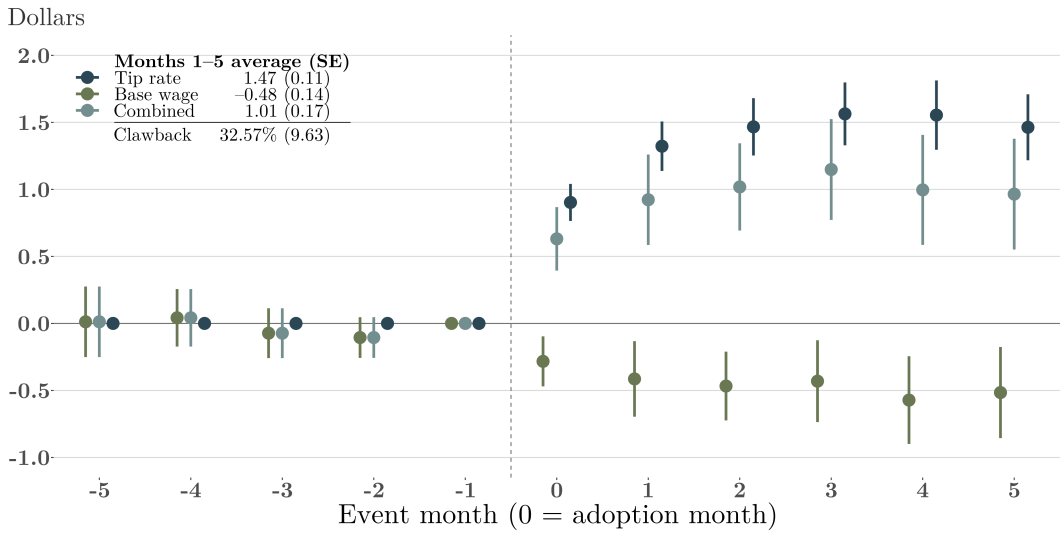
(d) Firm-level average, combined rate



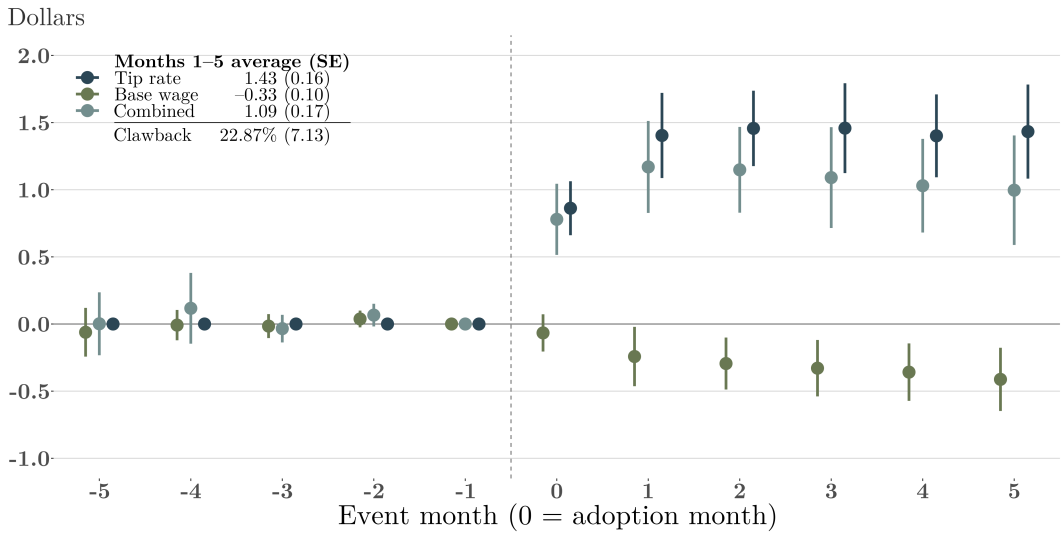
Notes: Mean outcome by event month at adopting firms and at the not-yet-treated firms that serve as controls, each arm normalized to zero in event month −1, in dollars per hour with 95 percent confidence bands. No fixed effects or calendar-month adjustment are applied, so these are the two arms that Figure 7 differences rather than event-study coefficients. The top row is the 880 incumbent workers and the bottom row the 300 adopting firms, matching panels (b) and (a) of Figure 7 respectively. In both arms the base wage rises over the window: in panel (a) the control path reaches \$0.70 above its month −1 level by month +5 while incumbents at adopting firms reach \$0.32, and in panel (c) the control path reaches \$0.46 while adopting firms fall \$0.18 in the adoption month. The combined rate separates the arms sharply: \$0.90 in the adoption month and \$1.76 by month +5 for incumbents (panel b) against \$0.12 and \$0.72 for controls, and \$0.74 and \$1.49 at adopting firms (panel d) against \$0.07 and \$0.46 for controls. Control firms have not turned on tipping, so their combined rate path is their base wage path. The estimated base wage decline is therefore measured against a rising control path.

Figure 7: Firms Claw Back 23 to 33 Percent of Each Dollar of Tips

(a) All workers at adopting firms

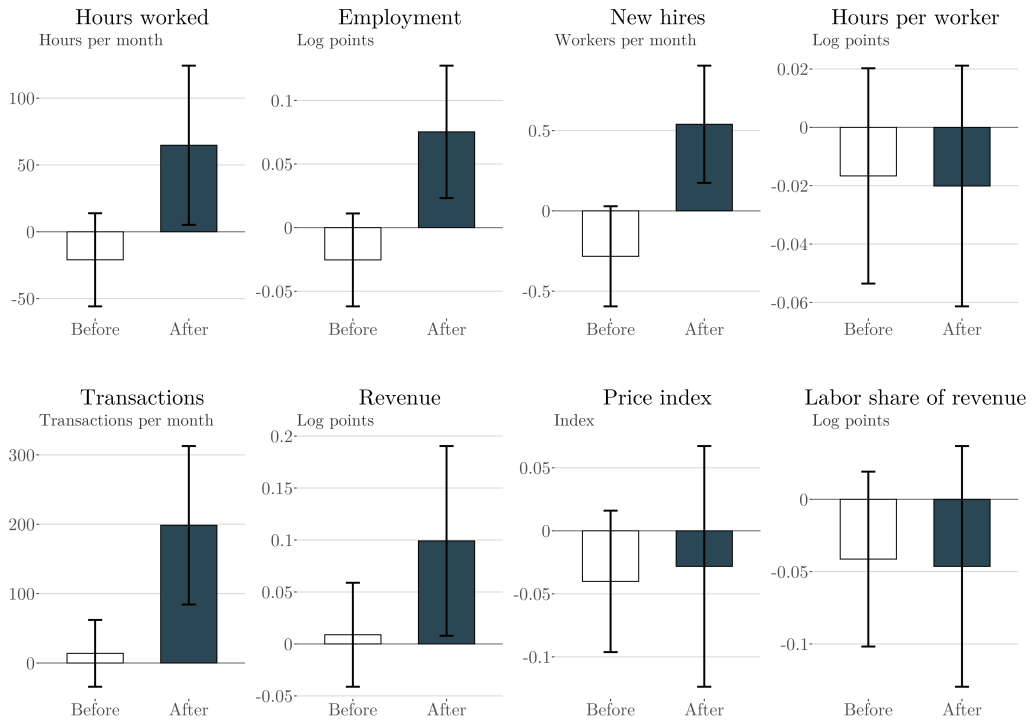


(b) Incumbent workers at adopting firms



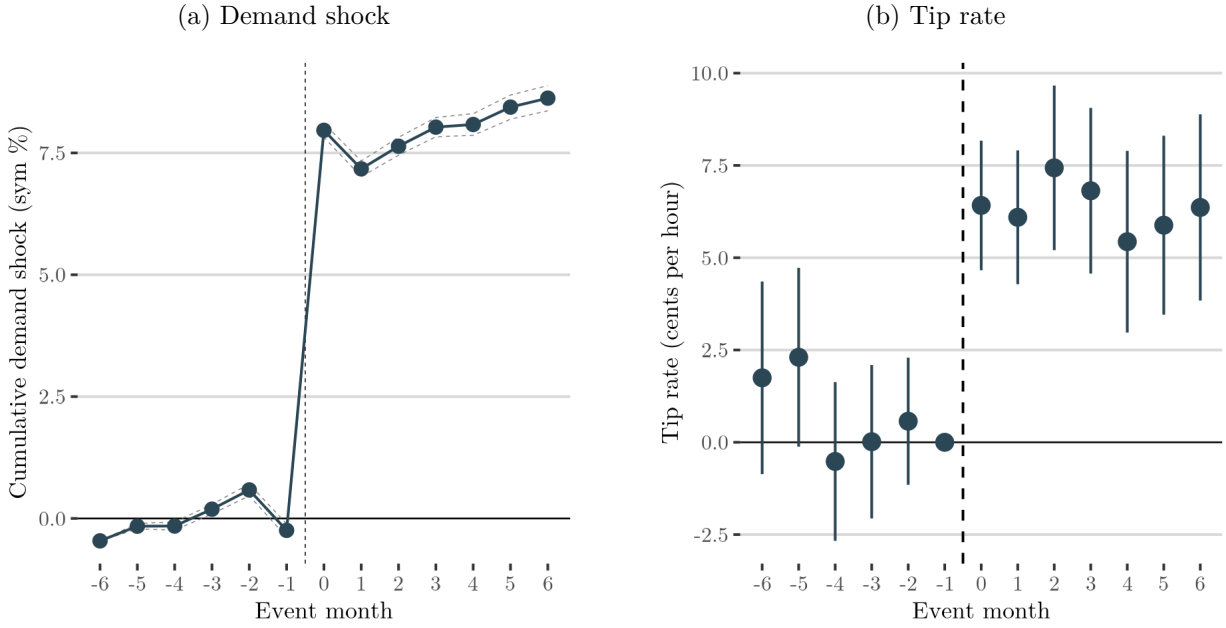
Notes: Base wage, tip rate, and combined hourly rate around tipping adoption, all in dollars per hour, with event month −1 as the reference and later-adopting firms as controls. Panel (a) is the firm level and covers all 300 adopting firms, including workers hired after adoption: averaged over months +1 to +5, the base wage falls \$0.48, the tip rate rises \$1.47, and the combined rate rises \$1.01, giving $\kappa = 0.33$ (standard error 0.10). The firm-level base wage falls \$0.28 in the adoption month itself. Panel (b) restricts to the 880 incumbent workers observed in all eleven months at the 230 adopting firms with at least one such worker: the base wage falls \$0.33, the tip rate rises \$1.43, and the combined rate rises \$1.09, giving $\kappa = 0.23$ (standard error 0.07). At full offset, the base wage would fall by the full tip rate; in both panels it falls by less than a third of it. The base wage does not move in the adoption month for incumbents and falls steadily thereafter, reaching \$0.41 below its pre-adoption level by month +5. Payroll records the pay date rather than the days worked, so payroll tip earnings can first appear a month after card tips do; the event is dated to the month payroll tip earnings begin.

Figure 8: Adopting Firms Expand and Sell More



Notes: Firm-level outcomes for the 300 adopting firms, with later-adopting firms as controls and the event dated to the month payroll tip earnings begin. Each bar is the average event-study coefficient before or after adoption. The before window is months -5 to -2 for payroll outcomes (hours worked, employment, new hires, hours per worker; reference month -1) and months -5 to -3 for point-of-sale outcomes (transactions, revenue, price index, labor share of revenue; reference month -2). The after window is months 1 to 5 for all outcomes. Top row: hours worked is in levels (hours per month); employment and hours per worker are in logs; new hires are in levels (per month). Bottom row: transactions are in levels (per month); revenue and labor share of revenue are in logs; the price index is in index units.

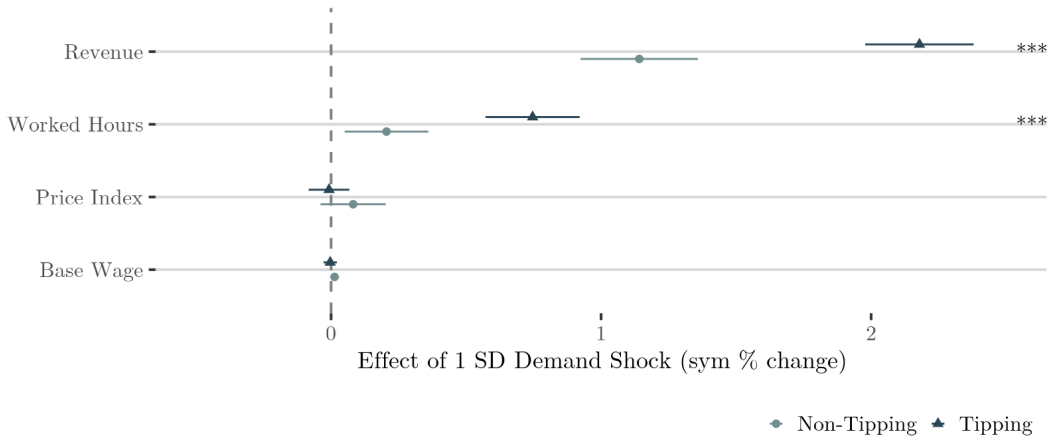
Figure 9: Demand Shocks Are Sharp and Tip Rates Respond Immediately



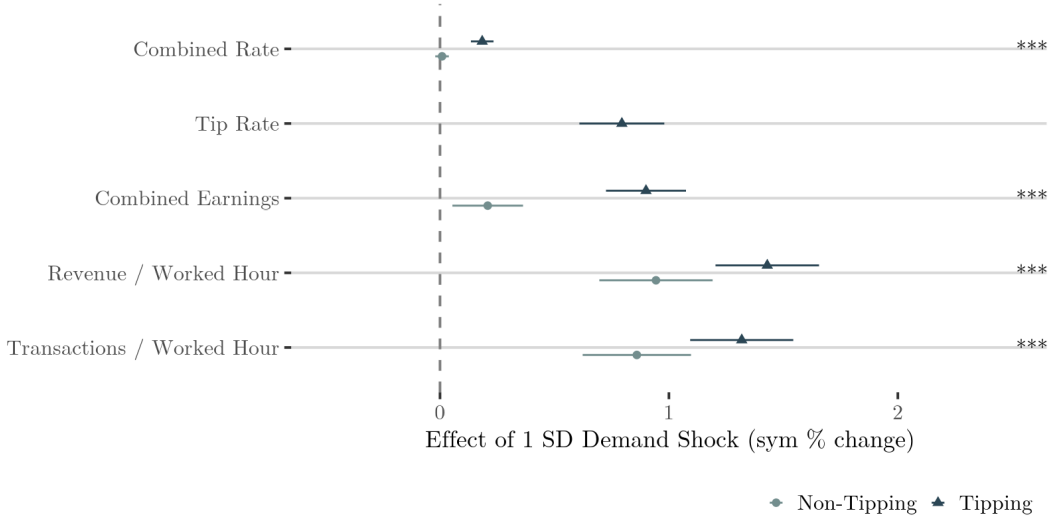
Notes: In each month, firms are split at the S_{jt}-residualized median demand shock following Lamadon, Mogstad and Setzler (2022). Panel (a) is the cumulated shock in symmetric percent, forward-summed from event time -6 and net of its mean over event times -6 to -2. Panel (b) is the tip rate in cents per hour, normalized to zero at event time -1. Bands are 95 percent confidence intervals from a 100-draw Bayesian bootstrap over firms. A month above the median raises the tip rate by about 6 cents per hour, and the gap does not close within six months.

Figure 10: Demand Shocks Raise Revenue and Tips, With Prices Unchanged

(a) Revenue, hours, prices, and base wages

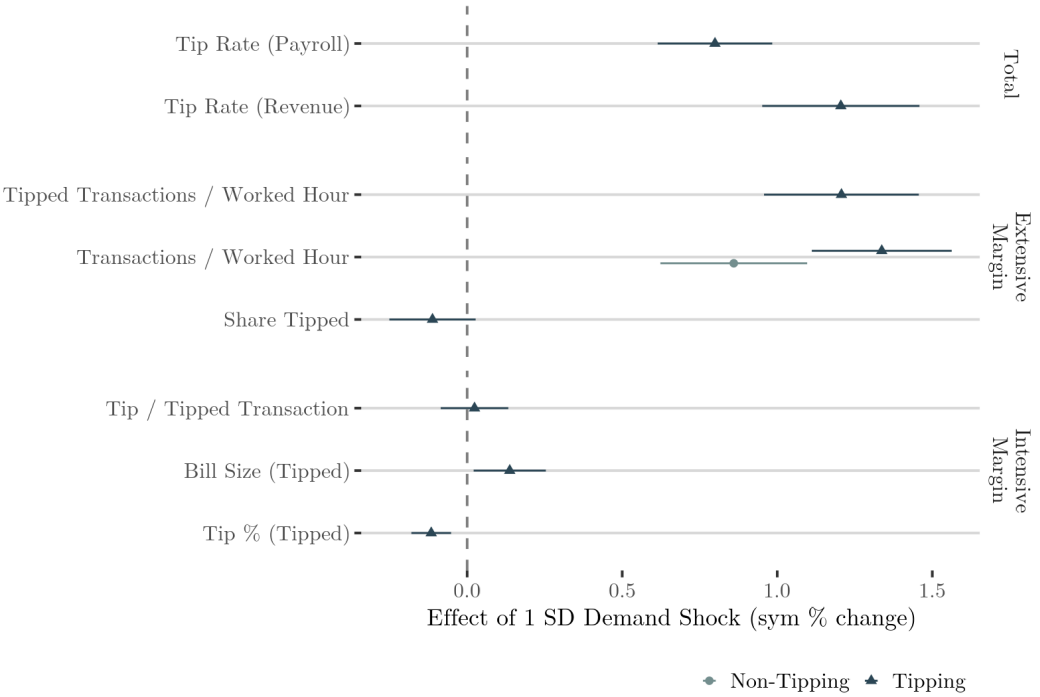


(b) Pay and output per worked hour



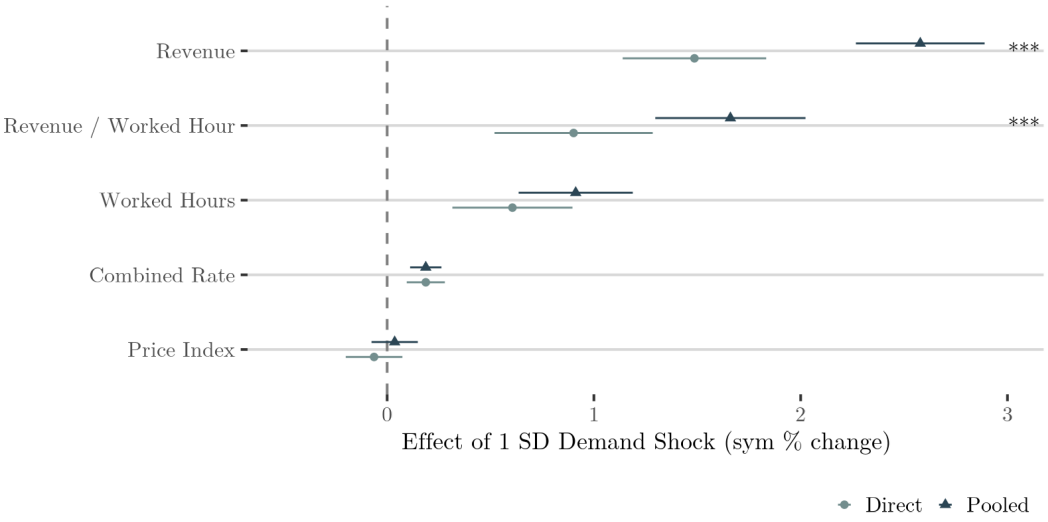
Notes: Firm-month panel, 2022 to 2025. Each row reports the response to a one-standard-deviation demand shock, along with the differential response at tipping firms. Estimates are in symmetric percent changes per one-standard-deviation demand shock; bars are 95 percent confidence intervals with standard errors clustered by firm. Controls are the recentering term S_{jt} and calendar-month effects, each interacted with compensation structure (Borusyak, Hull and Jaravel, 2022). Panel (a): revenue rises 2.18 sym pp at tipping firms against 1.14 at non-tipping firms, and worked hours 0.75 against 0.21; prices do not respond at either, and base wages are rigid at both compensation structures. Panel (b): the combined rate rises at tipping firms, driven by the tip rate. The combined rate rises 0.18 sym pp against 0.01, and combined earnings 0.90 against 0.21. Revenue per worked hour rises 1.43 sym pp at tipping firms (SE 0.12) against 0.94 at non-tipping firms (SE 0.13).

Figure 11: Tips Rise from More Customers, Not More Generous Customers



Notes: The tip rate response is split into an extensive margin (tipped transactions per worked hour) and an intensive margin (tip per tipped transaction). The response is driven by more transactions per worked hour. Tipped transactions per worked hour rise 1.21 sym pp; tip per tipped transaction is 0.02 (SE 0.06). Estimated on tipping firms only, using card transactions. Bill sizes and tip percentages move slightly in offsetting directions.

Figure 12: Pooled Firms Respond More, Mainly Through Transaction Volume



Notes: Tipping firms only, firm-month panel, 2022 to 2025. Responses to a one-standard-deviation demand shock at direct and pooled firms, in symmetric percent changes, using the same demand shock and controls as Figure 10 with an indicator for pooled firm-months in place of the tipping indicator; bars are 95 percent confidence intervals with standard errors clustered by firm. A firm-month is direct when every location pays each worker the tips from the transactions that worker serves, pooled when every location shares tips among workers, and excluded when locations disagree or the setting changed from the previous month (Section C.2). Revenue rises 2.58 sym pp at pooled firms against 1.49 at direct firms; transactions 2.25 against 1.66; revenue per worked hour 1.66 against 0.90; prices do not move under either setting.

Main Tables

Table 1: Compensation Summary, Tipping vs. Non-Tipping Firms

	Tipping		Non-tipping		
	Mean	Median	Mean	Median	Difference
Base wage ($/hour)	14.45	14.00	17.61	15.88	−3.16
Total hours	83.4	72.5	88.4	77.0	−4.9
Base earnings ($)	1,296	904	1,638	1,151	−342
Tip earnings ($)	453	234	1	0	452
Total earnings ($)	1,817	1,361	1,741	1,187	76
Total rate ($/hour)	21.09	19.36	18.53	16.00	2.56
Tip share (%)	26.11	23.82	0.07	0.00	26.04

Notes: Worker-month sample (4.0 million worker-months), restricted to worker-months with at least 5 paid hours. Base wage and total rate are hourly; hours, base earnings, tip earnings, and total earnings are monthly. Tip share is tip earnings as a percent of total earnings.

Table 2: Mean Absolute Deviation of Pay, Within Worker Spells

	Tipping	Non-tipping
Panel A: Hourly rate (base + tips per hour)		
Median MAD ($)	0.85	0.00
Median MAD (% of mean)	4.7%	0.0%
25th to 75th percentile of MAD ($)	0.41–1.64	0.00–0.50
Panel B: Monthly earnings (base + tips)		
Median MAD ($)	356.12	276.39
Median MAD (% of mean)	30.0%	28.0%
25th to 75th percentile of MAD ($)	207.31–558.97	151.75–451.17

Notes: Mean absolute deviation computed within worker spells of at least three consecutive months. Combined rate is base wage plus tips per hour. Monthly earnings are base plus tip earnings, excluding overtime, bonuses, and paid leave. Worker-months with fewer than 5 paid hours are excluded.

Table 3: Calibrated Parameters and Fixed Inputs

Parameter	Interpretation	Value
Panel A: Calibrated parameters		
ν	Joint returns to hours and effort	1.85136
θ	CES curvature	5.59148
α	CES hours weight	0.24373
$c'(1.05e_0)$	Marginal effort cost at $e = 1.05e_0$	8.51459
$\bar{a}$	Mean demand	1.54493
λ	Recruitment scale, dollars per month	3,518.29
Panel B: Derived parameters		
γ	Exponent of marginal effort cost	0.86136
k	Effort-cost scale in $c(e) = kx^{1+\gamma}/(1+\gamma)$	9,324.72
Panel C: Fixed inputs and normalizations		
e_0	Effort-unit normalization, reference NT hours per worker	82.95
τ_T	Worker share of total customer payments	0.088
N_{NT}	Reference employment	5
s_a	Standard deviation of log demand	0.28

Notes: The six parameters in Panel A are fitted by weighted minimum distance to the nine moments in Table A7. Effort cost is $c(e) = c_0 + kx^{1+\gamma}/(1+\gamma)$ with $x = (e - e_0)/e_0$, and the fixed effort cost c_0 is set to zero. The recruitment intercept ξ_N is set so that non-tipping employment equals five at each candidate parameter vector. $\tau_T = 0.088$ corresponds to tips equal to 9.7 percent of pre-tip sales.

Table 4: Compensation and Labor Allocation: Model and Cross-Section

	Non-tipping		Tipping	
Outcome	Model	Data	Model	Data
Base wage ($/hour)	15.60	17.61	15.23	14.45
Tip rate ($/hour)	0.00	0.00	5.53	6.64
Combined rate ($/hour)	15.60	18.53	20.76	21.09
Monthly hours per worker	93.40	88.4	107.17	83.4
Employment	5.00	—	6.15	—
Total monthly hours	467.0	—	659.1	—
Revenue per hour, including tips ($)	61.64	—	62.63	—
Base wage gap, tipping minus non-tipping ($/hour)				
Model		−0.37		
Data		−3.16		

Notes: Model columns are the solution at $\tau_T = 0.088$ and $\tau = 0$ at the calibrated parameters (Table 3); non-tipping employment is normalized to five workers. Data columns are raw means by compensation structure from Section 2, not the calibration targets: the targeted ratios are reweighted to the sector composition of tipping firms, and the targets and fit are reported in Section E.1. In the data columns, the combined rate is total earnings per paid hour, including overtime, bonuses, and paid leave, and the tipping tip rate is that rate minus the base wage, so the non-tipping combined rate exceeds the base wage. Revenue per hour is $\mathbb{E}[y]/\mathbb{E}[h]$ and the tip rate is $\tau\mathbb{E}[y]/\mathbb{E}[h]$. The data base wage gap is a difference in levels across firms, not a within-firm response; Table 5 reports the within-firm clawback.

Dashes mark data moments not reported in Section 2.

Table 5: Clawback: Model and Adoption Estimates

		Adoption estimates	
	Model	Firm level	Incumbent workers
Tip percent (share of pre-tip revenue)	2.66	2.66	
Clawback, κ	0.34	0.33	0.23
Standard error	—	(0.10)	(0.07)
For reference, at each column's tip percent			
Base wage change ($/hour)	−0.54	−0.48	−0.33
Tip rate change ($/hour)	1.59	1.47	1.43

Notes: The model column is the tipping-minus-non-tipping difference at the calibrated parameters, solved at $\tau = 0.0259$ of tip-inclusive revenue (2.66 percent of pre-tip revenue) with the recruitment intercept held at its calibrated value. The adoption columns are the stacked difference-in-differences estimates from Section 3 for the 300 adopting firms and their 880 incumbent workers at 230 adopting firms, averaged over months one through five after adoption; tips at adopters are 2.66 percent of pre-tip revenue, the firm-level post-adoption estimate across the 300 adopting firms. The clawback is the comparison across columns.

Table 6: The Value of Tipping to Workers and Firms

Panel A: Contributions to the change in worker value	Dollars per worker per month	Share of change in V	Per dollar of tips
Wage change	−40	−0.055	−0.067
Hours change	+215	0.295	+0.363
Tips	+592	0.814	+1.000
Effort cost	−39	−0.054	−0.066
Total change in V	+728	1.000	+1.229
Panel B: Firm outcomes			
	Non-tipping	Tipping	Change
Operating profit per worker ($/month)	4,300	4,488	+187
Employment	5.000	6.150	+1.150
Total firm operating profit ($/month)	21,502	27,597	+6,096

Notes: All changes are tipping minus non-tipping. Dollar values are rounded to the nearest dollar; calculations precede display rounding. The symmetric base-earnings decomposition assigns $(w_T - w_{NT})(\mathbb{E}[h_T] + \mathbb{E}[h_{NT}])/2$ to the wage change and $(\mathbb{E}[h_T] - \mathbb{E}[h_{NT}])(w_T + w_{NT})/2$ to the hours change. The effort-cost contribution is $-\{\mathbb{E}[c(e_T)] - \mathbb{E}[c(e_{NT})]\}$. Shares divide each contribution by $V_T - V_{NT}$; the per-dollar column divides by expected monthly tips in the tipping regime, $\mathbb{E}[\tau_T y_T]$. Panel B reports firm outcomes separately from the worker-value decomposition; operating profit is revenue net of tips and base payroll.

A Data Appendix

This appendix defines each variable, documents the sample restrictions, the quality screens, and the compensation-structure windows that produce the analytic sample, reports the classification under alternative denominators and cutoffs, and describes the demand-shock sample. All counts refer to the analytic sample unless stated otherwise.

A.1 Variable definitions

Payroll records hours by type on each paystub. *Worked hours* are regular, overtime, double-time, and contractor hours; they exclude paid sick leave and paid time off. *Total hours* add paid leave to worked hours. Output per hour is measured per worked hour throughout.

The *base wage* is regular earnings divided by regular hours, the hourly rate associated with the job. The *tip rate* is tip earnings divided by worked hours. The *combined rate* is the base wage plus the tip rate. We aggregate paystubs to the worker-firm-month. Paystubs are dated by pay date, which can fall up to four days after the end of the period worked, so a paycheck dated in month t can cover days worked in month $t-1$. Headcount in a month is the number of workers who received a paycheck in that month.

We set the base wage, the tip rate, and the combined rate to missing in firm-months where the base wage, the tip rate, or total earnings per hour lies outside its 0.1st to 99.9th percentile across all firm-months before the quality screens. This removes firm-months with implausible hours or earnings records, such as months with earnings but no recorded hours. These firm-months are ineligible for the compensation-structure windows (Section A.3), so none enters the analytic sample.

Revenue is the value of recorded transactions net of tips and sales taxes. *Card revenue* includes card transactions only; *revenue* without a qualifier includes all tender, cash among it. Both measures cover only transactions recorded at the point of sale, so cash sales never entered at the register appear in neither. Card transactions account for 83 percent of revenue overall, 84 percent at tipping firms and 80 percent at non-tipping firms.

Tips are the tips recorded at the point of sale, on all tender; *card tips* are those on card transactions. *Tip earnings* are the tips recorded in payroll. The *tip percent* is tips divided by revenue, both pre-tax and pre-tip. Tip percents reported in the paper use all-tender tips and revenue; the classification in Section A.4 uses a card-based measure.

The firm-level price index is a chained geometric Laspeyres index built from item-level price changes. For firm j in month t,

$$\ln P_{jt} = \ln P_{j,t-1} + \sum_{i \in \mathcal{I}_{jt}} s_{ij,t-1}(\ln p_{ijt} - \ln p_{ij,t-1}) \qquad P_{j0} = 1 \tag{22}$$

where $\mathcal{I}_{jt}$ is the set of items the firm sells in both $t-1$ and t, p_{ijt} is item i's revenue divided

by its quantity, and $s_{ij,t-1}$ is item i's share of revenue among those items in $t-1$. Weighting by lagged shares holds the basket fixed within each link, so the index moves only when the prices of continuing items move, not when the mix of items sold shifts.

A.2 Sample restrictions and quality screens

Before the quality screens, we restrict to U.S. firms with both payroll and transaction records and trim each firm's months to the period in which both are present. A firm must run payroll in at least eight months and pay an average of at least three workers in those months. We drop firms with a base wage above $100 per hour in any month and firms with any worker-month that records hours but no earnings. The adoption sample in Section 3 is drawn after these restrictions and before the steps below.

We apply three quality screens in sequence to firms with both transaction and payroll records: a revenue coverage screen, followed by the wage bill and tip-match screens described in Section 1. Of all screened-out firms, the revenue coverage screen removes 9 percent, the wage bill screen 29 percent, and the tip-match screen 62 percent. Applied independently, the three screens flag 9, 36, and 68 percent of screened-out firms, respectively.

The revenue coverage screen removes firms whose transaction records are missing for most of the months in which they run payroll.

The wage bill screen removes firms whose wage bill exceeds revenue in more than half of their months. A firm cannot sustain a wage bill above revenue for most months, so a firm that fails this screen is missing transaction records.

The tip-match screen compares payroll tip earnings with point-of-sale tips in each firm-month. A month passes when the ratio of payroll tip earnings to point-of-sale tips is between 0.6 and 1.5. Months with no tips in either record also pass, as do months with tip earnings but no point-of-sale tips. Months in which either record is missing are not scored. A firm is retained if at least half of its scored months pass.

The band is two-sided because each bound catches a different failure. A ratio below 0.6 means tips collected at the register do not appear in payroll. A ratio above 1.5 means payroll records more tips than the register, which indicates transaction records we do not capture rather than workers paid more than customers tipped. Because months without tips pass, the screen removes firms that consistently collect tips at the register and consistently do not report them through payroll. Among firms that pass the first two screens and fail the tip-match screen, the median firm reports 1.8 percent of its point-of-sale tips through payroll, and the firm at the 75th percentile reports 41 percent. The data do not record how these firms distribute tips outside payroll.

The firm-level ratio sums payroll tip earnings and point-of-sale tips over all of a firm's months before dividing. Summing removes the pay-period timing mismatch that the month-level band tolerates,

since payroll can record a tip the month after the register does. Before screening, the firm-level ratio has a median of 0.78 and an interquartile range of 0.01 to 0.98. In the analytic sample, the median is 0.96, the 25th percentile is 0.85, and the 75th percentile is 1.00. The analytic-sample distribution is one-sided, with a tail that runs left, so firms rarely report more tips through payroll than the register records.

The adoption sample in Section 3 does not apply the quality screens or the compensation-structure windows. Adopting firms change compensation structure by definition, and the tip-match screen would select them on how tips are recorded after adoption, a post-treatment outcome. Section B.3 reports the pass-through ratio at adopting firms.

A.3 Compensation-structure windows

We observe each firm over a window in which its compensation structure is stable. A month is eligible when the base wage, the combined rate, and revenue are all positive. A month is removed when card tips exceed one percent of card revenue and payroll records no tip earnings. Windows break at calendar gaps and at removed months. Within an unbroken stretch, a window also ends where a month with no tips meets a run of two or more consecutive months with a tip percent of at least one percent; runs of one month are treated as transitory. For each firm we keep the longest window of at least eight months, taking the later window when two are equally long. Firms with no such window leave the sample.

This step removes 2,100 firms and 100,700 firm-months from firms that pass the quality screens. Of the firm-months, 49,800 are ineligible, 7,600 are removed by the tip-reporting rule, and 43,300 fall outside the selected window. Of the firms, 720 lose every month to the eligibility and tip-reporting rules, and 1,400 have no window of eight months or more.

Each firm enters the sample in the first month in which both its payroll and its transaction records are present, so firms do not share a common observation window. Of the 19,700 firms, 9,200 enter in January 2022. The median firm enters in March 2022, the 75th percentile in May 2023, and the 90th percentile in February 2024.

A.4 Classification

A firm is *tipping* when its median monthly tip percent over its window is at least one percent. For classification, the tip percent is the larger of card tips and tip earnings, divided by card revenue. Taking the larger of the two records classifies a firm as tipping whether its tips are recorded mainly at the register or mainly in payroll. Classification is fixed over each firm's window.

The binary classification barely moves: a net difference of 40 firms. The finer categories are less stable. Because card revenue is the smaller denominator, the card measure places firms higher: 1,900 firms move from the top category on the card measure to the middle one on all tender, and

520 move from the middle to the lowest tipping category. Tip percents reported by category in Section 2 use all tender within categories defined on the card measure.

Raising the cutoff from one to five percent would classify 11,600 firms as tipping on the card measure; raising it to fifteen percent would classify 4,400.

Tips are rare at non-tipping firms. Tip earnings are positive in 2.5 percent of non-tipping firm-months and point-of-sale tips in 3.9 percent. Tip earnings average about one cent per hour, with a median of zero, against a base wage near $17. Tips are 10.4 percent of revenue at tipping firms and 0.01 percent at non-tipping firms. The combined rate at non-tipping firms is therefore effectively the base wage.

A.5 Sample description

The analytic sample contains 19,700 firms, 12,600 tipping and 7,100 non-tipping, over 505,900 firm-months from January 2022 through March 2025. It contains 500,000 distinct workers. The median firm pays 7 workers in a month (interquartile range 4.6 to 11; 90th percentile 16.5). Over a complete firm-year, the median firm pays 13 distinct workers (interquartile range 8 to 21; 90th percentile 35), across 30,800 complete firm-years at 15,200 firms. Food and beverage firms are 53 percent of firms and retail firms 23 percent; the remaining 23 percent are spread across other small-business sectors. Firms are located in all 50 states and the District of Columbia. Table A1 reports the industry composition by compensation structure and worker characteristics by group.

A.6 Demand-shock sample

The demand-shock sample in Section 5 starts from the analytic sample and applies two further restrictions. We restrict to sectors representative of the counter-service businesses that make up most of the analytic sample, removing 1,200 firms. Requiring the demand shock and the outcomes to be non-missing after differencing, over a minimum consecutive spell, removes about 120 more. The resulting sample contains 18,400 firms, 11,800 tipping and 6,500 non-tipping, over 436,800 firm-months from February 2022 through March 2025. Food and beverage and retail firms are 82 percent of this sample.

Because classification uses the median tip percent over each firm's window, a small number of firms cross the one percent threshold within their own window. Firms that switch to tipping within their window contribute about 6,000 of the 436,800 firm-months. At most 2,300 post-crossing firm-months carry a non-tipping classification, at most 1.5 percent of non-tipping firm-months. Misclassification in this direction shrinks the difference between tipping and non-tipping firms rather than inflating it.

B Clawback Appendix

B.1 Adoption definition

We apply the adoption rule to the panel before the quality screens and compensation-structure windows (Sections A.2 and A.3). A firm adopts tipping in month m when four conditions hold. First, each of the six consecutive calendar months before m records no tip earnings in payroll and positive card revenue. Positive card revenue ensures the firm is operating and recording transactions in those months, so zero tip earnings do not reflect missing records. Second, card tips over those six months are at most 0.05 percent of card revenue,

$$\frac{\sum_{k=1}^{6} \text{card tips}_{j,m-k}}{\sum_{k=1}^{6} \text{card revenue}_{j,m-k}} \leq 0.0005 \tag{23}$$

which excludes firms that already collect tips at the register but pay them outside payroll. Third, card tips are positive in month m. Fourth, payroll records tip earnings in month m or in month $m+1$, since payroll can record a tip in the month after the register does. Each firm takes the first month that satisfies all four conditions.

The event month is the first month with tip earnings in payroll: m when payroll records tips in m, and $m+1$ otherwise. Every month in the five months before the event therefore has no tip earnings in payroll. The rule uses no information after the event month, so it does not select firms on whether or how much they tip after adopting.

B.2 Slice construction

To estimate the effect of adoption, we compare adopting firms to those that have not yet adopted in their pre-adoption months. To estimate all cohort-by-relative-time effects, we construct *slices* which compare one adopting cohort to later-adopting ones in clean comparisons. We then *stack* these slices to estimate Equation (2) (Deshpande and Li, 2019; Wing, Freedman and Hollingsworth, 2024).

The firm-level slice for adoption month s keeps the following firm-month observations:

$$\underbrace{\{(i,t) : c(i) = s\}}_{\text{Treatment}} \quad \text{and} \quad \underbrace{\{(i,t) : c(i) > s \text{ and } t < c(i)\}}_{\text{Control}} \tag{24}$$

The treatment observations are the firms that adopt in month s. The control observations are the firms that adopt after month s, in every month before their own adoption, so every control observation is untreated. We truncate each slice to the eleven months from $s-5$ to $s+5$. Every firm in a slice meets the adoption definition in Section B.1, so firms that never adopt do not enter as controls. At the firm level, we drop firms with a missing base wage, tip rate, or combined rate in any month from the start of their run without payroll tips through five months after their own

adoption, and keep treated firms only if they are observed in the month before adoption. Within each slice, we keep the largest set of control units and months connected through shared months, so that the calendar-month effects are identified.

The worker-level slice for month s is built from the same firms, but selects those workers who appear in all eleven months around their own firm's adoption with the base wage, the tip rate, and the combined rate recorded. The treatment observations are the incumbents at firms that adopt in month s. The control observations are the workers at firms that adopt after month s, in each month before their firm adopts. Because the presence requirement is the same for treated and control workers, both groups are alike in staying at their firm throughout the eleven months around their own firm's adoption.

We stack the slices into one dataset and estimate Equation (2) on it. The unit and calendar-month effects are specific to each slice, so the stacked regression estimates the same $\beta_{\ell s}$ as estimating each slice separately: each compares firms that adopt in month s only with controls in slice s. The stacked estimation returns the full variance-covariance matrix for all slice-by-relative-time estimates $\beta_{\ell s}$. This makes it straightforward to calculate the standard errors for linear combinations of the slice-specific treatment effects. A firm that adopts late is a control in every earlier slice, so its observations appear more than once; clustering standard errors by firm accounts for the repetition.

The event-study coefficient averages these slice-specific effects,

$$\beta_\ell = \sum_s \omega_{\ell s} \beta_{\ell s}, \qquad \omega_{\ell s} = \frac{N_{\ell s}}{\sum_{s'} N_{\ell s'}} \tag{25}$$

where $N_{\ell s}$ is the number of treatment observations in slice s at relative month ℓ.

B.3 Adopting firms pass customer tips through payroll

After adoption, adopting firms pass nearly all customer tips through payroll. We measure pass-through as payroll tip earnings divided by point-of-sale tips, summed over each firm's months from the month after adoption onward. We start in the month after adoption because payroll can record a tip in the month after the register does, which leaves the adoption month incomplete. The median adopting firm passes 96 percent of its point-of-sale tips through payroll, against 97 percent at tipping firms in the analytic sample. The interquartile range, 75 to 101 percent, is wider than in the analytic sample, consistent with a shorter average spell length. If adopting firms recorded part of base pay as tip earnings, payroll tip earnings would exceed point-of-sale tips, but even the top of the interquartile range is near parity.

B.4 Mover clawback

In the analytic sample, we are able to observe individual workers across employers using a tokenized identifier. Using this identifier, we are able to construct job spells at different employers for an

individual, and combine these spells into job moves. Destination and origin spells are allowed to overlap by at most one month, with a gap of at most six months. We only allow spells to be a part of a single move. Since each firm is classified consistently across the sample period as tipping or non-tipping, there are three types of moves: tipping to non-tipping, the reverse, and same-type moves. Each mover is observed with non-missing base wage and tip rate for the full pre- and post-move period, from 4 months before to 4 months after the move. In this movers sample, 120 workers enter tipping, 90 leave for non-tipping employers, and workers make 300 non-tipping-to-non-tipping moves and 2,400 tipping-to-tipping moves.

Figures A6a and A6b report the stacked difference-in-differences on workers moving between firm types, and Figures A6c and A6d the triple difference that nets out same-type moves. Controls are workers making the same type of move later, in the months before their own move; the triple difference subtracts same-type moves that start from the same firm type. We normalize to month -2, the last full month at the origin employer.

The clawback is 0.23 for workers entering tipping firms in the difference-in-differences and 0.39 net of same-type moves; for workers leaving, it is 0.51 and 0.44, respectively, averaging about 0.40 across directions in the triple difference. These estimates are of the same order as the adoption clawback of 0.23 to 0.33.

C Demand Shock Appendix

C.1 Demand shock diagnostics and card share distributions

Figure A4 reports the shock and exposure share distributions by firm type and the recentering diagnostic.

C.2 Tip pooling: classification, sample, and estimates

Classification. The point-of-sale system records a tip-distribution setting for each of a firm's locations, with the dates over which each setting was in effect. A location is *direct* when each worker receives the tips their own customers leave. It is *pooled* under any other setting: tips shared among the workers on a transaction, or pooled by hours or by percentage over a workday or a workweek. We drop settings in effect for fewer than 14 days, which reflect configuration changes that were reversed, and merge adjacent spells with the same setting. We also drop locations with no transactions in the sample window and firms whose recorded settings are unreliable because of a change in how the setting was recorded. A firm-month is direct when every location is direct in that month and pooled when every location is pooled. We exclude firm-months in which locations disagree and firm-months in which the setting differs from the previous calendar month, so that every month in the sample follows a full month under the same setting.

Sample. The sample is the tipping firms in the demand-shock sample of Section A.6 with a classified setting: 212,100 firm-months. About 60 percent of tipping firms pool. Pooling is more common at quick-service firms (74 percent) than at full-service firms (47 percent), and both settings are common in each. Direct and pooled firms are similar in size: the average direct firm-month has 11.2 workers and 926 hours, against 10.2 workers and 784 hours at pooled firms. The switcher sample is the subset of firms observed for at least three months under each setting, with the same exclusions: 14,400 firm-months.

Specification. We estimate Equation (17) on this sample with an indicator for pooled firm-months in place of T_j. The controls are those of Section 5, with the month effects and the coefficients on S_{jt} varying by tip-distribution setting instead of by compensation structure. Because these controls are estimated on tipping firms only, the direct and pooled coefficients are comparable with each other but not in level with the estimates in Section 5.2. Standard errors are clustered at the firm level.

Estimates. The baseline specification is in the text. Restricting to firms observed under both settings, so the comparison is between pooled and direct months of the same firms, the revenue-per-worked-hour response is -0.07 in direct months versus 0.86 in pooled ($p = 0.33$) and the transactions response is 0.61 versus 2.37 ($p = 0.01$). Adding industry-by-month effects to the baseline, so that firms are compared with others in the same industry and month, the revenue response is 1.37 versus 2.34 ($p < 0.001$), revenue per worked hour is 0.78 versus 1.43 ($p = 0.01$), and transactions are 1.57 versus 2.05 ($p = 0.04$). In all three specifications, the pooled response exceeds the direct response

for revenue, transactions, and revenue per worked hour.

Pay across coworkers. If the recorded setting describes how tips are actually paid, a demand shock should make earnings more equal across a firm's workers at pooled firms, where the additional tips are split among coworkers, and more so the longer the period over which tips are pooled. At direct firms a demand shock does not change the coefficient of variation of earnings across a firm's workers (-0.01 sym pp, standard error 0.18). The point estimate falls as the period over which tips are pooled lengthens: -0.27 (standard error 0.22) when tips are pooled per transaction, -0.44 (standard error 0.30) per workday, and -0.57 (standard error 0.47) per workweek (Figure A5). The ordering matches this prediction, but no pooling period is distinguishable from direct firms ($p = 0.36, 0.21$, and 0.26).

D Model Appendix

This appendix derives the effort and hours conditions, reduces the state problem to a system that can be solved numerically, establishes existence and uniqueness, and proves the propositions in Section 4.5. Under the cost in Equation (5), effort exceeds e_0 at every tipping firm with positive hours; the floor $e = e_0$ binds only at non-tipping firms. The formulas below use the positive-θ CES specification with $\alpha \in (0, 1)$ and $k > 0$.

D.1 Effort and hours

Define the CES hours share

$$s_h = \frac{\alpha h^{-\theta}}{\alpha h^{-\theta} + (1-\alpha)e^{-\theta}} \tag{26}$$

The elasticities of g with respect to hours and effort are νs_h and $\nu(1 - s_h)$. In particular,

$$\frac{g_{eh}}{g_e} = \frac{(\nu+\theta)s_h}{h} \qquad \frac{g_{ee}}{g_e} = \frac{\nu - 1 - (\nu+\theta)s_h}{e} \tag{27}$$

Under the cost in Equation (5), $c'(e_0) = 0$. At a tipping firm with $h > 0$, the marginal return to effort at e_0 is $\tau a g_e(h, e_0) > 0$, which strictly exceeds $c'(e_0) = 0$. The worker therefore sets $e^\star > e_0$ at every demand state with positive hours: effort always exceeds the minimum at tipping firms. At non-tipping firms, $\tau = 0$, so effort is e_0.

Differentiating the interior worker condition $\tau a g_e = c'(e^\star)$ and dividing by the common value gives

$$\left.\frac{\partial \log e^\star}{\partial \log a}\right|_h = \frac{1}{1 + \frac{\gamma e^\star}{e^\star - e_0} - \nu + (\nu+\theta)s_h} \tag{28}$$

$$\left.\frac{\partial \log e^\star}{\partial \log h}\right|_a = \frac{(\nu+\theta)s_h}{1 + \frac{\gamma e^\star}{e^\star - e_0} - \nu + (\nu+\theta)s_h} \tag{29}$$

The term $\gamma e^\star/(e^\star - e_0)$ is the elasticity of $c'(e)$ with respect to e, evaluated at $e^\star$. It exceeds γ for all finite $e^\star > e_0$ and approaches γ as effort grows large. The restriction $1 + \gamma > \nu$, together with $\nu + \theta > 0$, guarantees that the denominator is positive throughout the positive hours–effort domain: the denominator exceeds $1 + \gamma - \nu > 0$, with two additional positive terms from $\gamma e_0/(e^\star - e_0)$ and $(\nu + \theta)s_h$. Under these restrictions, both derivatives are positive. These are conditional effort derivatives; the total response to demand also includes the firm's hours adjustment.

To obtain these formulas, write the worker condition as $\log(\tau a g_e) = \log c'(e^\star)$ and differentiate. For a change in hours at fixed demand, the differential is

$$(\nu+\theta)s_h \; d\log h + [\nu - 1 - (\nu+\theta)s_h] \; d\log e = \frac{\gamma e^\star}{e^\star - e_0} \; d\log e$$

where the left side uses Equation (27) and the right side is the elasticity of c' at $e^\star$. For a demand

change at fixed hours, replace the first term with $d \log a$. Rearranging gives Equations (28) and (29).

The induced-effort ratio in the firm's hours condition is

$$\frac{g_e}{g_h}\frac{\partial e^\star}{\partial h} = \frac{(\nu+\theta)(1-s_h)}{1+\frac{\gamma e^\star}{e^\star - e_0} - \nu + (\nu+\theta)s_h} \tag{30}$$

The base wage bill as a share of tip-inclusive revenue satisfies

$$\frac{wh}{y} = (1-\tau)\nu s_h \left(1 + \frac{g_e}{g_h}\frac{\partial e^\star}{\partial h}\right) \tag{31}$$

where the ratio is given by Equation (30). At non-tipping firms, $e = e_0$ and the induced-effort ratio is zero. At tipping firms, complementarity between hours and effort makes the ratio positive: an additional hour raises the return to effort, inducing the worker to exert more, which raises the marginal product of hours above g_h alone.

D.2 Analytical reduction of the state problem

For non-tipping firms, $e = e_0$. The CES share implies

$$h = e_0 \left[\frac{\alpha(1-s_h)}{(1-\alpha)s_h}\right]^{1/\theta} \tag{32}$$

Substitution into $ag_h = w$ gives a scalar equation for $s_h \in (0,1)$:

$$w = a\nu e_0^{\nu-1}\alpha^{-1/\theta}s_h^{1+1/\theta}\left(\frac{1-s_h}{1-\alpha}\right)^{(\nu-1)/\theta} \tag{33}$$

This equation implicitly characterizes hours. Stationary points must be checked against each other. For example, when $\nu > 1$, the right-hand side need not be monotone over the entire share interval, so solving one first-order condition is insufficient to establish a global maximum.

For a tipping firm, effort exceeds e_0 at every demand state (Section D.1). The worker condition and the firm's hours condition form a pair of equations in the share s_h and effort e. For a candidate share s_h, the worker condition

$$\tau a\nu(1-s_h)e^{\nu-1}\left(\frac{1-s_h}{1-\alpha}\right)^{\nu/\theta} = \frac{k}{e_0}\left(\frac{e-e_0}{e_0}\right)^{\gamma} \tag{34}$$

determines effort as the unique $e > e_0$ satisfying the equation (Section D.4). The firm's hours condition, evaluated at this effort, provides a scalar equation in s_h:

$$w = (1-\tau)a\nu e^{\nu-1}\alpha^{-1/\theta}s_h^{1+1/\theta}\left(\frac{1-s_h}{1-\alpha}\right)^{(\nu-1)/\theta}\frac{1+\frac{\gamma e}{e-e_0}+\theta}{1+\frac{\gamma e}{e-e_0}-\nu+(\nu+\theta)s_h} \tag{35}$$

where e is the solution to Equation (34) at the candidate s_h. The state problem thus reduces to a

scalar search over s_h, with effort determined implicitly at each candidate share.

These representations follow from $g(h,e) = e^{\nu}[(1-s_h)/(1-\alpha)]^{\nu/\theta}$, the worker condition, and Equation (30). Expected worker value and expected profit integrate the state solution over demand.

D.3 Effort when tips are pooled

Suppose the N workers at a firm pool tips and worker i receives a share s_i of the pool, with $\sum_i s_i = 1$. Worker i's payoff is $wh + s_i\tau \sum_j ag(h,e_j) - c(e_i)$.

Separate choices.

When each worker chooses effort taking the effort of coworkers as given, the interior condition is

$$s_i \tau a g_e(h,e_i) = c'(e_i) \tag{36}$$

With equal shares, $s_i = 1/N$, this is Equation (19). A unit of worker i's effort adds $\tau a g_e$ to the pool, of which worker i receives the fraction s_i. Because $s_i < 1$ for every worker under any division of the pool, effort is lower than under direct attribution. As s_i falls toward zero, the worker's problem approaches that of a worker at a non-tipping firm, and the firm's responses to demand approach the non-tipping responses in Equations (42) and (43), evaluated at the firm's hours share. Away from that limit, the response of revenue per hour to demand can be larger or smaller than under direct attribution. Lower effort raises the elasticity of marginal effort cost, $\gamma e^{\star}/(e^{\star} - e_0)$, which damps the response of effort to demand in Equation (28), while the firm's hours and the hours share s_h also change; Equation (44) gives no general sign for the net effect.

Joint choice.

When the workers choose effort to maximize their combined payoff, $Nwh + \tau \sum_j ag(h,e_j) - \sum_i c(e_i)$, the division of the pool drops out and the condition for each worker is

$$\tau a g_e(h,e_i) = c'(e_i) \tag{37}$$

which is Equation (8). Production is independent across workers, so under direct attribution a worker's effort has no effect on coworkers' payoffs and the separate and joint choices coincide. The model therefore implies the same allocation at pooled firms whose workers choose jointly as at direct firms.

D.4 Existence of the state solution

For $\theta > 0$, $\nu > 0$, and $1 + \gamma > \nu$, the worker problem has a unique solution $e^{\star} > e_0$ at every $a, h > 0$ with $\tau > 0$. The ratio of marginal reward to marginal cost, $\tau a g_e(h,e)/c'(e)$, has elasticity

with respect to e

$$\nu - 1 - (\nu + \theta)s_h - \frac{\gamma e}{e - e_0}$$

which is negative for all $e > e_0$: the first three terms sum to at most $\nu - 1$, while $\gamma e/(e - e_0) > \gamma > \nu - 1$. The ratio is therefore strictly decreasing on (e_0, ∞). It diverges to $+\infty$ as $e \to e_0^+$ (because $c'(e_0) = 0$ and $g_e(h, e_0) > 0$) and tends to zero as $e \to \infty$. By the intermediate value theorem, there is a unique crossing.

Moreover, $g \leq (1-\alpha)^{-\nu/\theta} e^{\nu}$ and $g_e \leq \nu(1-\alpha)^{-\nu/\theta} e^{\nu-1}$. At an interior solution, the worker condition and these inequalities give an upper bound on effort: for $e > 2e_0$, $(e - e_0)^{\gamma} > (e/2)^{\gamma}$, so

$$e^{1+\gamma-\nu} \leq \frac{2^{\gamma} \tau a \nu (1-\alpha)^{-\nu/\theta} e_0^{1+\gamma}}{k}$$

The left side grows without bound because $1 + \gamma > \nu$, so effort is bounded above. Together with e_0, this bounds the worker's response for fixed demand. For a positive wage, the firm's revenue is consequently bounded as $h \to \infty$, whereas payroll tends to infinity. As $h \downarrow 0$, revenue and payroll tend to zero. Continuity on $h > 0$, together with the option of zero hours, gives a profit-maximizing state allocation.

D.5 The wage first-order condition and labor supply

At fixed a, the worker's effort rule depends on assigned hours and the tip fraction, not directly on the base wage. Applying the envelope theorem to the firm's optimized state profit gives

$$\pi_w(a, w; \tau) = -h(a, w; \tau) \tag{38}$$

This equation applies at every demand state under regularity. It does not require the worker's hours to be fixed when the firm changes its wage. Differentiating Equation (11), using $N_w/N = V_w/\lambda$ and Equation (38), gives Equation (12). At a smooth interior wage optimum,

$$\frac{d \log N}{d \log w} = \frac{w V_w}{\lambda} = \frac{w \mathbb{E}_a[h]}{\mathbb{E}_a[\pi]} \tag{39}$$

This is the headcount elasticity with respect to the base wage, allowing assigned hours and effort to adjust and holding the recruitment intercept fixed. It differs from a ratio of adoption responses and from a within-job hours-supply elasticity. At the calibrated solution the elasticity is 0.339 at non-tipping firms and 0.364 at tipping firms; a \$100 increase in expected monthly job value, holding the rest of the recruitment environment fixed, raises headcount by $100[\exp(100/\lambda) - 1] = 2.88$ percent.

To see the adjustment inside V_w, differentiate the worker's payoff along the firm's hours rule. At tipping firms, effort is interior and the terms multiplying e_w cancel by the worker's first-order condition. At non-tipping firms, $e_w = 0$ because effort is e_0. In either case, within a demand state

the derivative is

$$h + (w + \tau a g_h) h_w \tag{40}$$

Taking expectations over demand gives V_w. The fixed-package approximation $w\mathbb{E}_a[h]/\lambda$ omits the hours adjustment and is not generally the employer elasticity.

D.6 Hours and revenue responses to demand

Let $\eta_h = \partial \log h / \partial \log a$ and $\eta_y = \partial \log y / \partial \log a$ denote the responses of hours and revenue to demand at a fixed posted wage and workforce. Revenue is $y = a g(h, e^\star(a, h; \tau))$, and the elasticities of g with respect to hours and effort are νs_h and $\nu(1 - s_h)$. Differentiating, and using Equations (29) and (30) to collect the terms in η_h,

$$\eta_y = 1 + \nu(1 - s_h) \frac{\partial \log e^\star}{\partial \log a}\bigg|_h + \frac{wh}{(1-\tau)y}\,\eta_h \tag{41}$$

The coefficient on η_h is the base wage bill as a share of the revenue the firm retains (Equation (31)). The revenue response is the sum of three terms: demand itself, the effort that demand induces at given hours, and the hours response scaled by the base payroll share. The ratio η_h/η_y mixes all three, which is why it is not a labor demand elasticity.

At non-tipping firms effort is fixed at e_0, so the second term is zero and the induced-effort ratio is zero. Differentiating the hours condition $a g_h = w$ at a fixed wage, with $\partial \log g_h / \partial \log h = -(1 + \theta) + (\nu + \theta) s_h$, gives

$$\eta_h = \frac{1}{(1 + \theta) - (\nu + \theta) s_h} \tag{42}$$

where the denominator is positive at a profit maximum. Substituting into Equation (41),

$$\frac{\eta_h}{\eta_y}\bigg|_{\tau=0} = \frac{1}{1 + \theta(1 - s_h)} \tag{43}$$

The ratio depends only on θ and the hours share, and it lies between the elasticity of substitution between hours and effort, $1/(1 + \theta)$, and one. When $\theta = 0$ (Cobb-Douglas), the elasticity of substitution is one, the ratio equals one, and revenue per hour is constant in demand. With effort fixed, how much the firm expands hours per unit of additional revenue reflects how readily hours substitute for the fixed input, not how labor demand responds to pay. This is the sense in which the non-tipping hours-to-revenue ratio informs θ in Section E.1.

At tipping firms with interior effort, all three terms in Equation (41) are positive, and η_h follows from differentiating the hours condition Equation (35). The tipping ratio therefore also reflects the curvature of effort costs, through the response of effort to demand in Equation (28), and the weight of effort in production. Revenue per hour rises with demand at both tipping and non-tipping firms (Section D.7).

D.7 Proofs

Proposition 1 (Effort).

At a non-tipping firm, $\tau = 0$, so the marginal return to effort is zero and $e^\star = e_0$ regardless of a. At a tipping firm, the conditional effort derivative $\partial \log e^\star / \partial \log a|_h$ in Equation (28) is positive under $\theta > 0$, $\nu > 0$, and $1 + \gamma > \nu$: the denominator exceeds $1 + \gamma - \nu > 0$ (Section D.1). An increase in a also raises the marginal revenue product of hours at any given h, so the firm increases hours ($\eta_h > 0$) by the second-order condition at the profit maximum. Since $\partial \log e^\star / \partial \log h|_a > 0$, the total effort response

$$\frac{d \log e^\star}{d \log a} = \frac{1 + (\nu + \theta) s_h \, \eta_h}{1 + \frac{\gamma e^\star}{e^\star - e_0} - \nu + (\nu + \theta) s_h}$$

is positive. □

Proposition 2 (Revenue per hour).

At a non-tipping firm, Equation (43) gives $\eta_h / \eta_y = 1/[1 + \theta(1 - s_h)] < 1$ when $\theta > 0$ and $s_h < 1$ (which holds at any interior hours optimum), so $\eta_y > \eta_h$ and revenue per hour rises with demand.

At a tipping firm, log-differentiate the hours condition Equation (10) with respect to $\log a$ at fixed w, using Equation (29) to collect terms in η_h. Because effort varies with demand, the elasticity of marginal effort cost, $\gamma e^\star / (e^\star - e_0)$, varies too: its elasticity with respect to e is $-e_0/(e^\star - e_0)$. Accounting for this variation and substituting the resulting hours response into Equation (41) gives

$$\eta_y - \eta_h = \frac{1 - s_h}{\left[1 + \frac{\gamma e^\star}{e^\star - e_0} - \nu + (\nu + \theta) s_h\right]^2 \left(-\frac{\partial \log (g_h + g_e \, \partial e^\star / \partial h)}{\partial \log h}\Big|_a\right)} \times \left[\theta \left(1 + \frac{\gamma e^\star}{e^\star - e_0} - \nu\right) \frac{\gamma e^\star}{e^\star - e_0} - \frac{(\nu + \theta)(1 + \theta s_h)}{\gamma \left(1 + \frac{\gamma e^\star}{e^\star - e_0} + \theta\right)} \frac{\gamma e^\star}{e^\star - e_0} \frac{\gamma e_0}{e^\star - e_0}\right] \quad (44)$$

The first factor in the denominator is the square of the denominator of Equation (28), which exceeds $1 + \gamma - \nu > 0$. The second factor is minus the elasticity of the marginal revenue product of hours with respect to hours, which is positive by the second-order condition at the firm's profit-maximizing hours. In the brackets, the first term is positive. The second term is positive for finite effort and comes from the variation of $\gamma e^\star / (e^\star - e_0)$ with effort; it vanishes as $e^\star / e_0 \to \infty$. Under Equation (45), which the third condition of Assumption 1 implies, the first term exceeds the second (Remark 1), so $\eta_y - \eta_h > 0$ and revenue per hour rises with demand at tipping firms as well. □

Remark 1 (A weaker sufficient condition)**.** *The third condition of Assumption 1 is sufficient but not necessary for $\eta_y - \eta_h > 0$ at tipping firms. A weaker sufficient condition, together with $\theta > 0$, $\nu > 0$, and $1 + \gamma > \nu$, is*

$$\theta \gamma \left(\sqrt{1 + \gamma - \nu} + \sqrt{1 + \gamma + \theta}\right)^2 \geq (\nu + \theta) \left(1 + \frac{\theta}{\max\{1, \nu\}}\right) \quad (45)$$

To derive it, use $\frac{\gamma e_0}{e^\star - e_0} = \frac{\gamma e^\star}{e^\star - e_0} - \gamma$*, divide the bracket in Equation* (44) *by* $\frac{\gamma e^\star}{e^\star - e_0}$*, and multiply it by* $\gamma\left(1 + \frac{\gamma e^\star}{e^\star - e_0} + \theta\right) / \frac{\gamma e_0}{e^\star - e_0}$*; both factors are positive. The bracket is therefore positive if and only if*

$$\theta\gamma\left[\frac{\gamma e_0}{e^\star - e_0} + \frac{(1+\gamma-\nu)(1+\gamma+\theta)}{\gamma e_0/(e^\star - e_0)} + (1+\gamma-\nu) + (1+\gamma+\theta)\right] \quad > \quad (\nu \ + \ \theta)(1 \ + \ \theta s_h)$$

On the left, by the inequality of arithmetic and geometric means, the first two terms in brackets sum to at least $2\sqrt{(1+\gamma-\nu)(1+\gamma+\theta)}$*, so at every* $e^\star > e_0$ *the bracket is at least*

$$\left(\sqrt{1+\gamma-\nu} + \sqrt{1+\gamma+\theta}\right)^2$$

On the right, nonnegative profit bounds s_h*. The firm can choose zero hours, so at its profit-maximizing hours the base wage bill does not exceed the revenue it retains,* $wh \leq (1-\tau)y$*. By Equations* (30) *and* (31)*, this is*

$$\frac{\nu s_h\left(1 + \frac{\gamma e^\star}{e^\star - e_0} + \theta\right)}{1 + \frac{\gamma e^\star}{e^\star - e_0} - \nu + (\nu+\theta)s_h} \leq 1$$

When $\nu > 1$*, rearranging gives*

$$s_h \leq \frac{1 + \frac{\gamma e^\star}{e^\star - e_0} - \nu}{\nu\frac{\gamma e^\star}{e^\star - e_0} + \theta(\nu - 1)} < \frac{1}{\nu}$$

where the last inequality is equivalent to $(\nu-1)(\nu+\theta) > 0$*. When* $\nu \leq 1$*,* $s_h < 1$*. In either case* $1 + \theta s_h < 1 + \theta/\max\{1, \nu\}$*, so Equation* (45) *makes the bracket positive at every* $e^\star > e_0$*. The third condition of Assumption 1 implies Equation* (45)*. The condition* $\theta(\gamma - 1) \geq 1$ *gives* $\theta\gamma \geq 1 + \theta$*, and* $1 + \gamma > \nu$ *gives* $\left(\sqrt{1+\gamma-\nu} + \sqrt{1+\gamma+\theta}\right)^2 \geq (1+\gamma-\nu) + (1+\gamma+\theta) > \nu + \theta$*. The left side of Equation* (45) *is therefore at least* $(1+\theta)(\nu+\theta)$*, which is at least its right side. Equation* (45) *holds at the estimated parameters (Section E.1).*

Proposition 3 (Combined rate).

At a non-tipping firm, $\tau = 0$, so the combined rate is w, which does not depend on a. At a tipping firm, w is posted before demand is realized and therefore fixed with respect to a. Revenue per hour y/h rises with a by Proposition 2, so $\tau \cdot y/h$ rises with a and the combined rate $w + \tau \cdot y/h$ rises with a. □

Convexity of optimized profit.

At non-tipping firms, $\pi(a, w; \tau)$ is the pointwise maximum of functions linear in a and therefore convex. At tipping firms, profit at given hours need not be convex in a because effort responds to demand. However, the firm's re-optimization of hours makes π convex. At any interior hours

optimum, the envelope derivative of per-worker profit is

$$\pi'(a) = (1-\tau)\, g(h, e^\star)\, \frac{1 + \frac{\gamma e^\star}{e^\star - e_0} + \theta s_h}{1 + \frac{\gamma e^\star}{e^\star - e_0} - \nu + (\nu + \theta) s_h} > 0$$

Log-differentiating along the optimal hours and effort paths and using the responses from Propositions 1–2, the positivity condition for $d \log \pi' / d \log a$ reduces to a polynomial in (s_h, $\gamma e^\star/(e^\star - e_0)$, θ, ν, γ) whose coefficients are all positive after substitutions that map the admissible parameter and allocation space to free positive variables. So π' is increasing and π is strictly convex on every branch where hours are interior. At boundaries where optimal hours jump between local maxima or to zero, π is the maximum of convex branches and its kinks bend upward, preserving convexity. $\square$

E Counterfactual Appendix

E.1 Calibration: moment counterparts and criterion

The demand distribution is lognormal,

$$\log a \sim \mathcal{N}\left(\log \bar{a} - \frac{s_a^2}{2}, s_a^2\right) \qquad \mathbb{E}[a] = \bar{a}$$

with $s_a = 0.28$. The recruitment intercept ξ_N is set so that non-tipping employment equals five at each candidate parameter vector and is then held fixed across regimes and wage perturbations. The effort unit $e_0 = 82.95$ is fixed when predicted hours change.

The model demand moments are local derivatives at reference demand $a = \bar{a}$, holding the posted wage and employment fixed, while their empirical counterparts are regression coefficients across firm-month observations. Writing $\eta_{h,R} = \partial \log h_R / \partial \log a$ and $\eta_{y,R} = \partial \log y_R / \partial \log a$ for regime R, the three demand targets are $\eta_{h,NT}/\eta_{y,NT}$, $\eta_{h,T}/\eta_{y,T}$, and $\eta_{y,T}/\eta_{y,NT}$. Because the model holds employment fixed within a demand realization, the local total-hours and total-revenue elasticities equal their per-worker counterparts. The within-regime ratios remove a common scaling of the demand shock; the cross-regime revenue-response ratio additionally requires the same scaling across regimes. These are joint responses to demand, not causal effects of revenue on hours or labor-supply elasticities. Section D.6 writes the three targets in terms of model primitives.

The model wage share is $\mathbb{E}[wh/y]$ and total hours are $N\mathbb{E}[h]$. Because hiring is deterministic and precedes demand, $\mathbb{E}[H] = N\mathbb{E}[h]$ and $\mathbb{E}[H/N] = \mathbb{E}[H]/N = \mathbb{E}[h]$ for a given model firm. In the data, a mean of within-firm ratios, a ratio of means, and a median target are different objects under firm heterogeneity and sample restrictions. Wage-share targets use the median of within-firm ratios, while model revenue per hour and tip rates use ratios of expectations. The non-tipping hours and wage targets are raw medians. The tipping/non-tipping hours and employment targets are ratios of sector-cell medians aggregated using common tipping-firm sector weights.

Let $r_j = \log(m_j/t_j)$ denote the model's log deviation from target j. The minimum-distance criterion is $r'Wr/9$, with

$$W = \frac{9D\widehat{\Sigma}^{-1}D}{\operatorname{tr}(D\widehat{\Sigma}^{-1}D)} \qquad D = \operatorname{diag}(t_1, \ldots, t_9)$$

where $\widehat{\Sigma}$ is the level covariance of the nine estimates across 4,000 firm-level Bayesian bootstrap repetitions. The criterion uses the inverse covariance of proportional deviations as a local approximation for weighting log deviations.

E.2 Counterfactual accounting

Within a fixed tipping regime, tip-exclusive revenue is $(1-\tau)Ny$, tips are τNy, and base payroll is wNh. Operating profit is therefore tip-exclusive revenue minus base payroll. Adding earnings net

of effort cost to operating profit per worker cancels the wage and tip transfers:

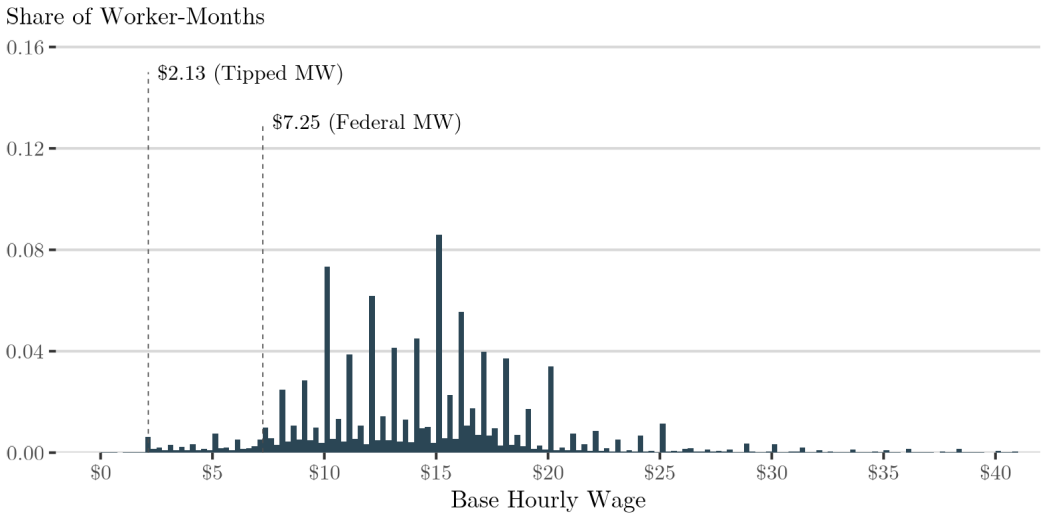
$$[wh + \tau y - c(e)] + [(1 - \tau)y - wh] = y - c(e) \tag{46}$$

This identity describes the joint value of the job and its operating profit, conditional on customer payments.

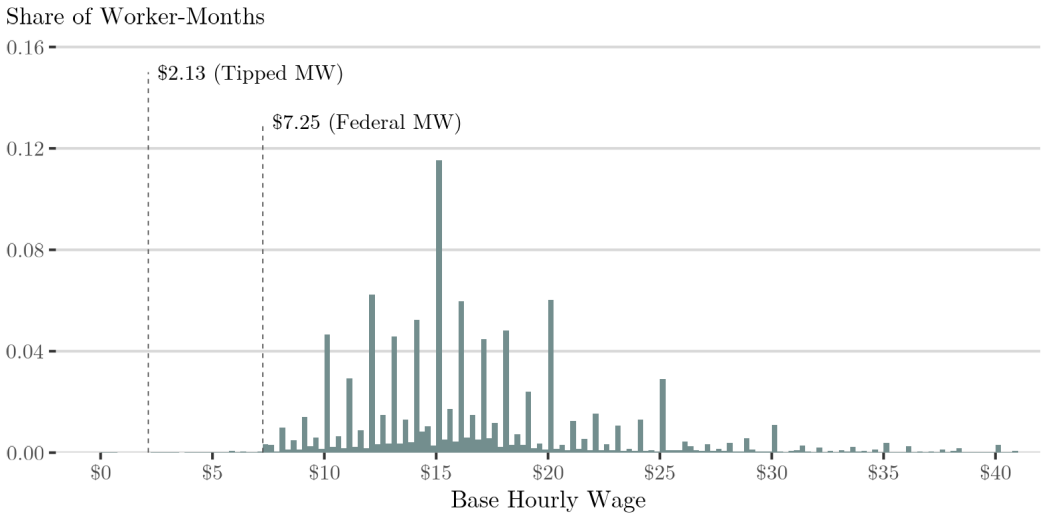
Appendix Figures

Figure A1: Base Wage Distribution, Tipping vs. Non-Tipping Firms

(a) Tipping firms

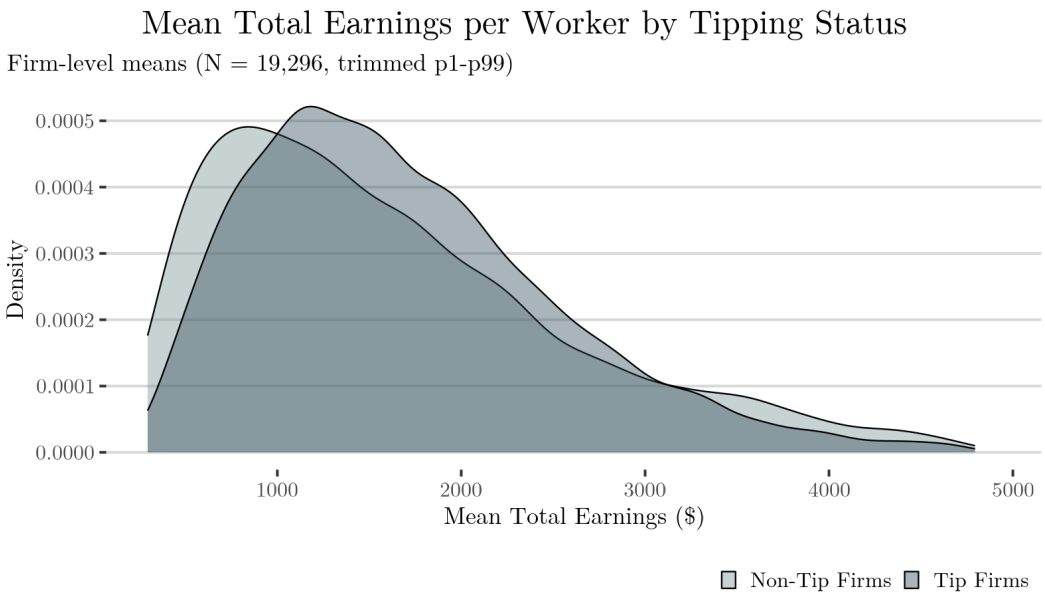


(b) Non-tipping firms



Notes: Share of worker-months by base hourly wage, in $0.25 bins, by firm type. Both distributions bunch at statutory minimums and round dollar amounts; the tipping distribution has more mass below $15 per hour.

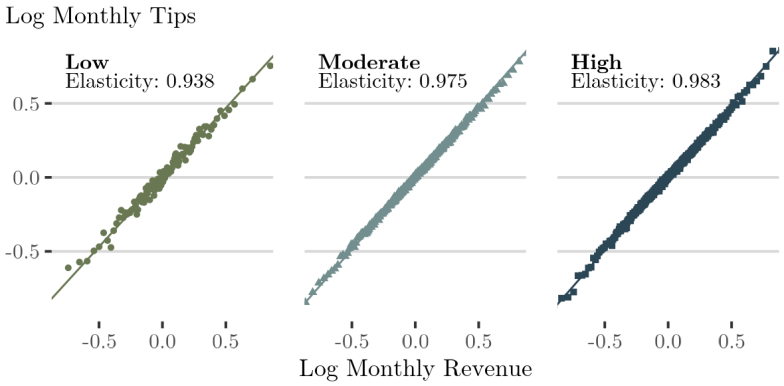
Figure A2: Total Earnings Density, Tipping vs. Non-Tipping Firms



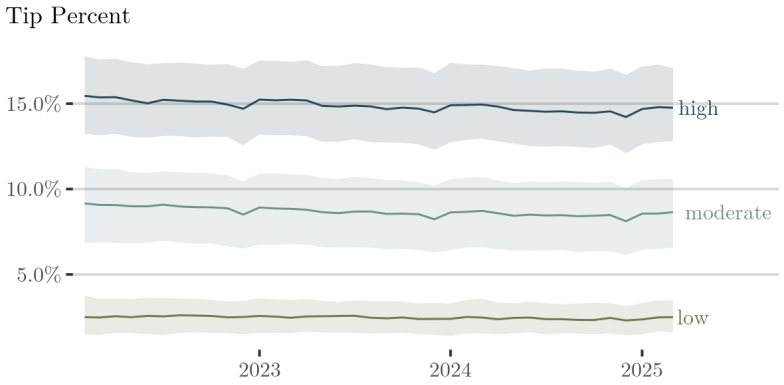
Notes: Firm-level density of mean monthly total earnings per worker (base plus tips), by firm type. Non-tipping firms have a thicker right tail of high-earnings workers, consistent with the larger gap in medians than in means.

Figure A3: Tip Percent by Tipping Category: Elasticity, Level, and Stability

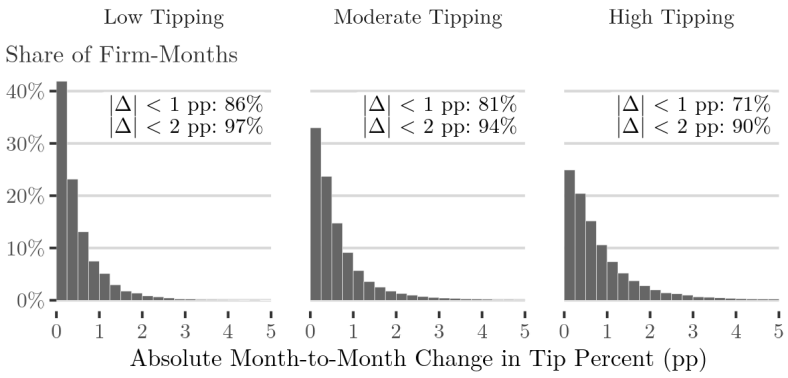
(a) Tip-revenue elasticity



(b) Median tip percent over time



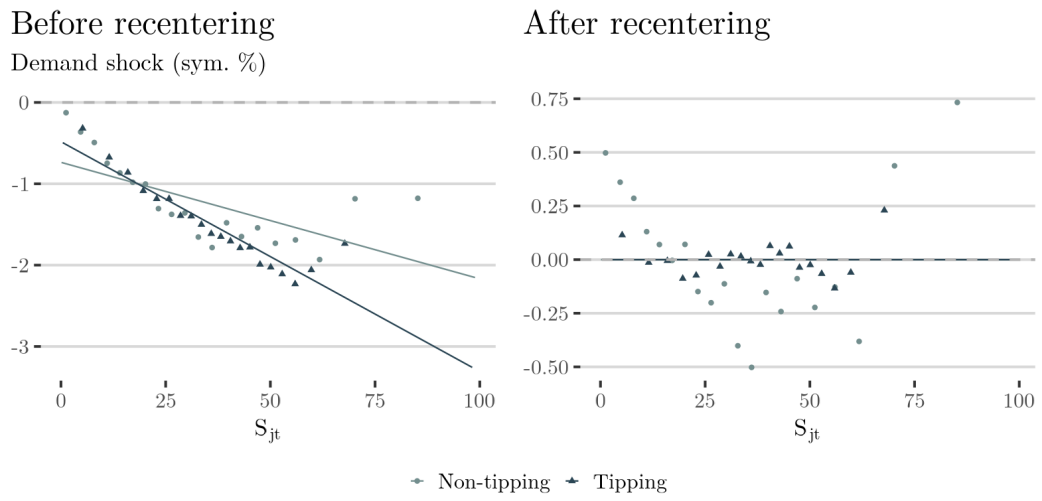
(c) Absolute month-to-month change in the tip percent



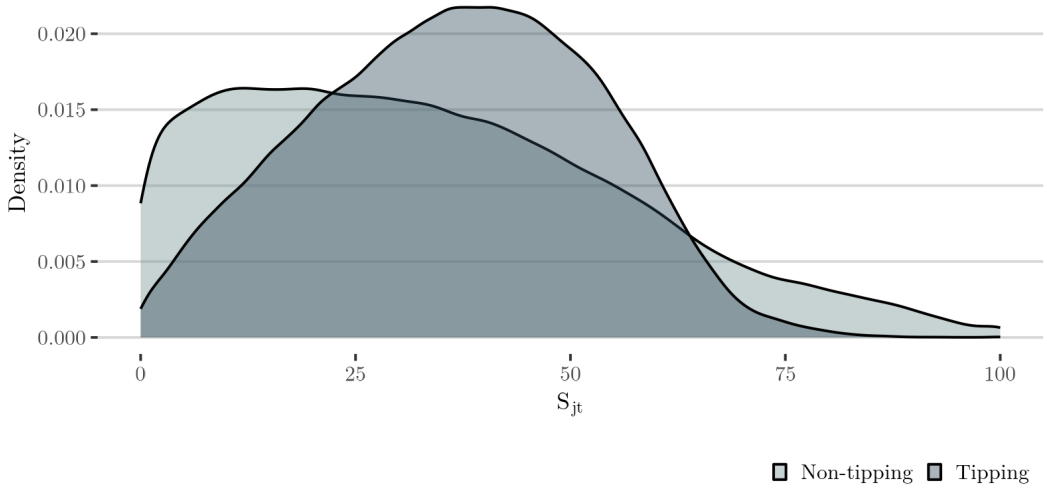
Notes: Categories are defined by each firm's median tip percent on the card-based measure used for classification (Section A.4): low (1 to 5 percent), moderate (5 to 15 percent), and high (15 percent or more). Median all-tender tip percents are 2.5, 8.6, and 14.9 percent. Panel (a): log monthly tips against log monthly revenue, residualized on firm and calendar-month fixed effects. The labeled elasticity is the slope of the fitted line. Panel (b): median tip percent by tipping category, 2022 through 2025. Shaded bands show the interquartile range. Panel (c): distribution of the absolute month-to-month change in the tip percent, by tipping category. The boxed statistics report the share of firm-months with changes below one and two percentage points. The horizontal axis is truncated at five percentage points.

Figure A4: Demand Shock Diagnostics

(a) Conditional mean of shock vs. card share (S_{jt})

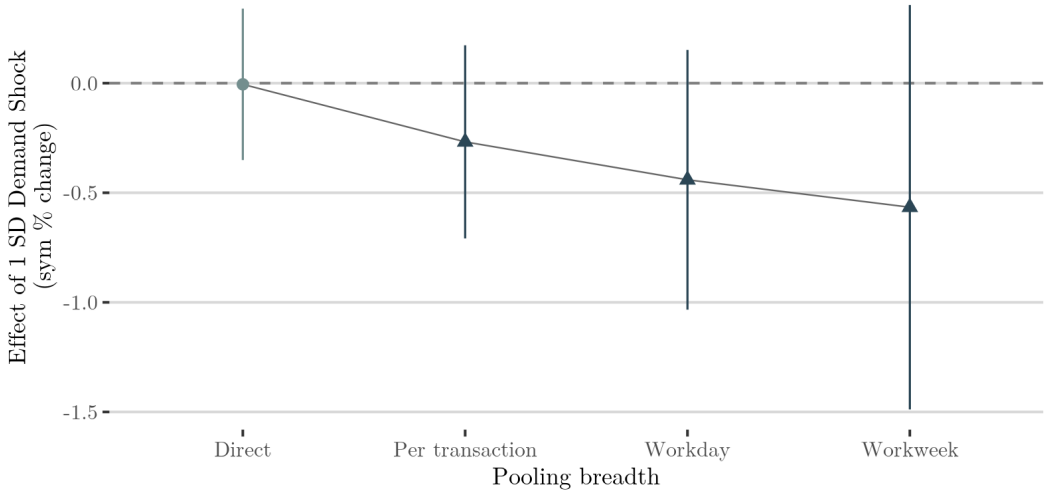


(b) Card share (S_{jt}) density at tipping and non-tipping firms



Notes: Panel (a) plots the conditional mean of the demand shock against card share S_{jt} before (left) and after (right) recentering by compensation structure following Borusyak, Hull and Jaravel (2022). Panel (b) shows the density of S_{jt} separately for tipping and non-tipping firms.

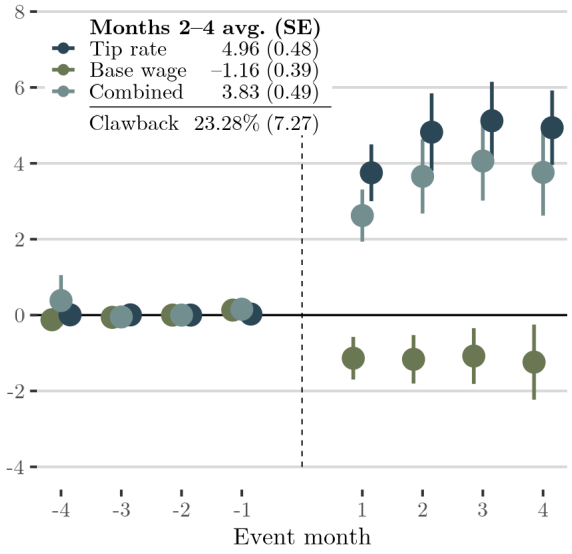
Figure A5: Wider Pooling Compresses the Earnings Response to Demand



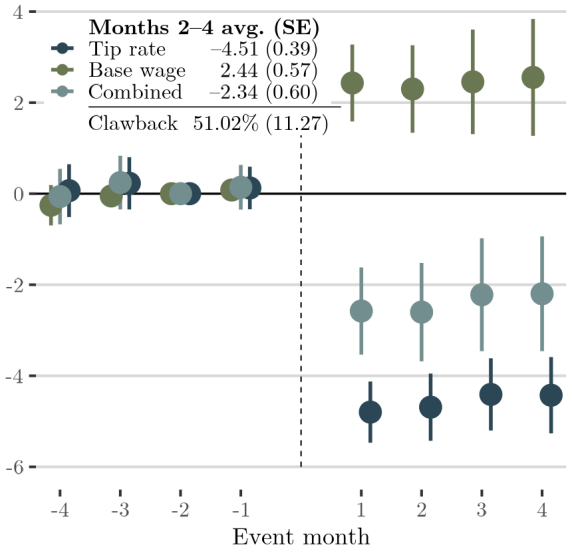
Tipping firms only, firm-month panel, 2022 to 2025. The outcome is the coefficient of variation of base-plus-tip earnings across a firm's workers in a month, in symmetric percent changes from the previous month. Each point is the response to a one-standard-deviation demand shock under one tip-distribution setting, ordered by the period over which tips are pooled: direct (-0.01), per transaction (-0.27), per workday (-0.39), and per workweek (-0.60). Bars are 95 percent confidence intervals with standard errors clustered by firm. No step from direct is distinguishable from zero.

Figure A6: Workers Moving Between Tipping and Non-Tipping Firms Face a Clawback

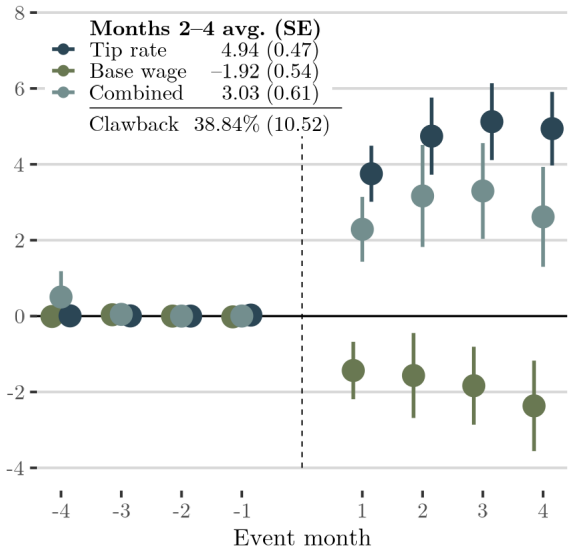
(a) Entering tipping



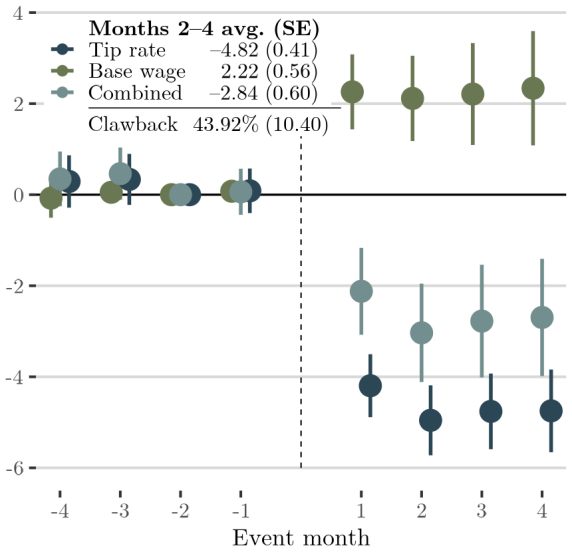
(b) Leaving tipping



(c) Entering tipping, net of job-switching gains



(d) Leaving tipping, net of job-switching gains



Notes: Panels (a) and (b): stacked difference-in-differences with later movers as controls. The sample restrictions and event-time definitions are in Section B.4; the clawback is averaged over months 2 to 4 after the move. The outcomes are the base wage, tip rate, and combined rate in dollars per hour, normalized to event month −2, the last full month at the origin employer. Entering tipping gives $\kappa = 0.23$; leaving tipping gives $\kappa = 0.51$. Panels (c) and (d): cross-type moves net of same-type moves, which establish the job-switching baseline that the triple difference removes. Clawback is 39 cents per dollar of tips entering tipping and 44 cents leaving.

Appendix Tables

Table A1: Industry Composition and Worker Characteristics

Panel A: Industry composition, percent of firms (number of firms)

	All firms	Tipping	Non-tipping	Adopters (300)	Demand-shock sample
Food and beverage	53.4 (10,500)	78.1 (9,800)	9.8 (700)	32.2 (100)	57.2 (10,500)
Retail	23.2 (4,600)	8.2 (1,000)	49.6 (3,500)	37.8 (120)	24.8 (4,500)
Other	23.4 (4,600)	13.7 (1,700)	40.6 (2,900)	29.9 (90)	18.0 (3,300)

Panel B: Worker characteristics

	Tipping	NT retail	NT other
Median workers per firm-month	8	5	5
Weekly hours, mean	19.1	19.6	21.5
Weekly hours, median	16.6	16.6	18.9
Base wage ($/hr), mean	14.47	16.40	20.04
Base wage ($/hr), median	14.00	15.00	17.00
Combined rate ($/hr), mean	20.61	16.41	20.06
Combined rate ($/hr), median	19.07	15.00	17.00
Monthly earnings ($), mean	1,819	1,559	2,077
Monthly earnings ($), median	1,353	1,069	1,475
Worker-months	3,302,300	586,200	428,000

Notes: Panel A counts firms by predicted business category. Food and beverage is the food and drink category, retail is the retail category, and other is every remaining category. All firms, tipping, and non-tipping are the analytic sample. Adopters are the 300 firms in the adoption sample. Panel B uses worker-months with at least five paid hours at analytic-sample firm-months. Weekly hours are monthly paid hours times 12/52. The base wage is regular earnings per regular hour, the combined rate adds tip earnings, and monthly earnings are total earnings.

Table A2: Sample Characteristics and External Benchmarks

Statistic	Sample	Benchmark
Share of firms with <20 employees, food and beverage	90.3%	SUSB 2022, NAICS 722: 77%
Share of firms with <20 employees, retail	96.8%	SUSB 2022, NAICS 44-45: 91%
Weekly hours, tipping (interior months)	21.4	CES food services: 23.9
Weekly hours, non-tipping retail (interior months)	21.6	CES retail: 30.3
Monthly earnings ($), tipping food and beverage (interior months, 2022)	$1,953	SUSB 2022, NAICS 722, 5-19 employees: $1,890
Base wage ($/hr), non-tipping retail (all months)	$16.40	OEWS May 2024, retail salespersons (reweighted): $17.90
Base wage ($/hr), non-tipping retail (all months)	$16.40	OEWS May 2024, cashiers (reweighted): $15.29

Notes: Benchmarks are from the Census Bureau's Statistics of U.S. Businesses (SUSB) for 2022 (U.S. Census Bureau, 2024), the BLS Occupational Employment and Wage Statistics (OEWS) for May 2024 (Bureau of Labor Statistics, 2025), and the BLS Current Employment Statistics (CES) for January 2022 to March 2025 (Bureau of Labor Statistics, 2026). CES production and nonsupervisory average weekly hours, not seasonally adjusted, January 2022 to March 2025. SUSB payroll includes Medicare wages and tips. OEWS wages are reweighted by the sample's state composition.

Table A3: Adopting Firms Before and After Adoption

	Adopters pre	Non-tipping	Adopters post	Tipping
Median workers per firm-month	6	5	6	8
Weekly hours, mean	20.0	20.2	20.6	19.1
Weekly hours, median	17.3	17.6	18.4	16.6
Base wage ($/hr), mean	16.90	17.68	16.72	14.47
Base wage ($/hr), median	15.50	16.00	15.00	14.00
Combined rate ($/hr), mean	16.90	17.70	18.35	20.61
Combined rate ($/hr), median	15.50	16.00	17.00	19.07
Monthly earnings ($), mean	1,622	1,739	1,788	1,819
Monthly earnings ($), median	1,129	1,183	1,302	1,353
Worker-months	11,100	1,146,200	11,200	3,302,300

Notes: Adopters pre covers the five months before adoption (300 firms). Adopters post covers the five months after adoption (the 290 firms observed after adoption). The non-tipping and tipping columns are the analytic sample, all months, worker-months with at least five paid hours. Definitions match Table A1.

Table A4: Revenue Volatility Gap Under Sequential Controls

	Tipping	Non-tipping	Difference	Std. error
(a) No controls	0.3509	0.4742	−0.1233	0.0054
(b) + Firm age	0.3512	0.4737	−0.1226	0.0054
(c) + Months in sample	0.3517	0.4727	−0.1210	0.0052
(d) + Seasonality	0.3603	0.4576	−0.0973	0.0054
(e) + State	0.3603	0.4575	−0.0972	0.0054
(f) + Sub-industry	0.3718	0.4372	−0.0653	0.0082
(g) + Firm size	0.3729	0.4352	−0.0623	0.0083
(h) + Transaction volume	0.3795	0.4235	−0.0440	0.0085
(i) + Bill-size SD	0.3806	0.4215	−0.0408	0.0076

Notes: Each row adds the listed control to the row above, on the same firms. The outcome is the firm-level standard deviation of log monthly revenue after removing firm and year-month fixed effects. The difference column is the coefficient on tipping status, with heteroskedasticity-robust standard errors; the tipping and non-tipping columns are adjusted means that differ by that coefficient, and row (a) reports raw group means. Firm age, months in sample, state, and sub-industry enter as fixed effects; the other controls enter linearly. Firm age is whole years from the firm's first transaction to its first sample month, top-coded at ten; months in sample is the number of months with revenue. Seasonality is the standard deviation, over the firm's sample months, of the mean outcome-residual revenue of the other firms in its sub-industry in the same month of the year; firms whose sub-industry has fewer than 20 other firms use their broader category. Firm size is log mean monthly revenue, transaction volume is log mean monthly transactions, and bill-size SD is the log standard deviation of the firm's monthly average bill. Sample: 12,600 tipping and 7,100 non-tipping firms. Visualized in Figure 4a.

Table A5: Revenue Variance Decomposition into Transaction Count and Bill Size

	Tipping	Non-tipping	Difference
N firms	12,600	7,100	
Panel A: Variance components			
$\mathrm{Var}(\ln N)$	0.2507	0.3659	−0.1152
$\mathrm{Var}(\ln P)$	0.0330	0.0873	−0.0543
$2\,\mathrm{Cov}(\ln N, \ln P)$	−0.0635	−0.0784	0.0149
$\mathrm{Var}(\ln R)$, total	0.2202	0.3748	−0.1546
Panel B: Shares of total variance			
$\mathrm{Var}(\ln N)/\,\mathrm{Var}(\ln R)$	113.9%	97.6%	16.2pp
$\mathrm{Var}(\ln P)/\,\mathrm{Var}(\ln R)$	15.0%	23.3%	−8.3pp
$2\,\mathrm{Cov}\,/\,\mathrm{Var}(\ln R)$	−28.8%	−20.9%	−7.9pp

Notes: $R = N \times P$, where N is transaction count and P is average bill size, both in logs; residuals from firm and calendar-month fixed effects. Shares are ratios of cross-firm mean variances and need not sum to 100 percent.

Table A6: Within-Spell Earnings Variance Decomposition, by Minimum Spell Length

	Var(ln rate)	Var(ln hours)	2 Cov(ln rate, ln hours)	Var(ln earnings)	Spells	Ratio
Panel A: Spells of 3 months or more						
Tipping	0.015	0.316	0.010	0.341	255,100	2.86
Non-tipping	0.005	0.290	0.001	0.296	80,200	
Panel B: Spells of 6 months or more						
Tipping	0.015	0.256	0.007	0.279	162,900	2.31
Non-tipping	0.006	0.232	0.001	0.239	53,100	
Panel C: Spells of 12 months or more						
Tipping	0.016	0.202	0.005	0.223	76,600	1.84
Non-tipping	0.009	0.179	0.002	0.189	26,800	

Notes: Worker-firm-month panel, paid hours ≥ 5. A spell is the longest consecutive run of months at a single employer. Variances are within-spell, averaged across spells. Ratio is the tipping-to-non-tipping ratio of Var(ln rate) at that minimum spell length, computed on the paid-hours ≥ 5 sample used throughout Section 2.

Table A7: Targeted Moments and Model Fit

Moment	Target	Bootstrap standard error	Model	Error (%)
Panel A: Demand responses				
NT hours/revenue demand response	0.1798	0.0705	0.1718	−4.46
T hours/revenue demand response	0.3454	0.0396	0.3025	−12.43
T/NT revenue demand response	1.8933	0.2099	1.1701	−38.20
Panel B: Levels and ratios				
NT base-payroll share	0.2717	0.0057	0.2718	+0.03
T base-payroll share	0.2644	0.0011	0.2644	−0.03
NT monthly worked hours	489.51	14.28	467.01	−4.60
T/NT worked hours	1.3517	0.0408	1.4113	+4.41
NT base wage ($/hour)	15.6744	0.1794	15.5986	−0.48
T/NT employment	1.3333	0.0851	1.2299	−7.76

Notes: T and NT denote tipping and non-tipping. Errors are 100(model/target − 1), calculated before display rounding. Bootstrap standard errors describe the empirical estimates, from 4,000 firm-level Bayesian bootstrap repetitions. All nine rows enter the criterion. Wage shares use tip-inclusive model revenue. Demand responses are local derivatives at $a = \bar{a}$. The NT hours and base wage targets are raw medians; the T/NT ratios are reweighted to the sector composition of tipping firms.